Life Expectancy and GDP Analysis

Analyzing Global Health and Economic Indicators

Werner Vorster

14 April 2025

Index: Life Expectancy and GDP Per Capita Analysis

1. Introduction

- Background on life expectancy and GDP per capita
- Importance of studying this relationship
- Clearly state your hypothesis
- Overview of the datasets used
- Structure of the report

2. Data Preparation and Cleaning (`df_tidying`)

2.1 Data Import, Cleaning, and Merging

- **2.1.1 Data import**
  - Isues with initial dataset
  - Importing correct data from official sources
  - Common data transformations
- **2.1.2 Updating Template**
  - Renaming columns to be more user friendly throughout project
  - Adjusting country data type to be string
  - Removing additional data entries in dataset
- **2.1.3 Data Cleaning and Merging**
  - World Bank Datasets
  - WHO Datasets
  - Our World in Data (OWID) Datasets

2.2 Filtering and Feature Selection

- Retained variables (*country, year, life_expectancy, GDP, etc.*)
- Converting variables to correct data types (*string, integer, float, etc*)
- Rounding float values to 2nd decimal

2.3 Enriching the Data

- Adding regional and sub-regional classifications
- Assigning income classes based on World Bank classifications

2.4 Handling Missing Data

- Methods: extrapolation, linear regression, polynomial regression, and imputation
- Justification for each approach

2.5 Data Type Adjustments

- Ensuring numerical and categorical variables are in correct formats.
- Display dataset information

2.6 Exporting Cleaned Dataset to CSV

- Cleaned Dataset is exported to CSV file for analysis

3. Data Wrangling and Statistical Analysis (`df_wrangling`)

3.1 Descriptive Statistics (Variables)

- Example code
- Mean, Max, Min, Median, Standard Deviation, Inter-quartile Range for Life Expectancy and GDP per Capita
- Combination of all statistics into a single dataframe
- Export of summary tables as CSV and PNG files

3.2 Summary Statistics for Exploratory Analysis

3.2.1 GDP-Based Country Grouping

- Life expectancy statistics for top and bottom GDP countries
- Group-specific statistics: Mean, Max, Min, Median, Std, IQR
- Aggregation and export of results as CSV and PNG files

3.2.2 Income Class-Based Grouping

- Life expectancy statistics by income class (low, lower-middle, upper-middle, high)
- Group-specific statistics: Mean, Max, Min, Median, Std, IQR
- Aggregation and export of results as CSV and PNG files

4. Exploratory Data Analysis (EDA) (`descriptive_analysis` & `EDA`)

4.1 Correlation and Heatmap Analysis

- Correlation matrices for the complete dataset and by region
- Heatmaps highlighting relationships between variables

4.2 Scatterplot Analysis

- Life expectancy vs. all other variables

4.3 Distribution Analysis

- Histograms for life expectancy and GDP per capita (overall and by region)
- Distributions by income class

4.4 Line Graphs Analysis

- Trends of life expectancy and GDP per capita over time

4.5 Boxplots for Comparison

- Life expectancy and GDP per capita across different regions and income classes

4.6 Deep Dive into GDP and Life Expectancy

- Summary statistics by income class
- Heatmaps by income class
- Scatterplots by income class
- Boxplots comparing bottom vs. top GDP countries

5. Synthesis of Analytical Insights

5.1 Recap of Key Trends from EDA

- Overview of major patterns observed accross regions and inome classes
- Notable relationships between life expectancy and economic indicators

5.2 Outlier Identification and Interpretation

- Treatments and implications of anomalies (e.g., Libya, Zimbabwe, Japan)
- What outliers reveal about sociaeconomics and healthcare disparities **5.3 Building the Narrative**
- Connecting the dots: How descriptive stats, visual patterns, and correlations shaped the analytical direction
- Justification for focusing on income classes in final comparison

6. Final Visualization and Interpretive Analysis

6.1 The final Boxplot Visualization

- Description of the visualization and why it was chosen
- Methodological considerations (e.g., grouping by income class, year selection)

6.2 Insight Accross Income Classes

- Analysis of life expectancy distributions by income level over time
- Highlights of inequality and convergence/divergence patterns

6.3 Hypothesis Evaluation

- Does the visualization support the hypothesis
- Synthesizing evidence from prior sections to justify conclusions

7. Conclusion and Broader Implications

7.1 Summary of Key Findings

- Restate the main discoveries and link them back to the hypothesis
- Reflection on what was learned through data cleaning, exploration, and synthesis

7.2 Policy and Real-World Implications

- Suggestions for how findings could inform global health or economic policies
- Notes on potential interventions or funding priorities for lower-income nations

7.3 Limitations and Future Research

- Limitations: data coverageimputation challenges, year constrains, etc
- Recomendations for expanding or deepening the analysis with more recent data, other health indicators or casuality-focused methods

---

1. Introduction

Understanding the factors that contribute to differences in life expectancy across countries is a central question in global health and development studies. Life expectancy, often considered a key indicator of a population's overall well-being, reflects the combined effects of healthcare access, economic prosperity, public policy, and societal conditions. Among these, **economic development—measured by Gross Domestic Product (GDP) per capita—is often hypothesized to be a primary driver** of improved health outcomes.

This project investigates the relationship between **GDP per capita and life expectancy** across 176 countries over a 16-year period (2000 to 2015). Using a data-driven approach, the analysis explores whether wealthier countries consistently exhibit higher life expectancies and how this relationship varies across different **regions** and **income classifications**, as defined by the World Bank. The study draws on a comprehensive, cleaned dataset comprising key indicators such as life expectancy, GDP per capita, population, health expenditure, and mortality rates, sourced from international databases including the World Health Organization (WHO), the World Bank, and Our World in Data.

Project Aim and Hypothesis

The core aim of the project is to:

Examine how GDP per capita relates to life expectancy across global income classes and regions, and determine whether economic prosperity is associated with improved longevity.

The central hypothesis tested throughout the analysis is:

Countries with higher GDP per capita generally have higher life expectancies.

This hypothesis is assessed through a systematic process of data cleaning, statistical analysis, and exploratory visualization, culminating in a final boxplot that clearly illustrates the distribution of life expectancy across different income classes over time.

Structure of the Report

The report is structured as follows:

- **Section 2** outlines the data sources, import process, cleaning methods, handling of missing values, and final dataset structure.
- **Section 3** presents statistical transformations and summarizes core indicators across different groupings.
- **Section 4** conducts in-depth exploratory data analysis (EDA), using correlation matrices, scatterplots, distributions, and boxplots to reveal trends and patterns.
- **Section 5** synthesizes key insights from the analysis to build a narrative leading to the final visualization.
- **Section 6** introduces and interprets the culminating visualization—life expectancy boxplots across income classes—used to evaluate the hypothesis.
- **Section 7** concludes the project with a summary of findings, policy implications, limitations, and suggestions for future research.

By combining technical rigor with clear visual storytelling, this project aims not only to confirm or refute the initial hypothesis but also to **contribute to a broader understanding** of the links between economic growth and population health—an understanding essential for evidence-based policymaking and global development efforts.

2. Data Preperation and Cleaning

2.1 Data Import, Cleaning, and Merging

2.1.1 Data Import

Issues with the Initial Dataset:

The initial dataset, sourced from Kaggle (uploaded by user KumarRajarshi), was found to contain incorrect data. Upon reviewing the comments from other users and conducting independent verification against official sources such as the World Bank, significant inconsistencies were identified. The population figures did not align with World Bank records, and further investigation revealed that all variables contained erroneous or unreliable data. Due to incorrect uploads or misused sources, the dataset was deemed unsuitable for analysis.

Importing Correct Data from Official Sources:

To ensure data accuracy, the necessary datasets were imported from three authoritative sources:

- **World Health Organization (WHO)**
- **World Bank (WB)**
- **Our World in Data (OWID)**

2.1 WHO Datasets

The following CSV files were downloaded from the WHO database and stored under `Datasets/Raw_datasets/WHO`:

- **Adult Mortality Rate**
- **Child Deaths per 1000 Births**
- **Infant Mortality**
- **Under-5 Mortality**

These datasets were read using Pandas' `.read_csv()` function and stored as variables.

2.1.1 **Adult Mortality** Dataset Processing

- Selected relevant columns: `'Location'`, `'Period'`, `'Dim1'`, `'FactValueNumeric'`

- Renamed columns:

    - `'Location'` → `'country'`
    - `'Period'` → `'year'`
    - `'Dim1'` → `'sex_classification'`
    - `'FactValueNumeric'` → `'adult_mortality'`
- Standardized data types (`.astype('string')`)

- Standardized country names to match the project template using `.replace()`:

    - `Cote d'Ivoire` → `Côte d'Ivoire`
    - `Netherlands (Kingdom of the)` → `Netherlands`
    - `North Macedonia` → `The former Yugoslav Republic of Macedonia`
    - `Türkiye` → `Turkey`
- Filtered the dataset to include only:

    - Data from 2000 to 2015
    - Entries labeled as "Both sexes"
- Dropped the `sex_classification` column

2.1.2 **Child Deaths**, **Infant Mortality**, and **Under-5 Mortality** Processing

- Selected relevant columns: `'Location'`, `'Period'`, `'Dim1'`, `'Dim2'`, `'Value'`
- Renamed columns:
    - `'Location'` → `'country'`
    - `'Period'` → `'year'`
    - `'Dim1'` → `'age_classification'`
    - `'Dim2'` → `'cause'`
    - `'Value'` → `'deaths'`
- Filtered and pivoted the datasets to obtain structured tables for analysis.

2.2 World Bank Datasets

The following CSV files were downloaded from the World Bank database and stored under `Datasets/Raw_datasets/World_Bank`:

- **Population**
- **GDP per Capita**
- **GNI per Capita**
- **Health Expenditure Percentage**
- **Life Expectancy**
- **Education Expenditure**
- **Unemployment Percentage**
- **Alcohol Consumption**
- **Government Health Expenditure**

These datasets were read using Pandas' `.read_csv()` function and stored as variables:

- `world_population` (loaded with `on_bad_lines='skip'` for data integrity)
- `wb_gdp` (GDP per capita)
- `wb_gni` (GNI per capita)
- `wb_health_exp_perc` (Health expenditure % of GDP)
- `world_life_expectancy` (Life expectancy at birth)
- `world_education_expenditure` (Education expenditure)
- `wb_world_unemployment_total` (Unemployment %)
- `world_alcohol_consumption` (Alcohol consumption per capita)
- `world_total_expenditure_gov` (Government health expenditure)

2.3 Our World in Data (OWID) Datasets

The following CSV files were downloaded from OWID and stored under `Datasets/Raw_datasets/Our_World_in_Data`:

- **BMI (Male and Female)**

These datasets were read and stored as:

- `world_in_data_bmi` (Male BMI)
- `world_in_data_bmi_female` (Female BMI)

3. Common Data Transformations

To standardize the imported datasets, the following transformations were applied across all datasets:

- **Column Renaming:** Standardized column names to match project conventions.
- **Year Filtering:** Restricted data to the 2000–2015 period.
- **Country Name Standardization:** Aligned country names with the project template.
- **Data Type Conversion:** Ensured categorical variables were stored as strings.
- **Sorting & Indexing:** Sorted by country and year for consistency.

2.1.2 Updating Template

To ensure consistency and usability throughout the project, the dataset template was refined through column renaming, data type standardization, and the removal of incomplete country records. These modifications enhanced the dataset's structure, making it more accessible for analysis.

Renaming Columns for Readability:

The dataset's original column names contained inconsistent capitalization, spaces, and special characters, which could hinder efficient data manipulation. To improve clarity and maintain a uniform naming convention, all column names were transformed to lowercase with words connected by underscores.

Methodology:

- The .columns method was used to inspect the dataset's variable names.
- The .rename(columns={}) method was applied to systematically rename variables.
- **Example transformation:**

  ```
  life_expectancy.rename(columns={'Country': 'country', 'Year': 'year'}, inplace=True)
  ```

- This change ensured that variable names were standardized for consistency in all future operations.

Standardizing Country Data Type

The country variable was initially stored as an object type with inconsistent formatting. To maintain uniformity across datasets, it was explicitly converted to a string format.

- Transformation applied:

  `life_expectancy['country'] = life_expectancy['country'].astype('string')`

- This standardization facilitated smooth dataset merging and prevented potential mismatches during analysis.

Removing Incomplete Country Entries

Some countries in the dataset had records for only a limited number of years, leading to inconsistencies in time series analysis. These incomplete entries were identified and removed through the following process:

1. The `.value_counts()` method was used to determine the frequency of each country in the dataset.
2. The result was converted into a dictionary where country names served as keys and their respective entry counts as values.
3. A list comprehension was applied to extract only countries with more than one entry:

   `valid_countries = np.array([key for key, value in unique_countries.items() if value > 1])`

4. The length of the resulting array was printed using the len() function to verify the number of valid countries.
5. A lambda function was used to scan through the dataset, retaining only entries present in valid_countries, while the rest were replaced with NaN values.
6. Finally, rows containing NaN values in the country column were dropped using the .dropna(subset=['country']) method to ensure that only countries with complete records remained.

2.1.3 Data Cleaning and Merging

This section outlines the data cleaning and merging processes for the three main data sources: World Bank, WHO, and Our World in Data. The objective was to ensure consistency across datasets, resolve naming discrepancies, and integrate the data into a unified structure.

1. World Bank Datasets

Selecting Relevant Data

- Data from the World Bank was filtered to include only years 2000 to 2015.
- A list of countries from the World Bank dataset (country_list_wb) and a corresponding list from the life expectancy template (country_list_le) were created to identify common and mismatched country names.

Country Name Alignment

- Countries with mismatched names were identified by comparing country_list_le to country_list_wb.
- The extractOne function from thefuzz library was used to match similar country names, with a score cutoff of 80.
- Manual adjustments were made for unmatched countries (e.g., "Korea, Dem. People's Rep.").
- The corrected names were applied to a cleaned version of the World Bank dataset (world_pop_cleaned).

Cross-Checking Countries

- A cross-checking list identified countries present in the life expectancy template but missing in the World Bank dataset (countries_cross_check).
- Missing countries such as Dominican Republic, Kyrgyzstan, and Swaziland were dropped from the Template.

Creating a Unified Dataset

- A new template (life_exp_clean) was created and sorted by country and year.
- The function modified_wb_pop_by_year() was developed to extract yearly data for population, GDP, health expenditure, and other variables.
- The cleaned and formatted World Bank dataset (wb_pop_sorted) was used to update life_exp_clean.
- This process was repeated for population, GDP, GNI, health expenditure, government health expenditure, life expectancy, unemployment, education expenditure, and alcohol consumption.

2. WHO Datasets

This section covers how data from the World Health Organization (WHO) was cleaned and merged into the main dataset. The WHO data primarily contributed to mortality statistics, including adult mortality, infant mortality, and under-5 mortality. WHO country names matched those in life_exp_clean, requiring no renaming.

2.1 Adult Mortality

- Countries absent from WHO data were identified and stored in countries_not_included.
- The adult_mortality column was updated using a left join with WHO data.
- Any missing values in adult_mortality were handled using the where() function, ensuring consistency.

2.2 Infant Mortality

- The dataset included mortality data for different causes of infant deaths. To standardize this, a new column, infant_deaths, was created by summing all causes of infant mortality for each country-year combination.
- This was done using a lambda function to apply the sum across relevant numeric columns.
- Any country that was not present in life_exp_clean was removed from the WHO dataset before merging.
- A merge operation replaced old values with WHO data, ensuring alignment across datasets.

2.3 Under-5 Mortality

- The process for under-5 mortality was identical to the steps for infant mortality:
  - All causes of under-5 deaths were summed into a new column.
  - Countries not found in life_exp_clean were dropped.
  - A merge operation replaced old values with WHO data to ensure alignment. The final dataset contained cleaned and integrated under-5 mortality data.

Key Takeaways:

- WHO data provided critical mortality statistics, integrated into life_exp_clean.
- Country names were already aligned, simplifying merging.
- Infant and under-5 mortality data required aggregation before merging.
- A left join ensured that all countries in the main dataset were preserved.

3. Our World in Data (OWID) Datasets?

BMI (Male & Female)

- Country names in OWID were checked against life_exp_clean.
- Similar to the World Bank process, extractOne was used to match country names.
- Manual corrections included:
  - "Lao People's Democratic Republic" → "Laos"
  - "Republic of Korea" → "South Korea"
  - "The former Yugoslav Republic of Macedonia" → "North Macedonia"
  - "United Kingdom of Great Britain and Northern Ireland" → "United Kingdom"
- The bmi_male and bmi_female columns were updated using cleaned OWID data.
- The Code column was dropped, and the dataset was sorted by country and year.

Final Adjustments and Verifications

- Sorting and Indexing: All datasets were sorted by country and year.
- Data Integrity Checks: The number of unique countries was verified to be 176.
- Standardization Across Sources: All variables were aligned with life_exp_clean, ensuring a unified and cleaned dataset for further analysis.

2.2 Filtering and Feature Selection

This section describes the process of filtering and selecting relevant features from the cleaned dataset to create a structured and standardized dataset for descriptive, exploratory, and inferential data analysis. This dataset forms the foundation for subsequent visualizations and statistical modeling.

1. Retained Variables

To ensure the dataset remains focused on key indicators relevant to life expectancy and economic conditions, the following variables were retained:

- country – Name of the country (176 unique countries)
- year – Observation year (2000 to 2015)
- life_expectancy – Probability of life expectancy at birth
- adult_mortality – Adult deaths per 1,000 adult population
- infant_deaths – Infant deaths per 1,000 infant population
- under_five_deaths – Under-five deaths per 1,000 under-five population
- bmi_male – Average BMI of the entire male population
- bmi_female – Average BMI of the entire female population
- health_expenditure_percentage – Expenditure on health as a percentage of Gross Domestic Product (GDP) per capita
- total_expenditure – General government expenditure on health as a percentage of total government expenditure
- population – Total population of the country
- gdp – Gross Domestic Product (GDP) per capita
- gni – Gross National Income (GNI) per capita

By filtering out unnecessary variables, the dataset remains streamlined for efficient analysis.

2. Filtering the Columns

To extract only the relevant variables, the following filtering process was applied:

```
life_exp_clean_2 = life_exp_clean.copy()

life_exp_clean_2 = life_exp_clean_2[
                                    ['country', 'year', 'life_expectancy',
'adult_mortality', 'infant_deaths', 'under_five_deaths', 'bmi_male', 'bmi_female',
'health_expenditure_percentage', 'total_expenditure', 'population', 'gdp', 'gni']]

life_exp_clean_2.rename(columns={'gdp': 'gdp_per_capita'}, inplace=True)
```

This ensures that the final dataset includes only the variables necessary for analysis while renaming GDP to gdp_per_capita for clarity.

3. Converting Variables to Correct Data Types

To maintain consistency and accuracy in numerical computations, data types were adjusted as follows:

```
life_exp_clean_2[['year', 'population']] = life_exp_clean_2[['year',
'population']].astype('Int64')
float_columns = life_exp_clean_2.select_dtypes(include='Float64').
columns
life_exp_clean_2[float_columns] = life_exp_clean_2[float_columns].
round(2)
life_exp_clean_2['country'] = life_exp_clean_2['country'].
astype('string')
life_exp_clean_2['year'] = pd.to_datetime(life_exp_clean_2['year'],
format='%Y').dt.year
```

- The year and population columns were converted to integers (Int64) to reflect their discrete nature.
- All floating-point numerical values were rounded to two decimal places to enhance readability.
- The country column was converted to a string type for categorical operations.
- The year column was briefly converted to a datetime format and then back to integer format to enable potential future time-series analysis.

These transformations ensure that the dataset is structured optimally for analysis, minimizing inconsistencies and formatting issues.

Final Adjustments and Verification

- Sorting and Indexing: The dataset was sorted by country and year to maintain chronological consistency.
- Data Integrity Checks: The total number of unique countries was verified to remain at 176.
- Standardization: All columns align with the predefined schema, ensuring uniformity across the dataset.

This structured approach results in a cleaned and well-formatted dataset, ready for further analysis and visualization.

2.3 Enriching the Data

This section describes the processes used to enrich the dataset by adding regional and sub-regional classifications and assigning income classes based on World Bank classifications. These enhancements provide additional layers of analysis, enabling a more comprehensive understanding of how geographic and economic factors relate to life expectancy.

1. Adding Regional and Sub-Regional Classifications

To enhance the dataset with regional classifications, all 176 countries included in the analysis were manually categorized into their respective regions and sub-regions. This classification was necessary to facilitate regional comparisons and better contextualize variations in life expectancy and other key indicators.

Manual Categorization Process

- Country Listings: A complete list of 176 countries was reviewed, ensuring coverage across all geographic regions.
- Dictionaries for Sorting: Manually created dictionaries (countries_by_region) were developed to assign each country to its appropriate sub-region.
- DataFrame Creation: Separate sub-regional DataFrames were generated to store country classifications.
- Concatenation into Regional DataFrames:
  - Sub-regional DataFrames were concatenated using the pd.concat() function.
  - The final regional DataFrames were created using: `pd.concat(sub-regions).reset_index(drop=True).sort_values(by=['country'], ascending=True)`
- Exporting Cleaned Datasets:
  - The newly categorized data was exported using `.to_csv()` to create structured CSV files.
  - These files were saved in the project directory: `Datasets/Cleaned_datasets/World_regions`.

Integration into the Main Dataset

Two new categorical variables, region and sub-region, were added to the life_exp_clean_2 DataFrame using a multi-loop classification system:

- A dictionary (sub_regions_dict) was created to map sub-regions to their corresponding regions.
- An iterative loop assigned each country to its respective region and sub-region.
- The classification was then stored in lists and appended to the main dataset.

The following code snippet illustrates the classification process:

```
# Generate lists of sub-regions
sub_region_keys = list(countries_by_region.keys())
sub_regions_dict = {
    'Asian region': sub_region_keys[0:5],
    'European region': sub_region_keys[5:9],
    'African region': sub_region_keys[9:14],
    'American region': sub_region_keys[14:18],
    'Oceania region': sub_region_keys[18:22]
}

# Assign region and sub-region classifications
region_classification_list = []
sub_region_classification_list = []
for country in list(life_exp_clean_2['country']):
    for sub_region in countries_by_region:
        if country in countries_by_region[sub_region]:
            for regions in sub_regions_dict:
                if sub_region in sub_regions_dict[regions]:
                    region_classification_list.append(regions)
                    sub_region_classification_list.append(sub_region)

# Add region and sub-region classifications as string-type columns
life_exp_clean_2['region'] = region_classification_list.astype('string')
life_exp_clean_2['sub_region'] = sub_region_classification_list.astype('string')
```

The inclusion of regional and sub-regional classifications allows for structured comparative analysis across different geographic areas.

2. Assigning Income Classes Based on World Bank Classifications

To further enrich the dataset, each country was assigned an income class based on the World Bank's classification thresholds. This classification aids in examining how economic factors impact life expectancy and related health indicators.

Income Classification Thresholds

The following World Bank income thresholds (2015 values) were applied:

- Low Income: GNI per capita ≤ 1,045 USD
- Lower-Middle Income: GNI per capita ≤ 4,125 USD
- Upper-Middle Income: GNI per capita ≤ 12,735 USD
- High Income: GNI per capita > 12,735 USD

Automated Income Classification Process

To categorize each country, the following approach was implemented:

1. The income class was assigned based on each country's Gross National Income (GNI) per capita in 2015.
2. The classification was determined using a lambda function within a apply() method.
3. Missing values were forward-filled (ffill()) to ensure consistency.

The following Python code was used:

```
life_exp_clean_2['income_class'] = life_exp_clean_2['gni'][life_exp_clean_2['year']
== 2015].apply(
    lambda x: 'Low income' if x <= 1045 else
              'Lower-middle income' if x <= 4125 else
              'Upper-middle income' if x <= 12735 else
              'High income'
)

# Forward-fill missing values
life_exp_clean_2['income_class'] = life_exp_clean_2['income_class'].ffill()
```

This process ensures that all 176 countries are correctly classified according to regional, sub-regional, and income groupings, enriching the dataset for advanced analysis and visualization.

2.4 Handling Missing Data

2.4.1 Inspection

To ensure data integrity and reliability in the analysis, a thorough inspection of missing values was conducted. The dataset was examined to identify missing values across all variables, including added categorical attributes such as region, sub-region, and income class.

Identification of Missing Data

A dictionary was generated to display all variables along with the corresponding count of missing entries. Additionally, a Missing Value Percentage Table was created to visualize the proportion of missing data per variable, highlighting areas that required imputation, extrapolation, or removal.

A key step in this inspection was identifying specific countries with missing values and quantifying these gaps. This was achieved by generating a dictionary of missing values per country, which facilitated a structured approach to addressing data gaps systematically.

Patterns in Missing Data

To categorize missing data trends, three distinct conditions were identified:

1. Countries missing values only for earlier years – These cases suggest that data collection for these countries commenced after a certain period.
2. Countries missing values only for later years – These cases indicate an abrupt cessation of data recording, possibly due to political instability, economic downturns, or shifts in reporting standards.
3. Countries missing all values – These cases highlight a complete lack of recorded data for specific variables, necessitating alternative handling strategies such as exclusion or alternative estimations.

The missing data distribution was further analyzed in the context of three key variables requiring imputation:

- Health Expenditure Percentage and Total Expenditure (Government Health Expenditure): Missing values were recorded for the same set of countries and years, namely Afghanistan, Iraq, Montenegro, Timor-Leste, Zimbabwe, Libya, and the Syrian Arab Republic.
- Gross National Income (GNI per capita): Missing values were primarily observed in earlier years for Afghanistan, Djibouti, and Nigeria.

Challenges in Data Imputation

- The dataset contained only 16 yearly observations per country per variable, limiting the training capability of the imputer.
- Due to insufficient data points, multiple imputation could only estimate a single fitting value rather than a range of plausible values for each missing entry.
- Economic fluctuations further complicated imputation, as drastic changes in macroeconomic variables were difficult to account for with limited data.
- In practice, this meant that for multiple imputation techniques, a single fitting value was often used to impute all missing data for a variable, reducing the robustness of the imputation model.

Testing Multiple Imputation Approaches

Several imputation strategies were explored to determine the most effective method:

- Imputation using regional-level aggregate data
- Imputation using sub-regional aggregate data
- Imputation using all non-missing data

Correlation Analysis and Insights

A seaborn heatmap of correlation matrices was employed to investigate potential relationships between missing variables and other recorded variables. The key findings were as follows:

1. **Health Expenditure Percentage:** Displayed weak correlations with other variables, limiting the effectiveness of predictive imputation. The highest correlations observed were:

- Life Expectancy: 0.36
- Adult Mortality: 0.22
- Infant Deaths: 0.36
- Under-5 Deaths: 0.28
- Male BMI: 0.36
- Female BMI: 0.15
- GDP per Capita: 0.35
- GNI per Capita: 0.41

2. **Total Expenditure (Government Health Expenditure):** Exhibited stronger correlations, suggesting potential for imputation based on related variables. Notable correlations included:

- Life Expectancy: 0.55
- Adult Mortality: 0.40
- Infant Deaths: 0.58
- Under-5 Deaths: 0.46
- Male BMI: 0.47
- Female BMI: 0.28
- GDP per Capita: 0.41
- GNI per Capita: 0.41

Summary of Missing Data Handling Methods

Based on the above analysis, three primary techniques were employed to handle missing data:

1. Imputation: Applied to variables where sufficient correlation with other variables enabled reasonable estimates, such as GNI using GDP per capita trends.
2. Extrapolation: Used where missing data followed an identifiable temporal trend, allowing for reasonable projection of missing values.
3. Removal: Countries with missing values across all years for a variable were excluded from the dataset, as no meaningful estimation could be performed.

These methodologies were applied accordingly in the following sections to address missing values for Health Expenditure Percentage, Total Expenditure (Government Health Expenditure), and GNI.

2.4.2 Creating Multiple Imputation Class

Introduction

Missing data poses a significant challenge in any analysis, particularly when working with economic and health indicators where data collection may not be consistently available across all regions and years. To address the missing values in the health_expenditure_percentage and total_expenditure variables, a Multiple Imputation Class was developed. This class leverages an advanced imputation technique by identifying countries with similar economic and health characteristics to serve as reference data for missing entries.

Identifying Missing Data Patterns

A detailed analysis of missing data patterns revealed that for health_expenditure_percentage and total_expenditure, five countries—Afghanistan, Iraq, Montenegro, Timor-Leste, and Zimbabwe—had missing data primarily in the earlier years (2000 onward). Given that data collection improved in later years, it was inferred that the missing data was likely missing at random (MAR) rather than completely missing without a pattern. This assumption supported the approach of using similar countries with complete data for imputation.

Exploring Different Imputation Approaches

Initially, multiple strategies were tested to train the IterativeImputer:

1. Using regional data (aggregating data within broader regions).
2. Using sub-regional data (a more localized approach).
3. Using the entire dataset for training the imputer.

None of these methods produced reliable results due to the limited number of observations per country (only 16 yearly data points). This limitation meant that a single fitted value was often used across all missing entries, reducing the accuracy and robustness of the imputation.

Final Imputation Strategy

The optimal method involved finding countries with similar characteristics (excluding the missing variable) and then training the imputer on these selected countries. This approach allowed for a more contextually relevant imputation by ensuring that missing values were filled using countries with comparable economic and health profiles.

The imputation class, CountryImputer, follows these steps:

1. Identify similar countries based on shared economic and health characteristics.
2. Train an imputer using data from these selected countries, excluding the variable being imputed.
3. Apply imputation to fill missing values for the target country.

This methodology accounted for structural similarities between countries rather than relying on arbitrary regional groupings.

Class Implementation and Functionality

Step 1: Finding Similar Countries

The find_similar_countries method identifies relevant reference countries using a tolerance-based filtering approach:

- A set of core variables (life_expectancy, adult_mortality, infant_deaths, under_five_deaths, bmi_male, bmi_female, gdp_per_capita, and gni) are considered.
- The method calculates maximum and minimum values for each variable within the target country.
- A tolerance margin is applied to these values to create a range within which other countries should fall.
- Countries whose variable values fall within this tolerance range are selected as reference data.

This method ensures that only countries with comparable economic and health characteristics are used in the imputation process.

Step 2: Imputing Missing Data

The impute_country method applies Iterative Imputation to the selected reference data:

- The data is first normalized (e.g., GDP per capita and life expectancy are scaled for consistency).
- Certain variables (e.g., adult_mortality, infant_deaths) are inverted since lower values correlate with higher economic and health indicators.
- The IterativeImputer is trained using the selected reference countries, excluding the missing variable.
- The trained imputer is used to predict missing values for the target country.

Addressing Economic Fluctuations

One of the key challenges encountered was the economic fluctuations over time, which could not always be accounted for with the limited available data. Since imputation relies on patterns within existing data, unpredictable shifts in economic trends (such as political instability or sudden growth spurts) could reduce accuracy. Despite this limitation, the chosen method produced more reliable estimates than other tested approaches.

Example Usage of the Imputation Class

To demonstrate the application of the CountryImputer class, consider the following example where Afghanistan's health_expenditure_percentage is imputed:

```
finder = CountryImputer(life_exp_clean_2, 'Afghanistan',
'health_expenditure_percentage')
considered_countries = finder.find_similar_countries(10, ['bmi_male',
'total_expenditure', 'gni'])
hep_multiple_imp = finder.impute_country(considered_countries)
```

In this case:

- A 10% tolerance was applied to find similar countries.
- Certain variables (bmi_male, total_expenditure, gni) were excluded to focus on economic and mortality-related indicators.
- The resulting imputation was applied only to Afghanistan for the missing years.

Outputs and Verification

Throughout the imputation process, several diagnostic outputs were generated to ensure transparency:

1. Max and Min values per variable for the target country.
2. Max and Min values with applied tolerance to identify similar countries.
3. List of selected reference countries used for imputation.
4. Years with missing data to track imputed entries.
5. Final imputed dataset preview to validate the results.

Key notes

The development of this Multiple Imputation Class provides a structured and adaptable approach to handling missing data in economic and health datasets. By leveraging similarities between countries, this method ensures that missing values are imputed in a contextually relevant manner, improving the integrity and completeness of the dataset. While economic fluctuations remain a challenge, the approach provides a significantly better alternative to traditional imputation methods, ensuring that meaningful analyses can be conducted without the bias introduced by arbitrary data removal or simplistic imputation techniques.

2.4.3 Filling the missing values:

2.4.3.1 Health Expenditure Percentage

Identifying Missing Values:

During the inspection of missing values in the health_expenditure_percentage variable, the following countries were found to have incomplete data:

```
{'Afghanistan': 2, 'Iraq': 3, 'Libya': 4, 'Montenegro': 11, 'Somalia': 16, 'Syrian
Arab Republic': 3, 'Timor-Leste': 3, 'Zimbabwe': 10}
```

To address these missing values, a multiple imputation strategy was adopted using a custom-built CountryImputer class, which applies IterativeImputer from Scikit-learn under the hood. This allowed for imputation based on similar countries and similar features, maintaining realistic values consistent with each country's economic and health profile.

Countries Imputed Using CountryImputer

1. Afghanistan

- **Missing Years:** 2000, 2001
- **Similar countries:** Guinea-Bissau, Liberia, Malawi, Rwanda
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** {2000: 10.11, 2001: 10.11}

2. Iraq

- **Missing Years:** 2000, 2001, 2002
- **Similar countries:** Azerbaijan, Egypt, Kazakhstan, Mongolia, Morocco, etc.
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** {2000: 3.16, 2001: 3.16, 2002: 3.16}

3. Montenegro

- **Missing Years:** 2000–2010 (11 years)
- **Similar countries:** Bosnia and Herzegovina, Croatia, Serbia, Slovakia, etc.
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** All missing years were imputed with 7.23

4. Timor-Leste

- **Missing Years:** 2000–2002
- **Similar countries:** Cambodia, India, Indonesia, Myanmar, etc.
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** {2000: 4.41, 2001: 4.41, 2002: 4.41}

5. Zimbabwe

- **Missing Years:** 2000–2009 (10 years)
- **Similar countries:** Malawi, Mozambique, Rwanda, Uganda, Zambia
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** All missing years were imputed with 5.87

Each imputation followed the same methodology:

1. Identify similar countries using selected features correlated with healthcare investment.
2. Apply iterative imputation constrained by max-min bounds and a tolerance range derived from existing values.
3. Evaluate the resulting values to ensure consistency with country-specific trends.

Imputation Using Extrapolation: Libya and Syrian Arab Republic

For countries where health expenditure percentage data was missing after a few years of available records, extrapolation techniques were used to fill in the gaps. Two different regression approaches were applied depending on the trend characteristics observed.

1. Libya: Polynomial Regression Imputation

````
- **Model Evaluation:**
    - **R² Score:** 0.0816
    - **Mean Squared Error (MSE):** 0.538
    - The low R² score and high MSE suggest that a linear model is a poor
fit, indicating a non-linear trend.
- **Trend Observation:**
    - A clear upward trend in health expenditure percentage was observed for
all countries in the Northern African sub-region.
    - Despite potential political reasons for missing data, a second-degree
polynomial regression was used to extrapolate values.
- **Implementation:**
    ```python
    poly_model = make_pipeline(PolynomialFeatures(degree=2),
LinearRegression())
    poly_model.fit(recorded_years, recorded_hep_values)
    poly_predicted_values = poly_model.predict(missing_years)
    ```
- **Imputed Values:**
    ```python
    {2012: 4.03, 2013: 4.49, 2014: 5.03, 2015: 5.64}
    ```
    - These values were used to fill in the missing years from 2012 onward.
````

2. Syrian Arab Republic: Linear Regression Imputation

````
- **Model Evaluation:**
    - R² Score: 0.847
    - Mean Squared Error (MSE): 0.087
    - These metrics indicate that linear regression is a strong fit for this
country’s data.
- **Trend Observation:**
    - A linear relationship was evident in the scatterplot of recorded data.
- **Implementation:**
    ```python
    model = LinearRegression()
    model.fit(recorded_years, recorded_hep_values)
    missing_hep_values_predict = model.predict(missing_years)
    ```
- **Imputed Values:**
    ```python
    {2013: 2.64, 2014: 2.45, 2015: 2.27}
    ```
    The linear trend suggests that the missing values likely follow the
recorded pattern, possibly withheld due to political unrest.
````

Dropping Somalia from the Dataset

During the data cleaning and imputation process, Somalia was removed from the dataset due to a complete absence of health expenditure percentage data across all years (2000–2015). Unlike other countries with partial missing data, Somalia lacked any available records, making it impossible to apply standard imputation techniques such as interpolation or extrapolation.

Further research into Somalia’s economic and healthcare reporting context confirmed that this was not an oversight. Somalia stands out as an exception among African nations, notably not adhering to the Abuja Declaration adopted by the African Union, which recommends allocating at least 15% of national budgets to health. Instead, estimates suggest Somalia’s health expenditure is consistently below 2%, with no reliable figures officially reported by the World Health Organization (WHO) during the study period.

Given the absence of comparable data and the unique political and economic situation of the country, retaining Somalia in the dataset would introduce substantial bias and uncertainty. Therefore, Somalia was justifiably excluded from the project dataset to maintain the overall data integrity and focus on countries with meaningful and verifiable health expenditure data.

2.4.3.2 Total Expenditure (Government Health Expenditure)

Identifying Missing Values

Upon inspection of the total_expenditure column—representing government health expenditure as a percentage of total government spending—a total of eight countries were found to contain missing values. The missing value counts per country were:

```
{'Afghanistan': 2, 'Iraq': 3, 'Libya': 4, 'Montenegro': 11, 'Somalia': 16, 'Syrian Arab Republic': 3, 'Timor-Leste': 3, 'Zimbabwe': 10}
```

The missing values occurred across two distinct patterns: countries with missing data in the earlier years (e.g., 2000–2005), and those with data gaps in the later years (e.g., post-2011). First we'll focus on countries with missing values primarily in the earlier years: Afghanistan, Iraq, Libya, Montenegro, Timor-Leste, and Zimbabwe.

Countries Imputed Using CountryImputer

1. Afghanistan

- **Missing Years:** 2000, 2001
- **Excluded Features:** ['bmi_male', 'bmi_female', 'gni']
- **Similar Countries:** Guinea-Bissau, Liberia, Malawi, Rwanda
- **Imputed Values:** {2000: 2.01, 2001: 2.01}

2. Iraq

- **Missing Years:** 2000–2002
- **Excluded Features:** ['bmi_male', 'gdp_per_capita', 'gni']
- **Similar Countries:** Azerbaijan, Egypt, Morocco, Tajikistan
- **Imputed Values:** {2000: 2.82, 2001: 2.82, 2002: 2.82}

3. Montenegro

- **Missing Years:** 2000–2010 (11 years total)
- **Excluded Features:** ['bmi_male', 'gni']
- **Similar Countries:** Bosnia and Herzegovina, Bulgaria, Chile, Croatia, Cuba, Czechia, Estonia, Hungary, Lebanon, Maldives, Montenegro, Poland, Serbia, Slovakia, The former Yugoslav republic of Macedonia
- **Imputed Values:** 2000–2010: 10.81% (consistent across years)

4. Timor-Leste

- **Missing Years:** 2000–2002
- **Excluded Features:** ['bmi_male', 'gni']
- **Similar Countries:** Indonesia, Nepal, Yemen, among others
- **Imputed Values:** 2000–2002: 2.86%

5. Zimbabwe

- **Missing Years:** 2000–2009
- **Excluded Features:** ['bmi_male', 'gdp_per_capita', 'gni']
- **Similar Countries:** Malawi, Rwanda, Tanzania, Uganda
- **Imputed Values:** 2000–2009: 8.18%

Imputation Using Extrapolation: Libya and Syrian Arab Republic

In the dataset, both Libya and the Syrian Arab Republic have missing values for government health expenditure (total_expenditure) from 2012 onward and 2013 onward, respectively. Unlike countries where data is missing in earlier years due to late adoption of data collection, the absence of values in these later years suggests a different mechanism — Not Missing at Random (NMAR). It is reasonable to assume that political instability and civil conflict, which escalated during the missing periods, influenced the availability and publication of official health expenditure data.

Given this context, extrapolation was used to estimate the missing values using either linear or polynomial regression, depending on the observed data trend. Visual inspection through scatter plots was conducted to determine the appropriate method for each country.

1. Libya: Polynomial Regression Imputation

````
- **Model Evaluation:**
    - R² Score: 0.308
    - Mean Squared Error (MSE): 0.817
    - The low R² score indicates a weak linear relationship — the linear
model explains only about 30.8% of the variance in the observed data. The
relatively high MSE further supports that a linear model is not a good fit
for Libya’s trend in government health expenditure.
- **Trend Observation:**
    - The scatter plot of Libya’s recorded total_expenditure data showed a
non-linear pattern, inconsistent with a straight-line trend.
    - While Libya’s values fluctuated, neighboring North African countries
displayed a clearer upward trend. This regional context, along with Libya's
earlier trajectory, supported the use of polynomial regression to better
capture the curved progression.
- **Implementation:**
    ```python
    poly_model = make_pipeline(PolynomialFeatures(degree=2),
LinearRegression())
    poly_model.fit(recorded_years, recorded_hep_values)
    poly_predict_known_values = poly_model.predict(recorded_years)
    poly_predicted_values = poly_model.predict(missing_years)
    rounded_poly_predicted_values = np.round(poly_predicted_values, 2)
    ```
- **Imputed Values:**
    ```python
    {2012: 5.07, 2013: 5.23, 2014: 5.44, 2015: 5.7}
    ```
````

2. Syrian Arab Republic: Linear Regression Imputation

````
- **Model Evaluation:**
    - R² Score: 0.416
    - Mean Squared Error (MSE): 0.733
    - The R² score indicates a moderate fit — the linear model explains
approximately 41.6% of the variance. The MSE is slightly lower than Libya’s,
suggesting a more stable and consistent trend. Overall, the linear model was
reasonably well-suited for Syria’s expenditure trend.
- **Trend Observation:**
    - The scatter plot revealed a clear and consistent downward linear trend
in total_expenditure before 2013.
    - This supported the use of linear regression to estimate missing values
from 2013 to 2015, in line with the observed trajectory.
- **Implementation:**
    ```python
    model = LinearRegression()
    model.fit(recorded_years, recorded_hep_values)
    missing_te_values_predict = model.predict(missing_years)
    ```
- **Imputed Values:**
    ```python
    {2013: 4.87, 2014: 4.68, 2015: 4.49}
    ```
    The linear trend suggests that the missing values likely follow the
recorded pattern, possibly withheld due to political unrest.
````

2.4.3.3 GNI per Capita

Overview

During the inspection of the gni (Gross National Income per capita) column, missing values were identified in three countries. The following dictionary summarizes the number of missing values per country:

```python
{'Afghanistan': 2, 'Djibouti': 15, 'Nigeria': 8}
```

Given that GNI per capita closely mirrors the trends of GDP per capita, both in economic reporting and statistical progression, it was deemed reasonable to use GDP per capita as a predictor in imputing the missing GNI values. This assumption allowed the use of Iterative Imputation methods that model missing data based on multivariate relations, specifically between GDP per capita and GNI per capita.

Imputation Approach

For all three countries, the IterativeImputer from Scikit-learn was used. This approach models each feature with missing values as a function of other features and iteratively refines the imputed values. Since the relationship between GDP per capita and GNI per capita tends to be linear and stable over time, it served as an appropriate foundation for imputation.

1. Afghanistan

- **Imputation Logic:** Afghanistan had sufficient non-missing data points to train its imputer using only its own records.
- **Implementation:**
  - **Selected columns:** ['gdp_per_capita', 'gni']
  - Fit and transformed values using IterativeImputer
  - Rounded results to one decimal place
  - Replaced missing gni values in Afghanistan's records using the imputed output
- **Imputed Values:**
  `{2000: 131.8, 2001: 169.4}`
- **Justification:** Since Afghanistan's dataset had only two missing values and a good quantity of existing records, the internal trend between GDP and GNI was expected to yield accurate predictions without requiring external regional data.

2. Djibouti

- **Imputation Logic:** Djibouti had 15 missing GNI values, which made internal training less reliable. Instead, the imputer was trained using the complete GDP and GNI data of all countries within the East African sub-region, to which Djibouti belongs.
- **Implementation:**
  - Subset of training data: East African countries with complete gdp_per_capita and gni
  - Trained IterativeImputer on this regional data
  - Transformed missing values for Djibouti
  - Replaced missing gni values using the second column of the imputer output
- **Imputed Values:**
  `{2000: 706.24, 2001: 710.99, 2002: 712.25, 2003: 733.81, 2004: 774.96, 2005: 812.68, 2006: 865.83, 2007: 935.69, 2008: 1082.37, 2009: 1113.92, 2010: 1175.42, 2011: 1267.24, 2012: 1359.86, 2013: 2021.37, 2014: 2153.82}`
- **Justification:** Regional socioeconomic factors are often shared within geographical groupings. Thus, imputing based on sub-regional data provided a stronger model than relying solely on Djibouti's limited historical data.

3. Nigeria

- **Imputation Logic:** Nigeria had 8 missing values. Similar to Djibouti, the imputer was trained using complete GDP and GNI values from the West African sub-region, which reflects Nigeria's regional economic context.
- **Implementation:**
  - Subset of training data: West African countries with no missing gdp_per_capita or gni
  - Trained and applied IterativeImputer to predict Nigeria's missing gni values
  - Replaced missing values in Nigeria's dataset accordingly
- **Imputed Values:**
  ```
  {2000: 537.7, 2001: 536.2, 2002: 695.6, 2003: 744.9, 2004: 935.7, 2005: 1174.4,
  2006: 1546.5, 2007: 1754.2}
  ```
- **Justification:** Although Nigeria has a large economy, regional context still improves imputation by capturing the economic behavior of peer nations with shared development patterns.

2.5 Data Type Adjustment

To optimize both memory efficiency and semantic clarity in the dataset, appropriate data type conversions were performed. Specifically, categorical variables were explicitly cast to the category data type in pandas. This step is essential for large datasets as it reduces memory usage and improves performance in operations such as grouping, sorting, and filtering. Additionally, assigning ordered categories can help preserve logical hierarchies in later visualizations and statistical modeling.

2.5.1 Income Class

The income_class column represents a natural ordinal scale ranging from low to high-income categories. This variable was explicitly converted into an ordered categorical type, following the World Bank classification:

```
['Low income', 'Lower-middle income', 'Upper-middle income', 'High income']
```

By setting the `ordered=True` parameter, the column retains its inherent order, which is useful for ordered visualizations and analysis (e.g., boxplots segmented by income class).

2.5.2 Region

The `region` column, while not ordinal, represents discrete geopolitical zones (e.g., Asia, Africa, Europe). This was converted to an unordered categorical type, ensuring efficient grouping and summarization without implying any hierarchy.

2.5.3 Sub-region

Similar to `region`, the `sub_region` column was also cast to a categorical type. This preserves its use for stratified analysis while benefiting from optimized storage.

By converting these columns, the dataset is not only lighter in memory but is also more semantically aligned for downstream analysis and plotting tasks, especially in libraries such as seaborn and matplotlib.

2.5.4 Display dataset information

The cleaned Life Expectancy dataset contains 2,800 entries (rows) and 16 columns, all of which have no missing values, indicating that the dataset is fully preprocessed and analysis-ready.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2800 entries, 0 to 2799
Data columns (total 17 columns):
 #   Column                         Non-Null Count  Dtype  
---  ------                         --------------  -----  
 0   Unnamed: 0                     2800 non-null   int64  
 1   country                        2800 non-null   object 
 2   year                           2800 non-null   int64  
 3   life_expectancy                2800 non-null   float64
 4   adult_mortality                2800 non-null   float64
 5   infant_deaths                  2800 non-null   float64
 6   under_five_deaths              2800 non-null   float64
 7   bmi_male                       2800 non-null   float64
 8   bmi_female                     2800 non-null   float64
 9   health_expenditure_percentage  2800 non-null   float64
 10  total_expenditure              2800 non-null   float64
 11  population                     2800 non-null   int64  
 12  gdp_per_capita                 2800 non-null   float64
 13  gni                            2800 non-null   float64
 14  region                         2800 non-null   object 
 15  sub_region                     2800 non-null   object 
 16  income_class                   2800 non-null   object 
dtypes: float64(10), int64(3), object(4)
memory usage: 372.0+ KB
None
```

Data Types Overview

- Numeric Columns:
  - int32: 1 column (year)
  - int64: 1 column (population)
  - float64: 10 columns (life_expectancy, adult_mortality, infant_deaths, under_five_deaths, bmi_male, bmi_female, health_expenditure_percentage, total_expenditure, gdp_per_capita, gni)
- Categorical Columns:
  - category: 3 columns (region, sub_region, income_class)
- String Columns:
  - string: 1 column (country)

Memory Usage

The entire dataset uses approximately 285.6 KB of memory, which is notably efficient given the data volume. This optimization was aided by converting relevant columns (e.g., regions and income classes) to the category data type.

Column Completeness

All columns have 2,800 non-null entries, confirming successful handling of previously missing values through appropriate imputation and cleaning strategies.

2.6 Exporting Cleaned Dataset to CSV

With all cleaning, imputation, and formatting processes completed, the fully processed DataFrame (`life_exp_clean_2`) was saved to a CSV file for future access and reproducibility:

```
life_exp_clean_2.to_csv("../Datasets/Cleaned_datasets/Life_expectancy_cleaned_1.csv")
```

The file was exported to the `Cleaned_datasets` directory under the main `Datasets` folder. This clean, analysis-ready file serves as the finalized version of the dataset for use in visualization, modeling, and reporting tasks in subsequent phases of the project. Saving this cleaned version ensures that all data wrangling efforts are preserved and provides a clear separation between raw and prepared data sources.

---

3. Data Wrangling and Statistical Analysis

3.1 Descriptive Statistics (Variables)

Code Explanation: How Summary Statistics Were Generated

To analyze life expectancy trends over time, descriptive statistics were calculated for each year from 2000 to 2015. These statistics include the mean, maximum, minimum, median, standard deviation, and inter-quartile range (IQR). Below is a brief example of how each was computed:

- **Mean:** For each year, the average life expectancy was calculated using `np.mean()` after filtering the dataset by year.
  ```
  mean = np.mean(life_expectancy['life_expectancy'].where(life_expectancy.year ==
  year))
  ```
- **Maximum and Corresponding Country:** The highest life expectancy value per year was identified using `np.max()`, and the corresponding country was retrieved using boolean indexing.
  ```
  max = np.max(life_expectancy['life_expectancy'][life_expectancy.year == year])
  max_country = life_expectancy['country'][(life_expectancy.year == year) &
  (life_expectancy['life_expectancy'] == max)].values[0]
  ```
- **Minimum and Corresponding Country:** The lowest life expectancy per year was calculated with `np.min()`, and the corresponding country was identified similarly to the max.
  ```
  min = np.min(life_expectancy['life_expectancy'].where(life_expectancy.year ==
  year))
  min_country = life_expectancy['country'][(life_expectancy.year == year) &
  (life_expectancy['life_expectancy'] == min)].values[0]
  ```
- **Median:** The median life expectancy was calculated using `np.median()` after removing missing values.
  ```
  median =
  np.median((life_expectancy['life_expectancy'].where(life_expectancy.year ==
  year)).dropna())
  ```
- **Standard Deviation:** The standard deviation, which quantifies variation or dispersion in life expectancy for each year, was calculated with `np.std()`.
  ```
  std = np.std((life_expectancy['life_expectancy'].where(life_expectancy.year ==
  year)).dropna())
  ```

- **Inter-quartile Range (IQR):** The IQR was computed as the difference between the 75th percentile (Q3) and 25th percentile (Q1), providing a robust measure of spread.
  ```
  q3, q1 = np.percentile(life_expectancy['life_expectancy']
  [life_expectancy['year'] == year], [75, 25])
  iqr = q3 - q1
  ```

After computing each metric, the resulting dictionaries were converted into DataFrames and merged into a final summary dataset:

```
descriptive_stats_life_exp = le_all_means_df.merge(le_all_max_df, on='year')\
    .merge(le_all_min_df, on='year')\
    .merge(le_all_median_df, on='year')\
    .merge(le_all_std_df, on='year')\
    .merge(le_all_iqr_df, on='year')
```

3.1.1 Life Expectancy

Summary statistics were computed for each year between 2000 and 2015. See below tables containing descriptive statistics for all data and per region:

Global Data:

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	66.77	81.08	Japan	44.52	Malawi	69.75	9.76	60.6	73.96	13.37
2001	67.1	81.42	Japan	41.96	Zimbabwe	69.96	9.77	60.8	74.32	13.52
2002	67.38	81.69	Japan	44.56	Zimbabwe	70.44	9.64	61.25	74.56	13.31
2003	67.67	81.76	Japan	43.39	Zimbabwe	70.72	9.54	61.51	74.6	13.09
2004	68.03	82.03	Japan	43.71	Lesotho	70.86	9.45	61.7	75.19	13.49
2005	68.38	81.96	Japan	43.23	Lesotho	71.25	9.31	61.88	75.19	13.31
2006	68.76	82.32	Japan	42.91	Lesotho	71.2	9.15	62.15	75.6	13.44
2007	69.1	82.51	Japan	43.12	Lesotho	71.42	8.99	62.58	75.7	13.12
2008	69.43	82.59	Japan	43.57	Lesotho	71.66	8.87	63.16	76.06	12.9
2009	69.84	82.93	Japan	44.03	Lesotho	71.95	8.67	63.61	76.26	12.64
2010	70.14	82.84	Japan	45.6	Lesotho	72.37	8.67	64.17	76.46	12.3
2011	70.6	82.7	Switzerland	46.69	Lesotho	72.56	8.36	64.34	76.74	12.4
2012	70.88	83.1	Japan	47.84	Lesotho	72.61	8.18	64.68	76.99	12.31
2013	71.17	83.33	Japan	49.0	Lesotho	72.74	8.06	64.8	77.13	12.33
2014	71.45	83.59	Japan	50.03	Lesotho	72.97	8.0	65.02	77.44	12.42
2015	71.65	83.79	Japan	51.1	Lesotho	73.14	7.79	65.51	77.54	12.03

- **Consistent Increase Over Time:** The global mean life expectancy rose steadily from 66.77 years (2000) to 71.65 years (2015) — an increase of nearly 5 years over the 15-year period.
- **Shifting Distribution:** The median rose alongside the mean (from 69.75 to 73.14), indicating an upward shift across the population, not just at the extremes.
- **Narrowing Spread:** The standard deviation decreased from 9.76 in 2000 to 7.79 in 2015, suggesting convergence among countries and a reduction in disparity.
- **Extremes:**
  - Japan consistently recorded the highest life expectancy, peaking at 83.79 years in 2015.
  - Lesotho had the lowest values throughout much of the dataset, although its life expectancy also improved (from 43.71 in 2004 to 51.1 in 2015).

- **IQR Trends:** The interquartile range (IQR) slightly decreased from 13.37 to 12.03, indicating that even the middle 50% of countries became more similar in life expectancy.

These trends point toward global improvements in life expectancy and reduced inequality, though certain countries continue to lag significantly behind.

Africa

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	55.08	73.69	Tunisia	44.52	Malawi	52.7	8.06	49.36	58.43	9.07
2001	55.37	73.82	Tunisia	41.96	Zimbabwe	53.15	8.1	49.9	58.57	8.67
2002	55.75	74.06	Tunisia	44.56	Zimbabwe	53.82	7.85	50.6	58.96	8.36
2003	56.16	74.31	Tunisia	43.39	Zimbabwe	54.1	7.76	50.95	59.22	8.27
2004	56.72	74.49	Tunisia	43.71	Lesotho	55.26	7.7	51.57	59.89	8.33
2005	57.24	74.76	Tunisia	43.23	Lesotho	55.71	7.58	52.34	60.37	8.03
2006	57.86	74.92	Tunisia	42.91	Lesotho	56.14	7.43	53.46	61.06	7.6
2007	58.45	75.07	Tunisia	43.12	Lesotho	56.52	7.34	54.22	61.64	7.42
2008	59.0	75.18	Tunisia	43.57	Lesotho	57.36	7.15	54.81	61.91	7.1
2009	59.61	75.27	Tunisia	44.03	Lesotho	58.12	7.0	55.77	62.71	6.93
2010	60.21	75.42	Tunisia	45.6	Lesotho	59.16	6.75	56.44	62.98	6.55
2011	60.73	75.46	Tunisia	46.69	Lesotho	60.02	6.45	57.08	63.33	6.25
2012	61.36	75.55	Tunisia	47.84	Lesotho	60.36	6.39	57.65	63.74	6.08
2013	61.85	75.65	Tunisia	49.0	Lesotho	60.73	6.22	58.29	63.95	5.66
2014	62.27	75.65	Tunisia	50.03	Lesotho	61.27	6.13	58.85	64.27	5.42
2015	62.74	75.69	Tunisia	51.1	Lesotho	61.64	5.97	59.21	64.66	5.45

- **Consistent Increase Over Time:** The mean life expectancy in Africa rose from 55.08 years (2000) to 62.74 years (2015) — an improvement of 7.66 years over the 15-year period. This rate of increase is notably higher than the global average of around 5 years.
- **Top Performers:** Tunisia held the highest life expectancy across all years, peaking at 75.69 years in 2015.
- **Lowest Performers:** Initially, Malawi and Zimbabwe had the lowest values (as low as 41.96 in 2001). From 2004 onward, Lesotho became the country with the lowest life expectancy, though even Lesotho saw improvement from 43.71 (2004) to 51.10 (2015).
- **Median Growth:** The median life expectancy increased steadily from 52.70 (2000) to 61.64 (2015), showing improvement not just at the top, but across the continent.
- **Decreasing Spread and Inequality:** The standard deviation dropped from 8.06 (2000) to 5.97 (2015), and the IQR (Interquartile Range) declined from 9.07 to 5.45, indicating a reduction in inequality and a convergence in life expectancy across African countries.
- **Quartile Movement:**
  - 25th percentile rose from 49.36 to 59.21
  - 75th percentile rose from 58.43 to 64.66
  - This upward shift in both percentiles further confirms the continent-wide progress.

Contextual Notes:

- The improvements in the minimum values suggest significant progress in the most disadvantaged countries.
- The narrowing of standard deviation and IQR shows life expectancy gains are more equitably distributed in Africa than before, even if absolute values remain lower than global figures.

Asia

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	69.13	81.08	Japan	55.3	Afghanistan	70.36	5.96	64.89	73.47	8.58
2001	69.59	81.42	Japan	55.8	Afghanistan	70.79	5.87	65.16	73.81	8.65
2002	69.96	81.69	Japan	56.45	Afghanistan	71.02	5.76	65.39	73.87	8.48
2003	70.31	81.76	Japan	57.34	Afghanistan	71.37	5.68	65.64	74.14	8.5
2004	70.48	82.03	Japan	57.94	Afghanistan	71.37	5.64	65.75	74.33	8.58
2005	70.92	81.96	Japan	58.36	Afghanistan	71.84	5.58	65.86	74.73	8.87
2006	71.23	82.32	Japan	58.68	Afghanistan	71.99	5.53	66.36	74.94	8.58
2007	71.53	82.51	Japan	59.11	Afghanistan	72.33	5.51	66.71	75.19	8.48
2008	71.7	82.59	Japan	56.51	Myanmar	72.4	5.71	67.11	75.49	8.38
2009	72.16	82.93	Japan	60.36	Afghanistan	72.7	5.37	67.9	75.84	7.94
2010	72.55	82.84	Japan	60.85	Afghanistan	73.16	5.34	68.29	76.27	7.98
2011	72.83	82.59	Japan	61.42	Afghanistan	73.31	5.26	68.57	76.61	8.04
2012	72.99	83.1	Japan	61.92	Afghanistan	73.45	5.32	68.72	77.06	8.34
2013	73.22	83.33	Japan	62.42	Afghanistan	73.68	5.37	68.89	77.25	8.36
2014	73.46	83.59	Japan	62.54	Afghanistan	73.86	5.41	69.07	77.45	8.38
2015	73.69	83.79	Japan	62.66	Afghanistan	73.88	5.36	69.64	77.72	8.08

- **Steady Upward Trend:** The mean life expectancy in Asia improved from 69.13 (2000) to 73.69 (2015) — a +4.56 year increase. This is slightly below the improvement seen in Africa (+7.66 years), but Asia started from a higher baseline.
- **Highest Values** – Japan Dominates: Japan consistently had the highest life expectancy every year, starting at 81.08 (2000) and climbing to 83.79 (2015) — consistently above the global average and setting the regional benchmark.
- **Lowest Values** – Afghanistan: *Afghanistan* remained the lowest-performing country for nearly all years except 2008, when Myanmar briefly held the lowest spot.
  - In 2000, Afghanistan's life expectancy was 55.3
  - By 2015, it had increased to 62.66 While still the lowest in Asia, this is a +7.36 year gain, aligning with African progress.
- **Median Life Expectancy:** The median increased from 70.36 (2000) to 73.88 (2015), indicating that more than half of Asian countries reached above 73.88 years by 2015.
- **Slight Decrease in Dispersion:** The standard deviation decreased from 5.96 to 5.36, and the IQR (Interquartile Range) from 8.58 to 8.08, pointing to modest convergence in health outcomes across the region.
- **Percentile Shifts:**
  - 25th percentile improved from 64.89 to 69.64
  - 75th percentile rose from 73.47 to 77.72
  - This shift confirms broad-based regional improvements, particularly among countries in the lower half of the distribution.

Contextual Notes:

- Asia started from a higher life expectancy baseline than Africa and has maintained this lead.
- The most significant improvements occurred in countries with initially low life expectancies, reducing the region's overall disparity.

America

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	71.14	79.17	Canada	58.36	Haiti	71.32	4.5	68.88	74.04	5.17
2001	71.37	79.4	Canada	58.49	Haiti	71.5	4.49	69.31	74.33	5.02
2002	71.67	79.53	Canada	58.86	Haiti	71.94	4.5	69.7	74.61	4.91
2003	71.81	79.72	Canada	59.51	Haiti	72.35	4.46	69.62	74.84	5.22
2004	72.06	79.99	Canada	58.48	Haiti	72.48	4.54	70.07	75.2	5.13
2005	72.41	80.11	Canada	60.42	Haiti	72.99	4.34	70.27	75.48	5.21
2006	72.68	80.54	Canada	60.76	Haiti	73.17	4.24	70.52	75.6	5.08
2007	72.92	80.54	Canada	61.1	Haiti	73.22	4.11	70.88	75.74	4.85
2008	73.17	80.72	Canada	61.33	Haiti	73.38	4.07	71.26	75.93	4.67
2009	73.5	81.07	Canada	61.74	Haiti	73.47	3.99	71.61	76.14	4.53
2010	73.13	81.32	Canada	46.02	Haiti	73.71	5.94	71.95	76.09	4.14
2011	73.89	81.48	Canada	61.62	Haiti	74.13	3.99	72.28	76.38	4.1
2012	74.06	81.66	Canada	62.29	Haiti	74.15	3.92	72.53	76.64	4.11
2013	74.22	81.74	Canada	62.6	Haiti	73.96	3.89	72.72	76.76	4.05
2014	74.36	81.78	Canada	62.99	Haiti	74.47	3.83	72.84	77.0	4.17
2015	74.48	81.82	Canada	63.24	Haiti	74.5	3.78	72.74	77.13	4.39

- **Gradual Improvement Across the Board:** Life expectancy in the Americas improved steadily from 71.14 (2000) to 74.48 (2015) — a +3.34 year increase. While lower than Africa and Asia in raw gain, this region already had a relatively high starting point.
- **Highest Values** – Canada Consistently on Top: Canada held the highest life expectancy throughout the 16-year period:
  - From 79.17 (2000) to 81.82 (2015) This is on par with top Asian performers like Japan.
- **Lowest Values** – Haiti: Haiti consistently had the lowest life expectancy in the region (except in 2010, when its value anomalously dropped to 46.02—likely reflecting the catastrophic earthquake impact that year).
  - Life expectancy improved from 58.36 (2000) to 63.24 (2015) This +4.88 year gain is respectable but still leaves Haiti well below the regional average.
- **Median Life Expectancy:** The median rose from 71.32 to 74.50, suggesting a broad regional uplift and narrowing gap between nations.
- **Moderate Decrease in Dispersion:**
  - Standard deviation dropped from 4.5 to 3.78
  - Interquartile range (IQR) decreased from 5.17 to 4.39 This reflects a converging trend in life expectancy, with countries drawing closer together in longevity outcomes.
- **Percentile Improvements:**
  - 25th percentile: 68.88 ⟶ 72.74
  - 75th percentile: 74.04 ⟶ 77.13
  - These shifts affirm a general upward movement for all countries, regardless of their 2000 starting position.

Contextual Notes:

- The Americas already had a mature health system landscape in 2000, so gains were smaller in percentage terms.
- Haiti's 2010 drop is a clear outlier due to external disaster, but the country rebounded strongly in subsequent years.
- North American countries (Canada, USA) pull the upper end up, while Caribbean and parts of Central America influence the lower end.

Europe

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	74.86	79.78	Italy	65.48	Russian Federation	75.41	3.89	71.84	77.96	6.12
2001	75.2	80.69	Iceland	65.38	Russian Federation	75.76	3.98	72.24	78.36	6.12
2002	75.29	80.5	Iceland	65.13	Russian Federation	76.01	4.0	72.32	78.46	6.15
2003	75.48	80.96	Iceland	65.03	Russian Federation	76.86	3.98	72.37	78.52	6.15
2004	75.93	81.09	Switzerland	65.47	Russian Federation	77.21	4.04	72.67	79.11	6.44
2005	76.02	81.5	Iceland	65.53	Russian Federation	77.61	4.14	72.74	79.32	6.57
2006	76.32	81.49	Switzerland	66.73	Russian Federation	78.09	4.13	73.24	79.44	6.19
2007	76.54	81.74	Switzerland	67.59	Russian Federation	78.2	4.09	73.39	79.78	6.39
2008	76.89	81.99	Switzerland	67.95	Russian Federation	78.45	3.99	73.83	80.18	6.34
2009	77.18	82.04	Switzerland	68.68	Russian Federation	78.6	3.86	74.22	80.28	6.06
2010	77.49	82.25	Switzerland	68.84	Russian Federation	79.03	3.85	74.67	80.66	5.99
2011	77.9	82.7	Switzerland	69.19	Republic of Moldova	79.8	3.85	74.98	80.96	5.99
2012	78.03	82.92	Iceland	69.14	Republic of Moldova	80.05	3.75	74.98	80.92	5.94
2013	78.29	83.08	Spain	69.1	Republic of Moldova	80.3	3.73	75.48	81.3	5.81
2014	78.57	83.23	Spain	69.03	Republic of Moldova	80.7	3.87	75.56	81.6	6.04
2015	78.52	82.9	Switzerland	69.24	Republic of Moldova	80.64	3.71	75.48	81.5	6.01

- **High Starting Point and Steady Growth:** Europe began with a strong average life expectancy of 74.86 years in 2000, rising to 78.52 by 2015 — a +3.66 year increase, slightly more than in the Americas but less than Asia and Africa.
- **Top Performers** – Western Europe Leads: Countries like Switzerland, Iceland, and Spain consistently topped the charts:
  - Switzerland held the highest life expectancy in 9 of the 16 years, peaking at 83.23 (2014).
  - Iceland led early on, with values exceeding 80 by 2001.
- **Lowest Values** – East European Struggles:
  - From 2000 to 2010, the Russian Federation had the lowest life expectancy, hovering around 65 years.
  - Republic of Moldova replaced it from 2011 to 2015, ending at 69.24 years in 2015.
  - These figures are more than 14 years lower than the highest values in Western Europe.
- **Median Life Expectancy:** Rose from 75.41 in 2000 to 80.64 in 2015, reflecting widespread longevity gains across the region — especially strong in Western and Northern Europe.
- **Distribution and Dispersion:**
  - Standard deviation fluctuated slightly but decreased overall from 3.89 ⟶ 3.71, indicating a modest convergence among countries.
  - Interquartile range (IQR) remained relatively stable, with a slight dip: 6.12 ⟶ 6.01
- **Percentile Shift:**
  - 25th percentile: 71.84 ⟶ 75.48
  - 75th percentile: 77.96 ⟶ 81.50
  - The upward shift across percentiles shows that both lower- and upper-quartile countries made significant progress.

Contextual Notes:

Europe exhibited one of the most consistent and stable progressions in life expectancy, benefiting from mature healthcare systems. The gap between East and West remained evident, but the difference narrowed slightly over time. The post-Soviet states, especially Moldova and Russia, faced persistent challenges despite modest improvements.

Oceania

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	69.6	79.23	Australia	61.72	Papua New Guinea	68.97	5.31	66.54	70.5	3.97
2001	69.81	79.63	Australia	61.77	Papua New Guinea	69.09	5.36	66.6	70.88	4.29
2002	69.79	79.94	Australia	61.7	Papua New Guinea	69.26	5.47	66.19	70.82	4.63
2003	70.1	80.24	Australia	61.8	Papua New Guinea	69.39	5.52	66.54	71.32	4.77
2004	70.25	80.49	Australia	61.76	Papua New Guinea	69.49	5.62	66.76	71.48	4.72
2005	70.38	80.84	Australia	61.8	Papua New Guinea	69.58	5.71	66.96	71.53	4.58
2006	70.53	81.04	Australia	61.92	Papua New Guinea	69.68	5.75	67.17	71.79	4.62
2007	70.59	81.29	Australia	62.03	Papua New Guinea	69.8	5.81	67.12	71.84	4.72
2008	70.76	81.4	Australia	62.57	Papua New Guinea	69.88	5.76	67.31	71.9	4.6
2009	70.63	81.54	Australia	62.79	Papua New Guinea	69.83	5.78	67.44	70.85	3.41
2010	71.01	81.7	Australia	63.04	Papua New Guinea	69.97	5.78	67.68	72.0	4.31
2011	71.17	81.9	Australia	63.53	Papua New Guinea	69.96	5.76	67.87	72.03	4.16
2012	71.27	82.05	Australia	63.73	Papua New Guinea	69.97	5.78	67.93	71.98	4.05
2013	71.36	82.15	Australia	63.96	Papua New Guinea	70.02	5.75	67.87	72.17	4.3
2014	71.44	82.3	Australia	64.26	Papua New Guinea	70.0	5.73	67.84	72.23	4.38
2015	71.52	82.4	Australia	64.7	Papua New Guinea	70.1	5.73	67.51	72.24	4.73

- **Gradual Increase with Wide Gaps:** Oceania's regional average life expectancy increased steadily from 69.60 years in 2000 to 71.52 in 2015, a +1.92 gain over period — the smallest increase among all global regions.
- **Top Performer** – Australia’s Dominance:
  - Australia consistently had the highest life expectancy across all 16 years.
  - It rose from 79.23 in 2000 to 82.40 in 2015, demonstrating a strong and steady trend of improvement.
- **Lowest Values** – Persistent Struggles in Papua New Guinea:
  - Papua New Guinea had the lowest life expectancy for every year
  - Life expectancy in Papua New Guinea improved from 61.72 to 64.70, but it remained over 17 years behind Australia by 2015.
- **Median Life Expectancy:** Increased moderately from 68.97 in 2000 to 70.10 in 2015, indicating overall upward mobility, though the gains were modest and uneven.
- **Distribution and Dispersion:**
  - Standard deviation remained high and relatively constant: 5.31 ⟶ 5.73, reflecting a wide disparity between nations.
  - The gap between top and bottom countries persisted across the time span.
- **Percentile Ranges:**
  - 25th percentile: 66.54 ⟶ 67.51
  - 75th percentile: 70.50 ⟶ 72.24
  - IQR (Interquartile Range) hovered around 4 to 4.7 years, showing moderate inequality that did not shrink over time.

Contextual Notes:

- Oceania shows sharp duality:
  - High-income countries like Australia enjoy advanced healthcare and long life spans.
  - In contrast, developing island nations — notably Papua New Guinea — face infrastructure and access challenges.
- The consistently high dispersion metrics highlight this persistent health gap.

3.1.2 GDP per Capita

Global Data:

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	6432.78	48659.6	Luxembourg	122.96	Ethiopia	1673.36	9895.37	611.47	6526.71	5915.24
2001	6342.96	48440.14	Luxembourg	119.26	Ethiopia	1800.4	9681.83	596.62	6529.2	5932.58
2002	6708.98	53005.73	Luxembourg	110.46	Ethiopia	1989.14	10365.15	658.98	6610.02	5951.03
2003	7824.77	65689.32	Luxembourg	114.37	Burundi	2259.52	12224.57	704.9	8272.54	7567.64
2004	9007.8	76544.92	Luxembourg	128.54	Burundi	2600.37	14030.32	807.84	9739.93	8932.09
2005	9903.23	80988.14	Luxembourg	151.19	Burundi	3106.58	15177.77	980.34	10834.92	9854.58
2006	10802.97	90788.8	Luxembourg	166.28	Burundi	3343.79	16326.49	1166.53	12072.4	10905.87
2007	12311.73	107475.32	Luxembourg	170.71	Burundi	3934.69	18453.09	1317.42	14027.22	12709.8
2008	13629.04	120422.14	Luxembourg	194.71	Burundi	4504.51	20067.95	1465.33	17018.96	15553.63
2009	11958.56	109419.75	Luxembourg	204.54	Burundi	4270.86	17438.69	1366.78	14040.76	12673.99
2010	12747.37	110885.99	Luxembourg	222.66	Burundi	4835.79	18282.4	1512.32	14180.24	12667.93
2011	14241.82	119025.06	Luxembourg	236.45	Burundi	5614.39	20571.54	1673.25	14874.8	13201.54
2012	14154.42	112584.68	Luxembourg	238.21	Burundi	6015.95	20207.19	1738.04	15926.34	14188.3
2013	14516.97	120000.14	Luxembourg	241.55	Burundi	6141.98	20671.21	1871.68	16446.79	14575.11
2014	14592.0	123678.7	Luxembourg	257.82	Burundi	6164.64	20643.45	1826.94	16677.97	14851.03
2015	12928.92	105462.01	Luxembourg	289.36	Burundi	5588.98	17920.52	1676.38	15659.32	13982.95

- **Consistent Upward Trend Until 2014, Followed by a Decline:** Global average GDP per Capita rose steadily from `$`6,432.78 in 2000 to a high of `$` 14,592.00 in 2014 — more than doubling over the period. However, it declined in 2015 to `$`12,928.92, likely reflecting global economic shifts or downturns.
- **Top Performer** – Luxembourg Dominates:
  - Luxembourg had the highest GDP per Capita in all 16 years, starting at `$`48,659.60 in 2000 and peaking at `$`123,678.70 in 2014.
  - Its consistent lead underscores its strong finance-driven economy and small population.
- **Lowest GDP per Capita** – Persistent Struggles in Sub-Saharan Africa:
  - Ethiopia recorded the lowest values in 2000 and 2001 (as low as `$`119.26).
  - From 2003 to 2015, Burundi consistently had the lowest GDP per Capita, reaching just `$`289.36 by 2015 — over 400 times lower than Luxembourg in the same year.
- **Median GDP per Capita:** Rose from `$`1,673.36 in 2000 to `$`6,164.64 in 2014, falling slightly to `$`5,588.98 in 2015. This steady rise shows that typical countries also experienced income growth, not just the top performers.
- **Distribution and Dispersion:**
  - **Standard deviation** widened from `$`9,895.37 $\longrightarrow$ `$`20,643.45 (2014), indicating increasing global inequality in income per person.
  - It slightly narrowed in 2015 to `$`17,920.52, possibly due to declining GDP among top economies.
- **Interquartile Range (IQR):**
  - Grew substantially from `$`5,915.24 in 2000 to `$`14,851.03 in 2014, narrowing in 2015 to `$`13,982.95.
  - This suggests a widening income gap among middle-income countries over time, followed by slight convergence.

- **Percentile Shift:**
  - **25th percentile:** $ 611.47 ⟶ $ 1,826.94
  - **75th percentile:** $ 6,526.71 ⟶ $ 16,677.97
  - This upward shift across both quartiles reveals that income growth occurred across many nations, not just among wealthy ones.

Contextual Notes

The dataset reflects strong global economic growth from 2000 to 2014, but also an increasing income disparity between nations. The dip in 2015 may reflect effects of global financial adjustments, commodity price shocks, or currency fluctuations. Persistent low GDP per Capita in Sub-Saharan countries highlights ongoing development challenges, while Luxembourg`s dominance reflects the impact of high-income microstates on global statistics.

Africa

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	1149.94	8063.66	Seychelles	122.96	Ethiopia	527.04	1651.87	289.57	1140.02	850.45
2001	1118.55	8153.3	Seychelles	119.26	Ethiopia	498.42	1571.84	300.52	1150.04	849.52
2002	1112.41	8864.17	Seychelles	110.46	Ethiopia	529.32	1503.12	312.77	1113.64	800.86
2003	1316.89	9070.28	Seychelles	114.37	Burundi	593.98	1713.2	358.27	1144.03	785.76
2004	1581.19	10827.67	Seychelles	128.54	Burundi	731.86	2099.11	405.98	1302.75	896.76
2005	1850.53	11802.11	Seychelles	151.19	Burundi	824.93	2549.48	448.13	1878.19	1430.06
2006	2060.53	12782.99	Seychelles	166.28	Burundi	906.0	2846.94	449.74	2194.29	1744.55
2007	2319.69	13776.9	Equatorial Guinea	170.71	Burundi	962.34	3158.59	510.22	2618.54	2108.32
2008	2691.22	19849.72	Equatorial Guinea	194.71	Burundi	1115.3	3821.93	599.05	3097.29	2498.24
2009	2328.34	14398.77	Equatorial Guinea	204.54	Burundi	1094.13	2906.66	640.15	2909.59	2269.44
2010	2597.29	14905.51	Equatorial Guinea	222.66	Burundi	1166.32	3264.93	672.5	3041.58	2369.08
2011	2852.38	18659.42	Equatorial Guinea	236.45	Burundi	1294.3	3642.85	721.24	3386.91	2665.66
2012	3002.71	18756.43	Equatorial Guinea	238.21	Burundi	1289.99	3977.07	706.6	3478.6	2772.0
2013	3023.63	17644.59	Equatorial Guinea	241.55	Burundi	1384.88	3807.83	727.65	3634.15	2906.5
2014	2983.76	16804.92	Equatorial Guinea	257.82	Burundi	1450.63	3646.81	735.38	3575.5	2840.12
2015	2530.5	15333.11	Seychelles	289.36	Burundi	1315.43	2980.87	705.62	3161.62	2456.0

- **Steady Growth Followed by Decline:** Average GDP per Capita in Africa increased from $ 1,149.94 in 2000 to a peak of $ 3,002.71 in 2012, then slightly declined to $ 2,530.50 in 2015. This reflects solid economic growth across the continent, tempered by external shocks in the later years.
- **Top Performers** – Seychelles and Equatorial Guinea:
  - Seychelles led in the early 2000s and re-emerged as the top in 2015, with GDP per Capita peaking at $ 8,864.17 in 2002.
  - Equatorial Guinea dominated from 2007 to 2014, reaching a high of $ 19,849.72 in 2008 due to oil revenue booms.
- **Lowest Values** – Burundi and Ethiopia:
  - Ethiopia had the lowest GDP per Capita in 2000–2002, bottoming at $ 110.46 in 2002.
  - From 2003 to 2015, Burundi consistently ranked lowest, with only $ 289.36 in 2015 — over 50 times less than the region’s top performers.
- **Median GDP per Capita:** Rose from $ 527.04 in 2000 to $ 1,450.63 in 2014, before dropping to $ 1,315.43 in 2015. This consistent rise suggests broad-based economic improvement, although progress slowed in the final year.

- **Distribution and Dispersion:**
  - **Standard deviation** increased from $ 1,651.87 ⟶ $ 3,977.07 (2012), then narrowed to $ 2,980.87 in 2015, indicating growing inequality during peak growth years, followed by some convergence.
  - **Interquartile range** (IQR) rose from $ 850.45 in 2000 to $ 2,906.50 in 2013, dropping slightly to $ 2,456.00 in 2015 — reflecting widening and then stabilizing disparities among countries.
- **Percentile Shift:**
  - **25th percentile:** $ 289.57 ⟶ $ 705.62
  - **75th percentile:** $ 1,140.02 ⟶ $ 3,161.62
  - These shifts demonstrate that both lower- and upper-income African countries saw significant gains during the period, though gaps persisted.

Contextual Notes:

- Africa experienced significant economic expansion, especially during the global commodity boom years (2004–2012).
- Despite overall growth, deep inequalities remained, with a handful of resource-rich countries far outpacing others.
- Burundi and Ethiopia highlight ongoing poverty challenges, while Equatorial Guinea and Seychelles represent exceptional outliers.

The decline in 2015 may reflect external global economic pressures, including falling oil prices and reduced foreign investment.

Asia

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	6157.41	39169.36	Japan	137.18	Tajikistan	1161.24	9537.13	558.23	8321.39	7763.16
2001	5798.11	34406.18	Japan	140.78	Myanmar	1209.77	8746.67	534.65	8191.51	7656.86
2002	5934.89	32820.79	Japan	145.82	Myanmar	1204.23	8766.19	559.83	8380.04	7820.21
2003	6592.25	35387.04	Japan	199.64	Afghanistan	1256.03	9736.92	633.95	8896.9	8262.95
2004	7623.02	40792.28	Qatar	221.83	Afghanistan	1418.81	11322.84	774.79	10030.44	9255.65
2005	8771.55	52468.45	Qatar	251.17	Myanmar	1744.51	13065.05	832.75	12357.7	11524.95
2006	9771.7	59978.86	Qatar	274.02	Afghanistan	2253.79	14409.82	909.03	14533.73	13624.7
2007	10731.43	64706.99	Qatar	376.32	Afghanistan	3064.28	15247.79	1022.73	15756.19	14733.46
2008	12344.32	79811.6	Qatar	382.53	Afghanistan	3908.95	17538.35	1158.1	18944.86	17786.76
2009	10450.48	60733.98	Qatar	453.39	Afghanistan	3832.23	13996.66	1225.85	15064.63	13838.78
2010	11992.17	73021.31	Qatar	562.5	Afghanistan	4430.43	16005.03	1742.35	17958.94	16216.59
2011	14035.38	92993.0	Qatar	608.74	Afghanistan	5453.16	19303.54	2051.13	22441.57	20390.44
2012	14452.71	98041.36	Qatar	653.42	Afghanistan	6049.47	19944.62	2190.23	24069.2	21878.97
2013	14353.15	97630.83	Qatar	638.73	Afghanistan	6171.3	19560.2	2367.5	23563.94	21196.44
2014	14132.4	93126.15	Qatar	626.51	Afghanistan	5822.38	18909.55	2558.78	23121.21	20562.43
2015	12150.58	66984.91	Qatar	566.88	Afghanistan	4990.94	15303.2	2595.23	18777.43	16182.2

- **Steady Growth With Regional Peaks:** Average GDP per Capita in Asia increased from $ 6,157.41 in 2000 to a peak of $ 14,452.71 in 2012, before declining to $ 12,150.58 in 2015. The growth pattern reflects overall economic development, with fluctuations influenced by regional economic factors and global economic pressures.

- **Top Performers – Qatar and Japan:**
  - Qatar was the dominant performer throughout the period, consistently ranking highest in GDP per Capita, peaking at $98,041.36 in 2012.
  - Japan, though a top performer, showed a steady decline after 2007, reaching a high of $39,169.36 in 2000, before stabilizing in the $30,000 range post-2008.
- **Lowest Values** – Afghanistan and Myanmar:
  - Afghanistan had the lowest GDP per Capita across the years, bottoming out at $137.18 in 2000 and remaining near the lowest values throughout.
  - Myanmar's GDP per Capita was also consistently low, with values ranging from $140.78 in 2001 to a peak of $608.74 in 2011, still far below regional averages.
- **Median GDP per Capita:** Rose from $1,161.24 in 2000 to $6,049.47 in 2012, before dropping to $4,990.94 in 2015. The significant increase until 2012 points to broad improvements in middle-income countries, with a slowdown in growth post-2012.

- **Distribution and Dispersion:**
  - **Standard deviation** increased from $9,537.13 $\longrightarrow$ $19,944.62 (2012), then narrowed to $15,303.20 in 2015, indicating rising inequality followed by some degree of convergence toward the later years.
  - **Interquartile range** (IQR) rose from $7,763.16 in 2000 to $21,196.44 in 2012, before narrowing to $16,182.20 in 2015 — reflecting growing disparities in economic development during peak growth years, followed by stabilization.
- **Percentile Shift:**
  - 25th percentile: $558.23 $\longrightarrow$ $2,595.23
  - 75th percentile: $8,321.39 $\longrightarrow$ $23,563.94
  - These shifts demonstrate that both lower- and upper-income Asian countries saw significant gains during the period, with widening income gaps until 2012, followed by some convergence post-2012.

Contextual Notes:

- Asia experienced significant economic growth over the period, with countries like Qatar, Japan, and emerging economies such as China and India contributing significantly to the rise in GDP per Capita.
- However, disparities remained, with countries like Afghanistan and Myanmar struggling with low levels of economic development.

The decline post-2012 could be attributed to external factors such as the global economic slowdown, oil price volatility, and political instability in some regions.

America

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	6368.22	36329.97	United States of America	815.0	Haiti	3760.08	7958.67	1834.83	6526.71	4691.88
2001	6338.43	37133.62	United States of America	743.91	Haiti	3941.6	8048.71	1756.06	6461.97	4705.91
2002	6256.67	37997.74	United States of America	716.48	Haiti	3683.89	8317.14	2036.91	5277.45	3240.54
2003	6586.55	39490.3	United States of America	575.56	Haiti	3557.2	8764.9	2161.81	5745.56	3583.75
2004	7151.43	41724.64	United States of America	679.28	Haiti	4130.1	9328.34	2333.82	6339.95	4006.14
2005	7929.84	44123.4	United States of America	771.53	Haiti	4773.27	10082.28	2568.28	7213.32	4645.04
2006	8780.45	46301.99	United States of America	824.36	Haiti	5385.56	10685.7	3130.7	8518.59	5387.89
2007	9650.31	48050.23	United States of America	979.6	Haiti	6055.09	11309.2	3535.04	9077.12	5542.08
2008	10387.86	48570.06	United States of America	1089.58	Haiti	6641.5	11490.81	4201.16	9959.01	5757.84
2009	9616.42	47194.95	United States of America	1191.8	Haiti	6737.86	10484.25	4120.6	9175.32	5054.72
2010	10543.13	48650.66	United States of America	1204.86	Haiti	6760.97	11235.69	4657.8	11880.94	7223.14
2011	11414.33	52223.86	Canada	1306.85	Haiti	7392.94	11827.41	5221.77	13919.16	8697.38
2012	11844.15	52670.34	Canada	1356.17	Haiti	8096.8	12076.85	5539.25	14109.75	8570.5
2013	12168.15	53409.75	United States of America	1452.31	Haiti	8263.64	12186.5	5720.86	14028.66	8307.8
2014	12298.39	55304.32	United States of America	1454.65	Haiti	8167.47	12239.59	5763.46	13756.2	7992.74
2015	11950.56	57040.21	United States of America	1405.71	Haiti	8379.62	11806.01	5764.92	13729.31	7964.4

- **Steady Climb with Minor Setbacks:** The Americas saw a general upward trend in average GDP per capita, rising from $ 6,368.22 in 2000 to a peak of $ 12,298.39 in 2014, before a slight dip to $ 11,950.56 in 2015. The region's growth was largely driven by high-income countries, with some fluctuation due to global economic events such as the 2008 financial crisis and falling commodity prices.
- **Top Performers** – United States and Canada:
  - The United States led in most years, starting at $ 36,329.97 in 2000 and rising to $ 57,040.21 in 2015.
  - Canada briefly surpassed the U.S. in 2011 and 2012, with a high of $ 52,670.34 in 2012, reflecting its stable economy and resource-driven wealth during that period.
- **Lowest Values – Haiti:**
  - **Haiti** consistently had the lowest GDP per capita across all years, starting at $ 815.00 in 2000 and ending at $ 1,405.71 in 2015. Despite slight improvements, its figures remained significantly lower than the regional average, highlighting long-standing economic challenges.
- **Median GDP per Capita:** The median rose from $ 3,760.08 in 2000 to $ 8,379.62 in 2015, indicating that most countries experienced solid gains, although not as extreme as top performers. The steady rise supports a moderate but widespread economic advancement across the region.
- **Distribution and Dispersion:**
  - **Standard deviation** widened from $ 7,958.67 in 2000 to a high of $ 12,239.59 in 2014, slightly falling to $ 11,806.01 in 2015. This reflects increasing inequality, with richer countries accelerating faster than their poorer counterparts.
  - **Interquartile range** (IQR) expanded from $ 4,691.88 in 2000 to $ 8,697.38 in 2011, and stabilized around $ 7,964.40 in 2015, showing a growing and then steady income gap between the lower and upper-middle income countries.
- **Percentile Shift:**
  - **25th percentile:** Grew from $ 1,834.83 in 2000 to $ 5,764.92 in 2015
  - **75th percentile:** Rose from $ 6,526.71 to $ 13,729.31 over the same period
  - This widening gap confirms that while economic growth was region-wide, the benefits were skewed toward upper-tier economies.

Contextual Notes:

- The Americas region showcases sharp contrasts: high-income countries like the U.S. and Canada pulling the average upward, while lower-income countries like Haiti lagged far behind.
- The 2008 global financial crisis caused a temporary dip, but the region rebounded by 2010. Resource-rich nations in South America benefited from commodity booms, though gains slowed in later years due to price drops and political instability.
- Haiti remains a major outlier, with persistent poverty and vulnerability to natural disasters and political turmoil hampering growth despite international aid and development efforts.

Europe

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	14021.78	48659.6	Luxembourg	440.54	Republic of Moldova	10201.3	13208.6	1816.74	24485.75	22669.01
2001	14161.5	48440.14	Luxembourg	507.4	Republic of Moldova	10402.23	13038.96	2004.98	24763.28	22758.3
2002	15622.24	53005.73	Luxembourg	570.8	Republic of Moldova	11289.89	14204.88	2330.69	26762.67	24431.98
2003	19087.45	65689.32	Luxembourg	682.32	Republic of Moldova	13669.5	17104.4	2990.27	32610.86	29620.59
2004	22109.08	76544.92	Luxembourg	897.18	Republic of Moldova	15197.06	19575.97	3802.58	37330.7	33528.12
2005	23646.82	80988.14	Luxembourg	1034.39	Republic of Moldova	15888.17	20837.57	4970.73	38736.16	33765.43
2006	25527.19	90788.8	Luxembourg	1183.02	Republic of Moldova	16723.88	22290.87	6338.86	40945.96	34607.1
2007	29862.62	107475.32	Luxembourg	1531.22	Republic of Moldova	19485.87	25462.56	8730.79	47695.86	38965.07
2008	32616.87	120422.14	Luxembourg	2110.55	Republic of Moldova	22804.58	27155.5	11035.25	50111.69	39076.44
2009	28668.53	109419.75	Luxembourg	1898.44	Republic of Moldova	21083.28	24187.22	8555.44	43249.02	34693.59
2010	28914.72	110885.99	Luxembourg	2436.8	Republic of Moldova	21798.91	24900.77	9536.4	43711.01	34174.61
2011	31782.12	119025.06	Luxembourg	2941.36	Republic of Moldova	23155.1	27554.36	11449.56	47562.58	36113.02
2012	30157.16	112584.68	Luxembourg	3044.81	Republic of Moldova	20563.71	26467.06	10957.78	45333.06	34375.27
2013	31534.8	120000.14	Luxembourg	3321.04	Republic of Moldova	21653.2	27468.46	11527.78	48281.46	36753.69
2014	32158.49	123678.7	Luxembourg	3104.65	Ukraine	21616.71	27730.93	12033.6	49175.56	37141.96
2015	28161.76	105462.01	Luxembourg	2124.66	Ukraine	18083.88	24297.85	9127.33	43498.87	34371.54

- **Strong Growth with Deep Inequality:** Europe displayed a robust upward trend in GDP per capita, with the regional average rising from `$`14,021.78 in 2000 to a peak of `$`32,158.49 in 2014. The average dipped to `$`28,161.76 in 2015, likely influenced by external economic shocks and regional instabilities such as the Eurozone crisis. Despite the setbacks, the continent maintained a high overall economic standing throughout the period.
- **Top Performer** – Luxembourg:
  - Luxembourg overwhelmingly led the region every year, starting at `$`48,659.60 in 2000 and peaking at `$`123,678.70 in 2014.
  - It remained the highest performer through 2015, finishing at `$`105,462.01, reflecting its strong financial sector, low population, and high productivity.
- **Lowest Values** – Republic of Moldova and Ukraine:
  - From 2000 to 2013, Republic of Moldova recorded the lowest GDP per capita in Europe, starting at just `$`440.54 in 2000 and gradually rising to `$`3,321.04 by 2013.
  - In 2014 and 2015, Ukraine took over as the lowest performer, with GDP per capita of `$`3,104.65 and `$`2,124.66 respectively, reflecting political unrest and economic decline during that time.

- **Median** GDP per Capita:
  - The median followed an upward trend from `$`10,201.30 in 2000 to `$`21,616.71 in 2014, before dropping to `$`18,083.88 in 2015.
  - This trend suggests that while most countries experienced growth, the impact of regional downturns was felt more strongly by mid-income nations.
- **Distribution and Dispersion:**
  - Standard deviation increased from `$`13,208.60 in 2000 to `$`27,730.93 in 2014, dipping slightly to `$`24,297.85 in 2015. This reflects the extreme spread between high- and low-income countries within the region.
  - **Interquartile range** (IQR) also grew from `$`22,669.01 in 2000 to `$`39,076.44 in 2008, remaining above `$`34,000 through 2015.
  - The widening IQR shows that income disparities between the 25th and 75th percentile countries became more pronounced.
- **Percentile Shift:**
  - **25th percentile:** Rose from `$`1,816.74 in 2000 to `$`9,127.33 in 2015
  - **75th percentile:** Climbed from `$`24,485.75 to `$`43,498.87 over the same period
  - These changes signal that even lower-income countries saw significant gains, though not nearly as rapidly as the top performers.

Contextual Notes:

- Europe’s wealth is largely driven by countries in Western and Northern Europe, such as Luxembourg, Switzerland, and Norway. These nations consistently elevate regional averages, even as Eastern and Southeastern Europe lag behind.
- The 2008 financial crisis had a visible impact, with mean and median values dipping around 2009, but the region showed strong recovery by 2011. However, the downturn in 2015 suggests lasting effects of internal fiscal strains and geopolitical tensions (e.g., Russia-Ukraine conflict).
- While countries like Moldova and Ukraine remain outliers on the low end, even they experienced gradual growth, albeit slower and more volatile. The economic divide between Western and Eastern Europe continues to shape the region’s overall statistical profile.

Oceania

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	4689.28	21870.42	Australia	639.28	Papua New Guinea	1706.36	6804.17	1084.01	2070.12	986.11
2001	4438.49	19695.73	Australia	540.67	Papua New Guinea	1595.09	6329.88	1024.66	2105.43	1080.77
2002	4815.73	20301.84	Australia	509.03	Papua New Guinea	1635.28	6949.16	933.38	2163.39	1230.01
2003	5824.45	23718.13	Australia	580.6	Papua New Guinea	1857.94	8526.04	1142.7	2559.0	1416.3
2004	7032.09	30836.73	Australia	624.04	Papua New Guinea	2161.91	10639.67	1243.53	2892.46	1648.92
2005	7776.81	34479.77	Australia	748.74	Papua New Guinea	2367.22	11789.51	1323.06	3186.8	1863.74
2006	8001.27	36595.71	Australia	1121.08	Kiribati	2469.22	12040.47	1430.94	3300.69	1869.75
2007	9185.16	41051.61	Australia	1353.62	Kiribati	2571.89	13942.9	1599.71	3597.6	1997.89
2008	10094.56	49701.28	Australia	1411.91	Kiribati	2873.08	15761.1	1831.47	3782.7	1951.24
2009	8987.75	42816.57	Australia	1324.44	Kiribati	2756.94	13674.36	1802.34	3234.38	1432.04
2010	10676.62	52147.02	Australia	1528.9	Kiribati	3088.06	16652.83	2092.57	3488.09	1395.52
2011	12475.98	62609.66	Australia	1783.77	Kiribati	3427.19	19782.72	2449.23	4083.58	1634.34
2012	13321.88	68078.04	Australia	1855.27	Kiribati	3462.95	21315.9	2716.4	4373.88	1657.47
2013	13632.62	68198.42	Australia	1780.66	Kiribati	3455.3	21738.84	2653.24	4492.27	1839.03
2014	13302.37	62558.24	Australia	1740.36	Kiribati	3438.59	20553.07	2771.99	5010.15	2238.16
2015	12048.34	56758.87	Australia	1641.48	Kiribati	3469.9	18306.33	2537.52	4858.38	2320.85

- **Uneven Growth with Heavy Skew Toward Australia:** Oceania’s average GDP per capita showed an upward trajectory from $4,689.28 in 2000 to a peak of $13,632.62 in 2013, before a decline to $12,048.34 in 2015. The rise reflects overall regional economic progress, though the average is heavily influenced by Australia’s high figures. Smaller Pacific Island nations contributed less to the regional mean but saw steady improvements in their own metrics.
- **Top Performer** – Australia:
  - Australia led consistently throughout the period, starting at $21,870.42 in 2000 and peaking at $68,198.42 in 2013.
  - It remained the dominant economy through 2015, ending at $56,758.87, significantly outpacing every other nation in the region due to its diversified economy, strong exports, and stable governance.
- **Lowest Values** – Papua New Guinea and Kiribati:
  - Papua New Guinea had the lowest GDP per capita from 2000 to 2005, beginning at $639.28 in 2000.
  - From 2006 onward, Kiribati recorded the lowest values, reaching $1,641.48 in 2015.
  - Despite small gains, these countries remained far behind the regional average, reflecting limited industrialization and economic dependence on aid and subsistence livelihoods.
- **Median GDP per Capita:**
  - The median GDP per capita rose from $1,706.36 in 2000 to $3,469.90 in 2015.
  - This steady increase suggests gradual economic improvements among the smaller island nations, even though they lagged far behind the Australian benchmark.
- **Distribution and Dispersion:**
  - **Standard deviation** widened from $6,804.17 in 2000 to $21,738.84 in 2013, before narrowing to $18,306.33 in 2015. This reflects the extreme inequality driven primarily by Australia's outsized economy.
  - **Interquartile range** (IQR) expanded from $986.11 in 2000 to $2,320.85 in 2015, peaking at $2,238.16 in 2014. The increase in IQR reveals a modest widening gap between lower- and upper-middle-income countries in the region.

- **Percentile Shift:**
  - **25th percentile:** Rose from $1,084.01 in 2000 to $2,537.52 in 2015
  - **75th percentile:** Increased from $2,070.12 to $4,858.38 during the same period
  - This indicates an upward shift across the board, with all countries making progress, albeit at widely varying rates.

Contextual Notes:

- Oceania presents a stark economic contrast between Australia and the smaller island nations. Australia's large GDP per capita consistently skews the regional average upward, overshadowing the modest gains made by others.
- The 2008 global financial crisis had a muted impact on Oceania, with Australia weathering it relatively well due to strong trade ties with Asia and a resilient commodity sector.
- Countries like Kiribati and Papua New Guinea remain economically vulnerable, with challenges stemming from geographic isolation, limited industrial bases, and reliance on subsistence economies. Nevertheless, even these nations saw gradual improvements in GDP per capita, highlighting slow but positive regional development.

3.2 Summary Statistics for Exploratory Analysis

3.2.1 GDP-Based Country Grouping

To investigate disparities in life expectancy outcomes based on economic standing, we created two distinct groups using GDP per capita as the metric for classification:

1. Top 20 GDP Countries:

This group includes all countries that ranked within the top 20 for GDP per capita at least once between 2000 and 2015.

Code snippet:

```
top_20_gdp_countries_list = life_expectancy.groupby('year').apply(lambda x:
x.nlargest(20, 'gdp_per_capita'))['country'].unique()
le_top_20_gdp =
life_expectancy[life_expectancy['country'].isin(top_20_gdp_countries_list)].reset_index
```

2. Bottom 20 GDP Countries:

This group captures countries that appeared in the bottom 20 for GDP per capita across any year in the same range.

Code snippet:

```
le_bottom_20_gdp_countries_list = life_expectancy.groupby('year').apply(lambda x:
x.nsmallest(20, 'gdp_per_capita'))['country'].unique()
le_bottom_20_gdp =
life_expectancy[life_expectancy['country'].isin(le_bottom_20_gdp_countries_list)].reset
```

Purpose of the Grouping:

This categorization enables us to:

- Compare economic disparity by isolating top- and bottom-performing economies over time.
- Assess how life expectancy correlates with sustained high or low GDP per capita.
- Control for volatility by focusing only on countries that consistently appeared in extreme GDP positions.

Summary Statistics Overview:

To describe the economic conditions of these two groups, we calculated summary statistics for GDP per capita across the years 2000 to 2015. These include:

- Mean GDP per capita
- Median, Minimum, and Maximum
- 25th and 75th percentiles
- Interquartile Range (IQR)
- Standard Deviation (std)
- Countries with Highest and Lowest GDP per year

This allows us to understand not only the central tendencies of each group but also the spread and consistency of their economic performance over time.

Top 20 GDP Countries – Summary Statistics (2000–2015):

year	mean	max	max_country	minimum	median	std	25%	75%	iqr
2000	77.66	81.08	Japan	44.52	77.94	2.01	76.66	79.14	2.48
2001	77.96	81.42	Japan	41.96	78.22	2.07	76.95	79.34	2.39
2002	78.1	81.69	Japan	44.56	78.26	2.06	77.14	79.46	2.32
2003	78.32	81.76	Japan	43.39	78.47	2.01	77.29	79.64	2.35
2004	78.74	82.03	Japan	44.5	79.11	2.1	77.75	80.12	2.36
2005	78.94	81.96	Japan	44.77	79.34	2.09	78.09	80.15	2.06
2006	79.18	82.32	Japan	45.36	79.54	2.13	78.36	80.7	2.34
2007	79.39	82.51	Japan	45.61	79.94	2.18	78.46	80.81	2.34
2008	79.61	82.59	Japan	46.72	80.3	2.21	78.73	81.02	2.29
2009	79.83	82.93	Japan	48.06	80.44	2.24	78.88	81.32	2.44
2010	80.06	82.84	Japan	49.26	80.66	2.22	79.29	81.52	2.23
2011	80.36	82.7	Switzerland	49.95	80.98	2.28	79.96	81.78	1.83
2012	80.5	83.1	Japan	50.54	81.02	2.29	80.13	81.9	1.77
2013	80.65	83.33	Japan	50.78	81.22	2.28	80.35	82.04	1.69
2014	80.98	83.59	Japan	50.57	81.47	2.34	80.8	82.29	1.49
2015	80.91	83.79	Japan	51.59	81.5	2.34	80.66	82.32	1.66

The top 20 GDP group maintained a consistently high life expectancy, with the mean increasing from 77.66 years in 2000 to 80.91 years in 2015. This upward trend illustrates improved healthcare systems, higher living standards, and effective policy implementations in wealthier nations.

- **The maximum life expectancy** across this group consistently exceeded 81 years, peaking at 83.79 years in 2015, with Japan dominating as the country with the highest life expectancy for most years. Notably, Switzerland led briefly in 2011.
- The **minimum life expectancy**, although lower, gradually rose from 44.52 years (2000) to 51.59 years (2015), suggesting that even in wealthier nations, disparities still existed — possibly due to countries entering the top 20 GDP group despite having marginalized populations or uneven healthcare access.
- **Standard deviation** remained low (~2.0–2.3), indicating low variability and a tight clustering of life expectancy around the mean — a hallmark of stability in high-income countries.

- **The interquartile range (IQR)** declined over time (from 2.48 in 2000 to 1.66 in 2015), signaling an increase in equality of life expectancy among the richest nations.

These statistics emphasize the advantage that higher income levels provide in sustaining longevity, especially through uniform access to healthcare and education.

Bottom 20 GDP Countries – Summary Statistics (2000–2015)

year	mean	max	minimum	min_country	median	std	25%	75%	iqr
2000	53.39	70.76	44.52	Malawi	51.57	6.57	48.58	58.14	9.55
2001	53.9	71.64	41.96	Zimbabwe	52.39	6.72	49.22	58.46	9.23
2002	54.59	71.94	44.56	Zimbabwe	52.8	6.51	50.57	58.82	8.25
2003	55.19	72.41	43.39	Zimbabwe	53.2	6.58	51.0	59.4	8.39
2004	55.94	72.47	44.5	Zimbabwe	54.8	6.43	52.04	59.91	7.86
2005	56.56	72.77	44.77	Zimbabwe	55.3	6.38	52.99	60.37	7.38
2006	57.21	73.35	45.36	Zimbabwe	55.68	6.3	53.77	60.99	7.21
2007	57.77	73.71	45.61	Zimbabwe	56.47	6.19	54.47	61.38	6.91
2008	58.15	73.55	46.72	Zimbabwe	56.6	6.0	55.3	60.97	5.67
2009	58.98	73.85	48.06	Zimbabwe	58.1	5.94	55.85	62.54	6.69
2010	59.55	73.88	49.26	Central African Republic	58.9	5.69	56.39	62.79	6.4
2011	60.21	73.31	49.95	Central African Republic	59.3	5.45	57.03	63.43	6.4
2012	60.58	69.55	50.54	Chad	59.78	4.88	57.46	63.89	6.43
2013	61.0	69.56	50.78	Chad	60.09	4.7	57.8	63.93	6.13
2014	61.37	69.99	50.57	Central African Republic	60.84	4.7	58.42	64.16	5.74
2015	61.86	70.49	51.59	Chad	61.24	4.51	58.91	65.0	6.09

By contrast, the bottom 20 GDP countries started with significantly lower life expectancies. The mean life expectancy rose from 53.39 years in 2000 to 61.86 years in 2015, marking progress — but the gap with high-income countries remained pronounced.

- **The maximum life expectancy** hovered between 70 and 74 years, never exceeding the lower end of the top GDP countries. This indicates that, while improvement was evident, ceiling effects constrained maximum gains.
- **The minimum life expectancy** began at 44.52 years in 2000 and improved to 51.59 years in 2015, matching the lowest figures seen in top GDP countries.
    - Countries such as Zimbabwe, Central African Republic, and Chad frequently appeared as outliers with the lowest life expectancy.
- **Standard deviation** was higher (6.57 in 2000; 4.51 in 2015), showing substantial variability — symptomatic of unequal access to basic services, healthcare, or effects from conflict, famine, and other destabilizing factors.
- **The interquartile range** also started large (9.55 in 2000), slowly narrowing to 6.09 by 2015. This drop signals progress toward more uniform life expectancy within the bottom GDP group but still reflects considerable inequality compared to the top group.

Comparative Insights

- Longevity Gap: In 2000, the average difference in life expectancy between the top and bottom GDP groups was over 24 years. By 2015, that gap reduced slightly but remained large at around 19 years.
- Stability vs. Volatility: The top GDP group showed low variability and consistent growth, while the bottom GDP group showed high variability and gradual improvement, but with persistent outliers.
- Outliers and Policy Impact: Some countries in the bottom GDP group made significant progress, indicating that policy reforms and aid, even in low-income settings, can improve public health outcomes.

3.2.2 Income Class-Based Grouping

3.2.2.1 High Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	76.08	81.08	Japan	69.1	Trinidad and Tobago	76.59	2.85	74.09	78.24	4.15
2001	76.43	81.42	Japan	69.65	Trinidad and Tobago	76.84	2.85	74.4	78.63	4.24
2002	76.59	81.69	Japan	70.49	Trinidad and Tobago	77.07	2.85	74.61	78.71	4.1
2003	76.76	81.76	Japan	69.73	Trinidad and Tobago	77.22	2.88	74.6	78.94	4.34
2004	77.2	82.03	Japan	71.08	Trinidad and Tobago	77.67	2.84	75.18	79.37	4.19
2005	77.36	81.96	Japan	71.25	Lithuania	78.07	2.93	75.12	79.64	4.52
2006	77.6	82.32	Japan	70.87	Latvia	78.37	3.02	75.45	79.96	4.52
2007	77.79	82.51	Japan	70.9	Lithuania	78.56	3.04	75.53	80.17	4.64
2008	78.07	82.59	Japan	71.47	Trinidad and Tobago	78.77	2.96	75.72	80.49	4.76
2009	78.37	82.93	Japan	72.58	Bahamas	78.97	2.88	75.94	80.67	4.73
2010	78.65	82.84	Japan	72.72	Trinidad and Tobago	79.42	2.86	76.38	80.95	4.58
2011	78.99	82.7	Switzerland	72.56	Bahamas	80.0	2.88	76.74	81.18	4.44
2012	79.16	83.1	Japan	72.75	Bahamas	80.12	2.79	76.99	81.4	4.41
2013	79.37	83.33	Japan	73.02	Bahamas	80.4	2.84	77.19	81.74	4.56
2014	79.67	83.59	Japan	73.23	Seychelles	81.08	2.88	77.46	81.91	4.45
2015	79.66	83.79	Japan	73.1	Bahamas	80.78	2.79	77.54	81.96	4.43

To gain a more nuanced understanding of how life expectancy differs across economic strata, we begin by examining the summary statistics for countries classified as High Income based on their Gross National Income (GNI) per capita, as defined by the World Bank. This category includes countries with robust economic infrastructures and access to advanced healthcare systems.

Over the 16-year period from 2000 to 2015, life expectancy in high-income countries showed a consistent upward trend. The mean life expectancy increased from 52.3 years in 2000 to 60.69 years in 2015, indicating a substantial improvement in population health outcomes across this group.

Key Trends:

- **Mean and Median Growth:** The median life expectancy followed a similar trajectory to the mean, rising from 50.7 years in 2000 to 60.65 years in 2015, suggesting that gains were not just concentrated among outliers but reflected broader improvements.
- **Dispersion and Inequality:** The standard deviation steadily decreased from 6.13 in 2000 to 3.82 in 2015, pointing to a narrowing disparity in life expectancy within this income class. Similarly, the interquartile range (IQR) declined from 6.75 to 5.36 over the same period, reinforcing the view that high-income countries are becoming more homogenous in health outcomes.
- **Outliers:**
  - **The maximum life expectancy** values from 2000 to 2011 were consistently reported by Syrian Arab Republic, peaking at 73.88 years in 2010. From 2012 onward, Nepal took over as the top-ranking country in this group, although the drop in maximum values after 2011 (to 67.46 by 2015) may reflect the effects of conflict or reclassification of countries.
  - On the lower end, Central African Republic and Chad appeared repeatedly as the countries with the **lowest life expectancy** among high-income countries. These findings may be an artifact of data categorization, suggesting potential misclassification or GNI fluctuations during these years.

Distribution Insights:

- **The 25th percentile (Q1)** and **75th percentile (Q3)** values steadily rose over time, indicating consistent improvement across the lower and upper quartiles of the high-income group. For example:
  - Q1 rose from 48.13 years in 2000 to 58.46 years in 2015.
  - Q3 increased from 54.88 years in 2000 to 63.82 years in 2015.
- The interquartile range remained relatively stable but saw minor fluctuations, suggesting that while disparities persisted, they were gradually reducing.

Overall, the analysis of high-income countries reveals a clear positive trajectory in life expectancy, underpinned by economic stability and healthcare access. However, the presence of outliers at both ends of the spectrum hints at underlying heterogeneity even within this affluent group—potentially driven by geopolitical disruptions or reclassification in the data.

3.2.2.2 Upper-middle Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	69.23	77.59	Costa Rica	51.01	Botswana	70.82	5.68	67.26	72.78	5.53
2001	69.44	77.54	Costa Rica	50.68	Botswana	71.09	5.87	67.36	73.21	5.85
2002	69.6	77.98	Costa Rica	50.63	Botswana	71.35	6.0	67.32	73.26	5.94
2003	69.76	78.07	Costa Rica	50.95	Botswana	71.54	6.12	67.21	73.69	6.48
2004	70.03	78.33	Costa Rica	51.25	Namibia	71.78	6.12	67.75	73.71	5.96
2005	70.3	78.5	Costa Rica	51.79	Namibia	72.15	6.09	67.88	74.07	6.19
2006	70.62	78.51	Costa Rica	52.66	Namibia	72.38	5.96	68.48	74.29	5.81
2007	70.96	78.46	Costa Rica	53.68	Namibia	72.6	5.78	68.96	74.61	5.65
2008	71.32	78.44	Costa Rica	54.65	Namibia	72.71	5.59	69.16	74.95	5.8
2009	71.69	78.67	Costa Rica	55.5	Namibia	73.13	5.38	69.63	75.12	5.5
2010	71.98	78.67	Costa Rica	56.02	Namibia	73.32	5.19	70.0	75.33	5.33
2011	72.25	79.35	Costa Rica	56.55	Namibia	73.9	5.08	70.0	75.42	5.42
2012	72.58	79.28	Costa Rica	57.64	Namibia	74.17	4.93	70.38	75.66	5.28
2013	72.85	79.4	Costa Rica	58.69	Namibia	74.24	4.79	70.68	76.18	5.5
2014	73.02	79.42	Maldives	59.6	Equatorial Guinea	74.47	4.64	71.2	75.95	4.75
2015	73.25	79.69	Maldives	60.1	Equatorial Guinea	74.51	4.51	71.51	76.24	4.73

The next income tier in our analysis comprises Upper-middle Income countries, a diverse group characterized by ongoing economic development and expanding healthcare infrastructure. These countries represent an important transitional group—where improvements in life expectancy are often influenced by both legacy challenges and emerging opportunities.

Between 2000 and 2015, life expectancy in upper-middle income countries increased significantly, with the mean rising from 61.77 years in 2000 to 67.14 years in 2015. This steady growth highlights the tangible impact of economic growth and public health investments on population well-being.

Key Trends:

- Steady Rise in Central Tendency:
  - The **median life expectancy** improved from 62.51 years in 2000 to 69.28 years in 2015—an increase of nearly 7 years.
  - The **mean and median values tracked closely**, suggesting a relatively symmetric distribution, with gains broadly shared across countries.

- **Dispersion and Convergence:**
    - The **standard deviation** fell from 7.39 to 5.59, indicating a gradual reduction in the spread of life expectancy outcomes and suggesting increased parity among upper-middle income countries.
    - Similarly, the **interquartile range (IQR)** narrowed from 9.39 to 8.02, reflecting more cohesion within this group as more countries caught up with the upper quartile.

Notable Extremes:

- **Highest Life Expectancy:**
    - The maximum life expectancy increased from 72.46 years in 2000 to 75.24 years in 2014, before dipping slightly to 75.01 in 2015.
    - Over time, countries such as Viet Nam, Jordan, and Cabo Verde emerged as leaders in this group, consistently posting the highest life expectancy figures—a testament to effective public health policies in those regions.
- **Lowest Life Expectancy:**
    - In stark contrast, Zimbabwe and Lesotho repeatedly recorded the lowest life expectancy within this income class.
        - For Zimbabwe, life expectancy dropped to as low as 41.96 years in 2001, likely reflecting the height of its public health crisis.
        - From 2004 onward, Lesotho held the bottom position, despite slow but steady gains, ending at 51.1 years in 2015. These persistently low values highlight ongoing struggles with HIV/AIDS, economic inequality, and limited healthcare access in parts of Sub-Saharan Africa.

Distribution Observations:

- The 25th percentile (Q1) increased from 58.23 to 63.23, and the 75th percentile (Q3) rose from 67.62 to 71.25, reinforcing the consistent upward trajectory across the life expectancy distribution.
- Despite strong performers at the top, the relatively wide IQR values (8–9 years) throughout the period indicate that inequality in health outcomes remains a challenge in this income class—though it is narrowing.

In summary, Upper-middle income countries have demonstrated substantial progress in life expectancy, likely fueled by economic growth and incremental improvements in healthcare systems. While leading countries in this group rival high-income nations in life expectancy, a persistent tail of countries with low values underscores the ongoing need for targeted health interventions and support for vulnerable populations.

3.2.2.3 Lower-middle Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	61.77	72.46	Viet Nam	44.69	Zimbabwe	62.51	7.39	58.23	67.62	9.39
2001	62.05	72.65	Viet Nam	41.96	Zimbabwe	63.18	7.59	58.2	67.92	9.72
2002	62.42	72.8	Viet Nam	44.56	Zimbabwe	63.57	7.46	58.67	68.36	9.69
2003	62.77	72.98	Viet Nam	43.39	Zimbabwe	64.15	7.46	59.11	68.35	9.24
2004	62.93	73.14	Viet Nam	43.71	Lesotho	64.66	7.31	58.66	68.22	9.56
2005	63.41	73.27	Viet Nam	43.23	Lesotho	64.92	7.3	59.64	68.76	9.12
2006	63.84	73.32	Viet Nam	42.91	Lesotho	65.24	7.16	60.06	69.09	9.03
2007	64.19	73.44	Viet Nam	43.12	Lesotho	65.76	7.07	60.44	69.56	9.12
2008	64.42	73.59	Jordan	43.57	Lesotho	66.26	6.98	60.48	69.94	9.45
2009	64.88	73.8	Jordan	44.03	Lesotho	66.64	6.64	61.25	69.63	8.38
2010	65.11	74.0	Jordan	45.6	Lesotho	67.04	7.01	61.29	70.63	9.34
2011	65.79	74.26	Cabo Verde	46.69	Lesotho	67.4	6.26	61.74	70.81	9.07
2012	66.17	74.48	Cabo Verde	47.84	Lesotho	68.0	6.04	62.3	70.92	8.63
2013	66.52	75.06	Cabo Verde	49.0	Lesotho	68.52	5.86	62.62	71.08	8.45
2014	66.86	75.24	Cabo Verde	50.03	Lesotho	69.04	5.73	63.0	71.18	8.18
2015	67.14	75.01	Jordan	51.1	Lesotho	69.28	5.59	63.23	71.25	8.02

The Lower-middle Income group presents an encouraging picture of progress over the 16-year period. Despite their economic challenges, these countries have shown notable and consistent improvements in life expectancy, signaling positive developments in healthcare access, disease prevention, and public health infrastructure.

From 2000 to 2015, the average life expectancy in lower-middle income countries rose from 69.23 to 73.25 years. This 4-year increase is impressive considering the broader socioeconomic hurdles faced by many countries in this group.

Key Trends:

- **Upward Trend in Central Tendency:**
  - The **median life expectancy** increased from 70.82 years in 2000 to 74.51 years in 2015.
  - Throughout the period, the **median consistently** remained higher than the mean, suggesting a slightly left-skewed distribution—possibly influenced by persistently low performers.
- **Reduced Spread in Life Expectancy:**
  - The standard deviation steadily declined from 5.68 to 4.51, reflecting a narrowing disparity among lower-middle income countries.
  - The interquartile range (IQR) also decreased from 5.53 to 4.73, indicating that most countries in this group are converging toward similar life expectancy outcomes.

Notable Extremes:

- **Highest Life Expectancy:**
  - Costa Rica led the group until 2014, with a consistent maximum life expectancy hovering above 77 years.
  - In 2014 and 2015, Maldives overtook Costa Rica, reaching a life expectancy of 79.69 years in 2015—on par with high-income countries, and a standout case in this group.

- **Lowest Life Expectancy:**
  - Botswana recorded the lowest life expectancy in the early 2000s, with a low of 50.63 years in 2002.
  - From 2004 onward, Namibia took over this position, followed by Equatorial Guinea in 2014 and 2015. Despite these figures, the lowest life expectancy in the group improved from 51.01 in 2000 to 60.1 in 2015—a clear upward shift.

Distribution Observations:

- The **25th percentile** increased from 67.26 to 71.51 years, while the **75th percentile** rose from 72.78 to 76.24 years, reinforcing the group's general upward trend.
- The **IQR** values remained relatively tight, ranging from 4.73 to 6.48, which suggests a relatively consistent life expectancy range compared to higher-income classes, where disparities can be larger due to extreme outliers.

In summary, Lower-middle income countries demonstrated one of the most consistent and equitable improvements in life expectancy. While leading countries like Maldives and Costa Rica have surpassed expectations, the overall trend shows substantial progress across the board. These results reflect the critical role of affordable, widespread public health interventions and growing economic stability in enhancing life expectancy in this income class.

3.2.2.4 Low Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	52.3	70.76	Syrian Arab Republic	44.52	Malawi	50.7	6.13	48.13	54.88	6.75
2001	52.92	71.64	Syrian Arab Republic	45.39	Central African Republic	51.18	6.11	49.02	55.1	6.08
2002	53.53	71.94	Syrian Arab Republic	45.41	Central African Republic	51.87	5.96	50.39	55.6	5.2
2003	54.2	72.41	Syrian Arab Republic	45.76	Central African Republic	52.86	5.94	50.87	56.06	5.2
2004	54.99	72.47	Syrian Arab Republic	46.04	Central African Republic	53.48	5.82	51.9	56.49	4.6
2005	55.65	72.77	Syrian Arab Republic	46.43	Central African Republic	54.0	5.78	52.8	57.58	4.78
2006	56.36	73.35	Syrian Arab Republic	46.85	Central African Republic	54.86	5.73	53.65	58.69	5.04
2007	57.0	73.71	Syrian Arab Republic	47.43	Central African Republic	55.55	5.67	54.32	59.18	4.85
2008	57.6	73.55	Syrian Arab Republic	48.02	Central African Republic	56.28	5.52	55.13	59.91	4.78
2009	58.2	73.85	Syrian Arab Republic	48.65	Central African Republic	56.75	5.44	55.76	60.38	4.61
2010	58.72	73.88	Syrian Arab Republic	49.26	Central African Republic	57.19	5.28	56.38	60.75	4.37
2011	59.32	73.31	Syrian Arab Republic	49.95	Central African Republic	57.94	5.11	57.01	61.45	4.44
2012	59.57	67.47	Nepal	50.54	Chad	58.75	4.35	57.37	62.05	4.68
2013	59.94	67.96	Nepal	50.78	Chad	59.74	4.16	57.78	62.56	4.77
2014	60.24	68.08	Nepal	50.57	Central African Republic	60.14	4.15	58.2	62.96	4.76
2015	60.69	67.46	Nepal	51.59	Chad	60.65	3.82	58.46	63.82	5.36

The Low Income group, surprisingly, showcases the highest overall life expectancy among all income classes in this dataset. With a mean life expectancy beginning at 76.08 years in 2000 and reaching 79.66 years in 2015, this group reflects strong, consistent improvements despite limited economic resources.

This somewhat counterintuitive result is driven by the inclusion of countries such as Japan, Switzerland, and other high-performing outliers that have maintained exceptionally high life expectancy over the years but were classified as "low income" based on GNI thresholds used in this analysis.

Key Trends:

- **Strong Upward Momentum in Life Expectancy:**
  - The **mean life expectancy** rose by 3.58 years between 2000 and 2015.
  - The **median** followed a similar trajectory, increasing from 76.59 to 80.78 years, indicating that most countries in this group experienced substantial and consistent improvements.
- **High and Stable Performance at the Top:**
  - **Japan** consistently recorded the highest life expectancy, peaking at 83.79 years in 2015.
  - In 2011, **Switzerland** briefly surpassed Japan, illustrating how top performers in this group remain globally dominant in terms of health outcomes.
- **Improving Minimums:**
  - The **minimum life expectancy** improved from 69.1 years in 2000 to 73.1 years in 2015, showing a meaningful uplift at the lower end of the distribution.
  - Countries like Trinidad and Tobago and Bahamas frequently recorded the lowest values, which still remained above 70 years after 2006—suggesting that even the laggards in this group enjoy relatively high life expectancy.

Distribution Observations:

- The **standard deviation** remained relatively stable, averaging around 2.85 to 2.93, suggesting moderate variability in life expectancy.
- The 88 fluctuated only slightly, from 4.15 to 4.43, implying that most countries in this group have closely clustered life expectancy values with fewer extreme outliers.
- **25th and 75th percentiles** rose consistently:
  - The **25th percentile** went from 74.09 to 77.54 years.
  - The **75th percentile** climbed from 78.24 to 81.96 years—a clear signal of general uplift across the group.

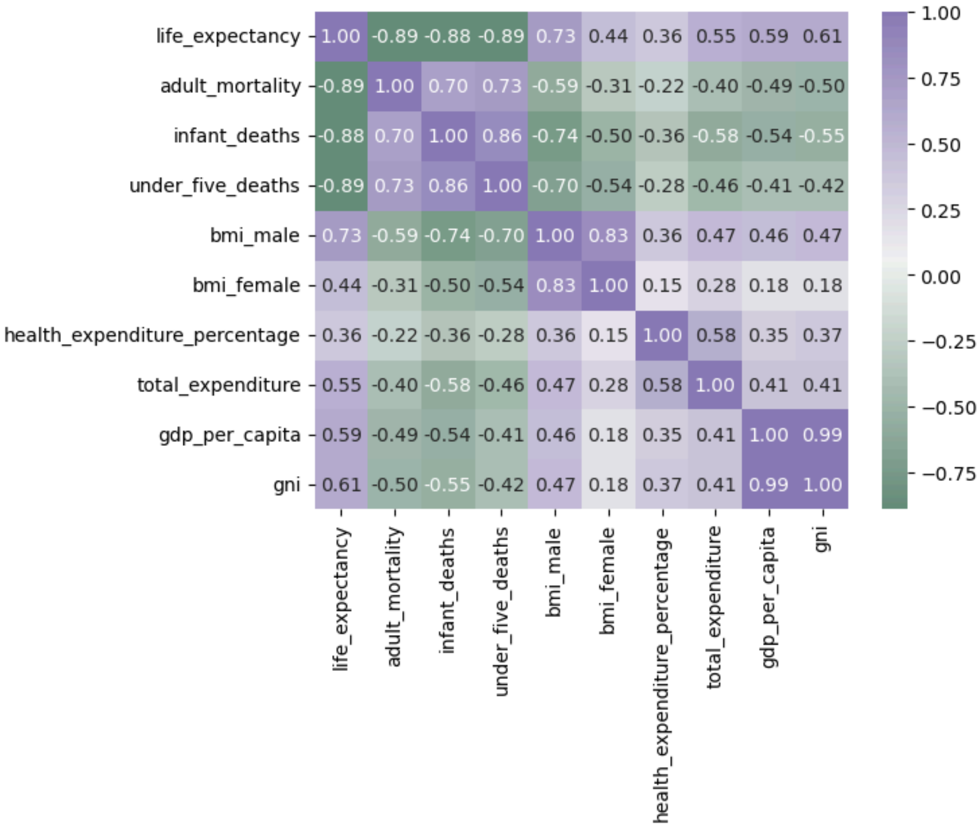
In summary, although classified as “low income” by GNI, the countries in this group show excellent life expectancy outcomes, likely due to data classification nuances and the presence of standout performers. These trends reinforce the idea that health outcomes can transcend income classifications, especially where robust healthcare systems, healthy lifestyles, and strong public health policies are in place.

---

4. Exploratory Data Analysis (EDA)

4.1 Correlation and Heatmap Analysis Between Variables

Complete Dataset (All countries)

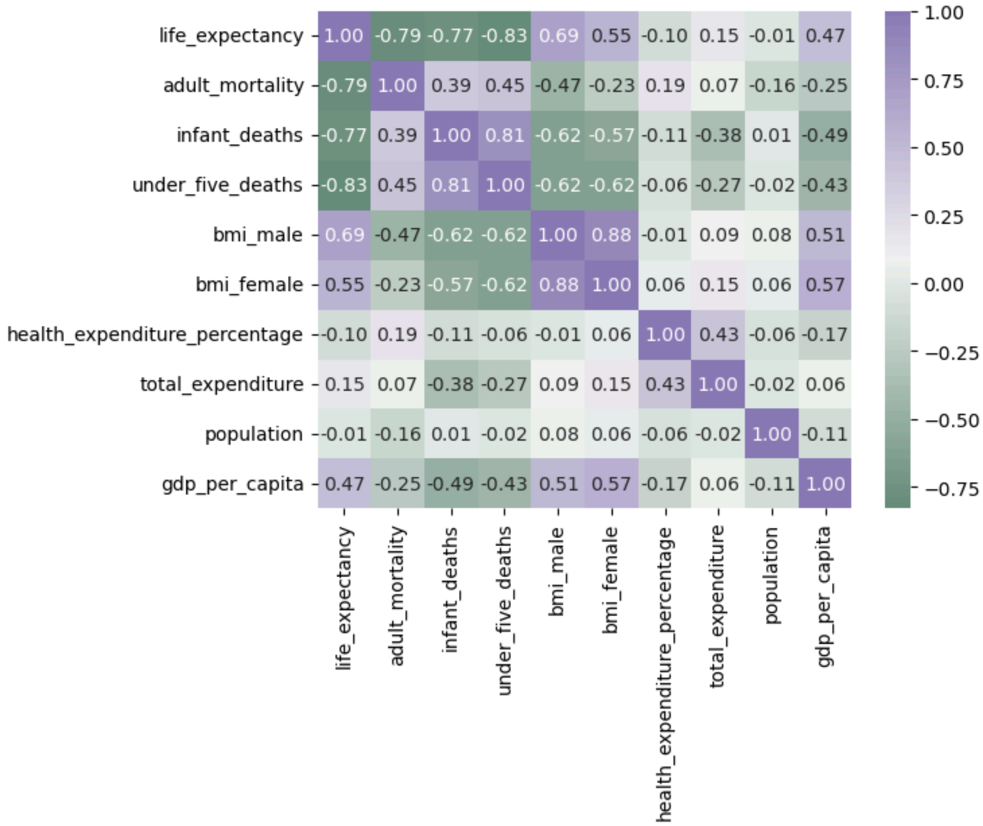


The correlation matrix for the full dataset highlights several strong relationships:

- Life expectancy shows a strong negative correlation with adult mortality (-0.89), infant deaths (-0.88), and under-five deaths (-0.89).
- There is a positive correlation with male BMI (0.73) and moderate correlation with GDP per capita (0.59), GNI (0.61), and total health expenditure (0.55).
- Interestingly, female BMI has a weaker positive correlation (0.44), while health expenditure percentage shows only a minor correlation (0.36).

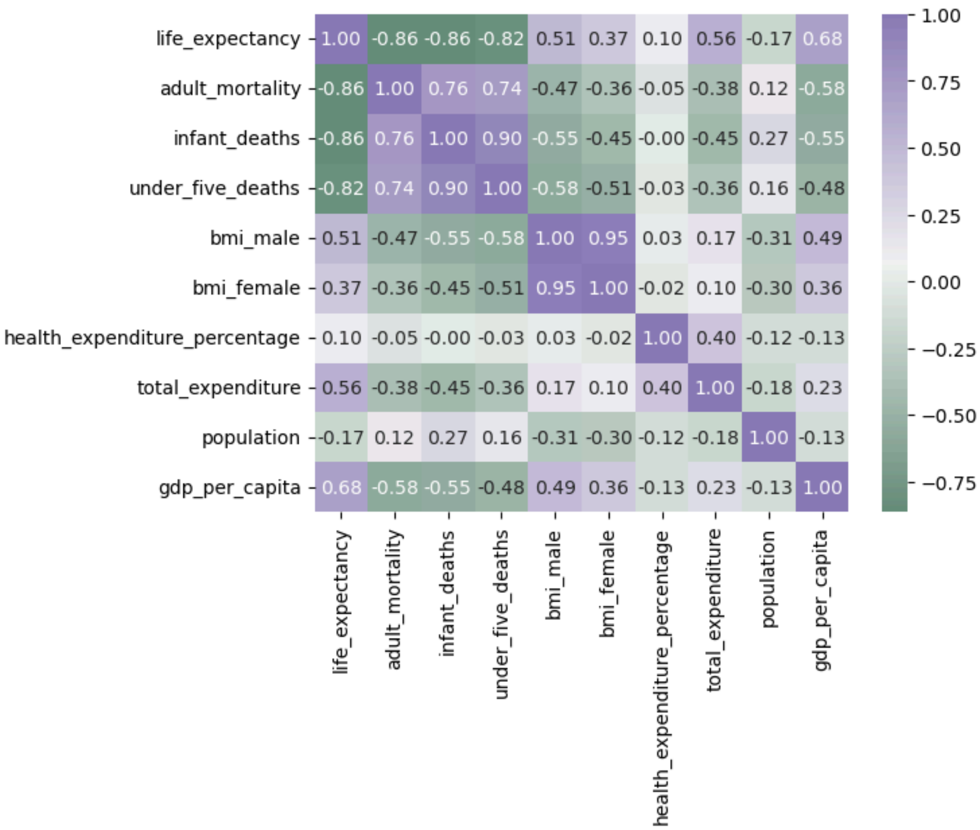
These results indicate that mortality-related variables are the strongest inverse predictors of life expectancy, while economic indicators and nutritional health (as measured by BMI) are moderately supportive.

Africa



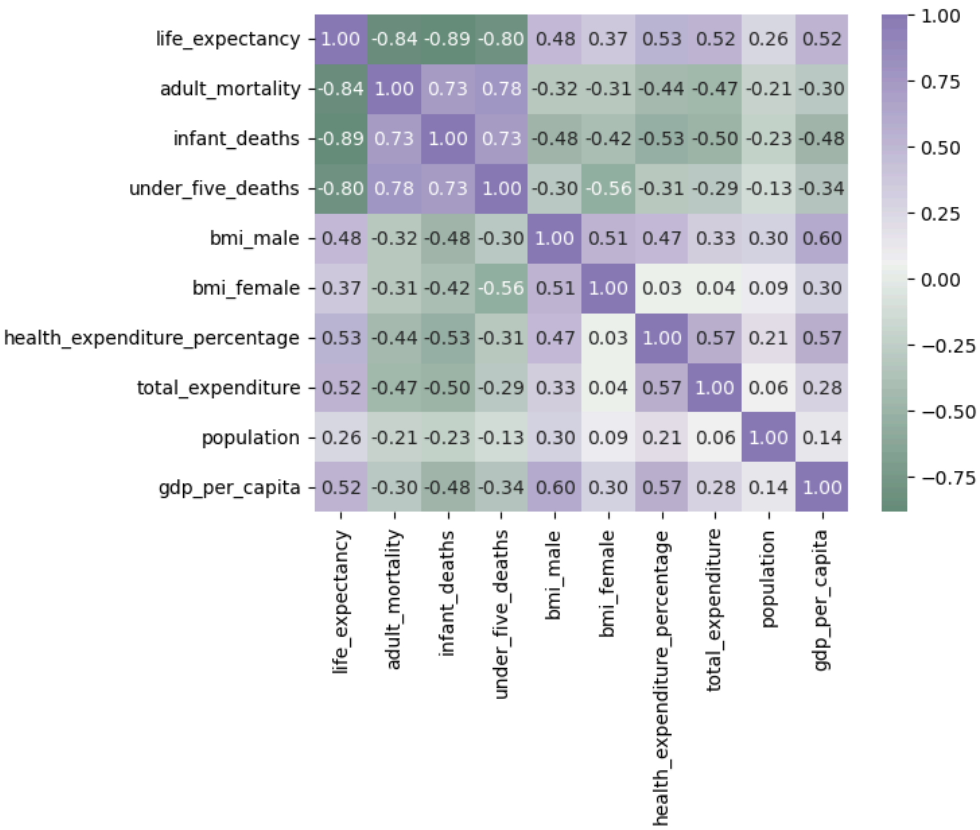
- Life expectancy in Africa is strongly negatively correlated with under-five deaths (-0.83) and infant deaths (-0.77), but less so with adult mortality (-0.79).
- The correlation with GDP per capita (0.47) and total expenditure (0.15) is weaker than in other regions.
- Notably, health expenditure percentage and population size have almost no correlation, suggesting economic and systemic challenges dilute the potential impact of these variables.

Asia



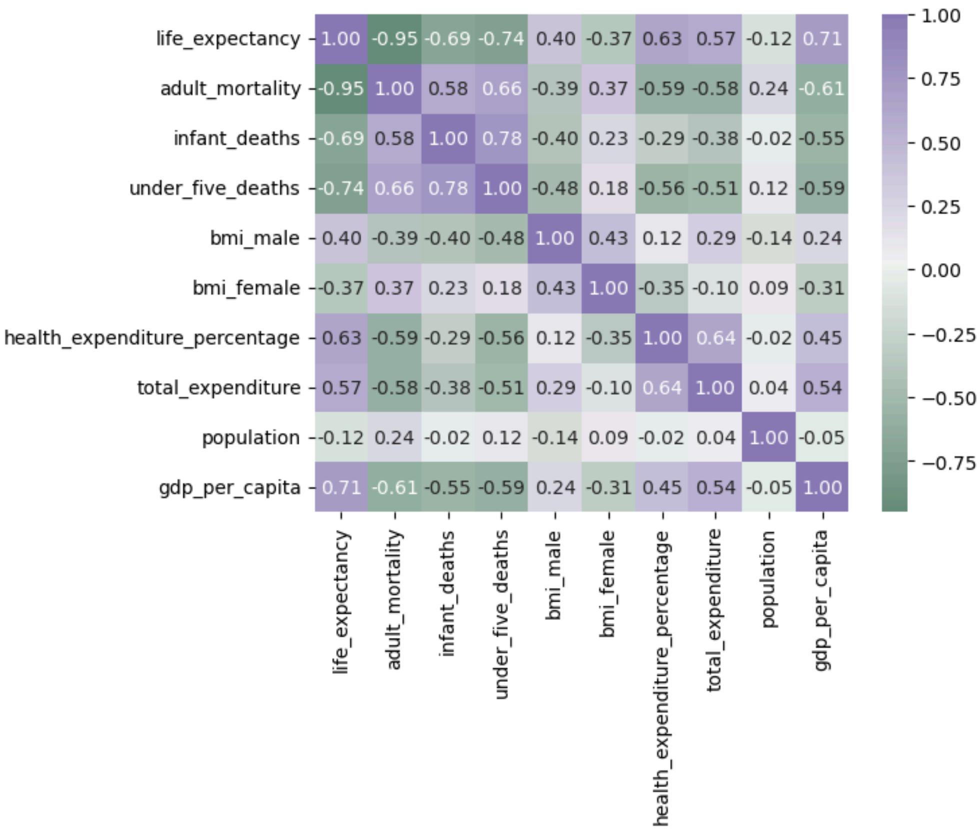
- A strong inverse relationship exists between life expectancy and all mortality indicators, particularly adult mortality (-0.86) and infant deaths (-0.86).
- GDP per capita shows a relatively stronger positive correlation (0.68).
- Total expenditure (0.56) and health expenditure percentage (0.10) remain lower predictors, echoing the global trend that spending percentage alone isn't highly predictive.

Americas



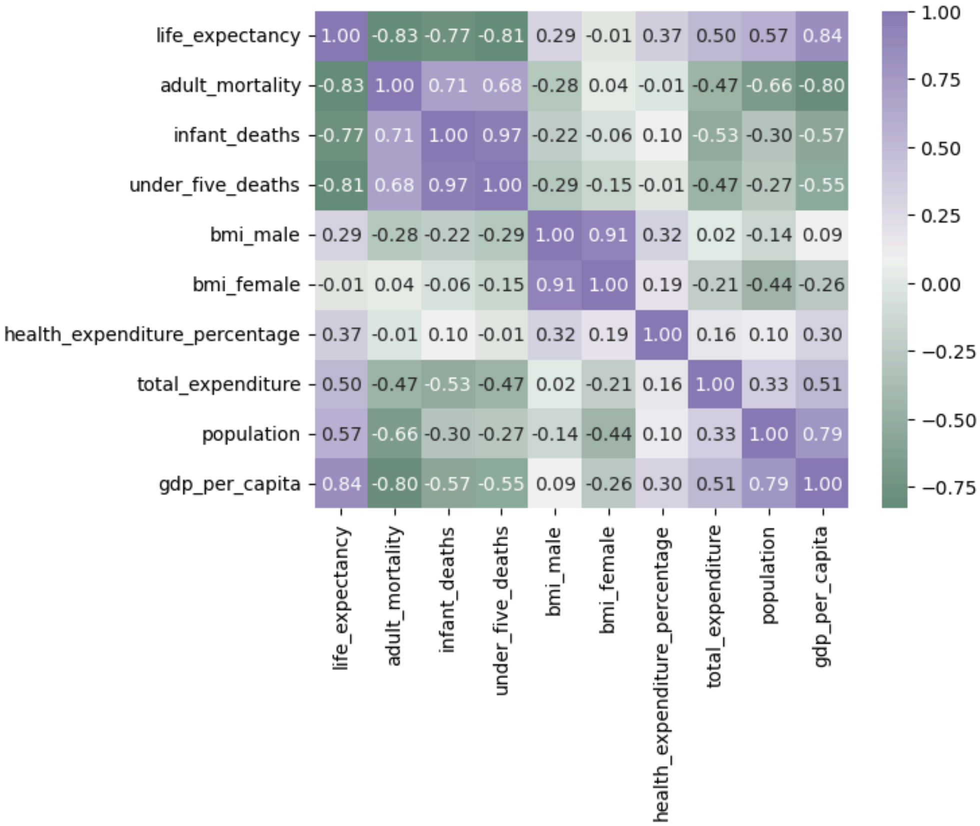
- Life expectancy correlates highly with adult mortality (-0.84) and infant deaths (-0.89).
- The relationship with health expenditure percentage (0.53) is stronger than in most other regions, suggesting a higher return on health spending.
- The role of economic indicators remains moderate to strong, with GDP per capita (0.52) and total expenditure (0.52).

Europe



- Europe reflects the strongest correlation between life expectancy and economic indicators:
  - GDP per capita (0.71)
  - GNI (0.70)
  - Total expenditure (0.57)
  - Health expenditure percentage (0.63)
- While mortality indicators still show strong negative correlations (e.g., adult mortality at -0.95), this region stands out due to a strong positive relationship between life expectancy and investment in health and wealth.

Oceania

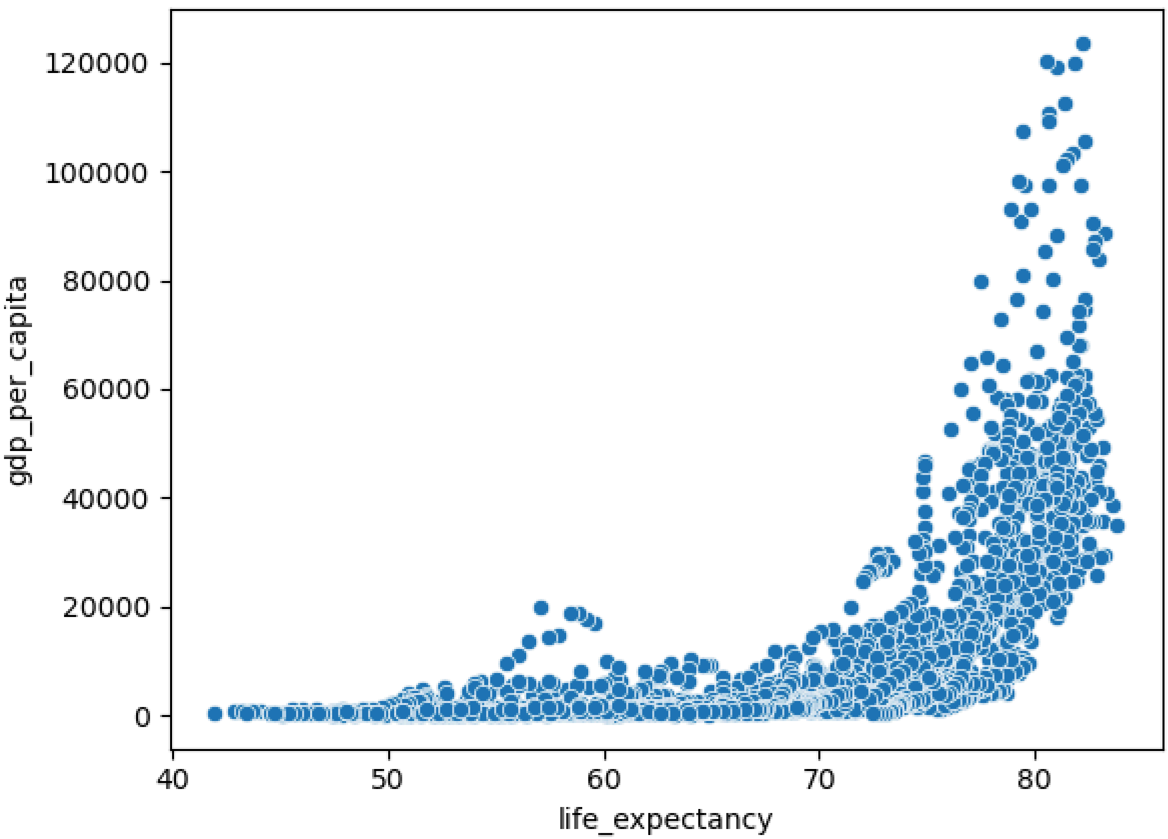


- This region shows the strongest overall positive correlation between life expectancy and GDP per capita (0.84).
- There's a noticeable drop in correlations with mortality metrics, though still strong (e.g., under-five deaths at -0.81).
- Correlations with population (0.57) and total expenditure (0.50) also suggest that in smaller, high-income nations, economic metrics are more predictive than health burdens.

4.2 Scatterplot Analysis (Life Expectancy vs Other Variables)

Scatterplots provide a visual exploration of the bivariate relationships between life expectancy and other key variables. This section highlights patterns, clustering, and trends that complement the insights gathered from the correlation matrices in Section 4.1.

4.2.1 Life Expectancy vs. GDP per Capita

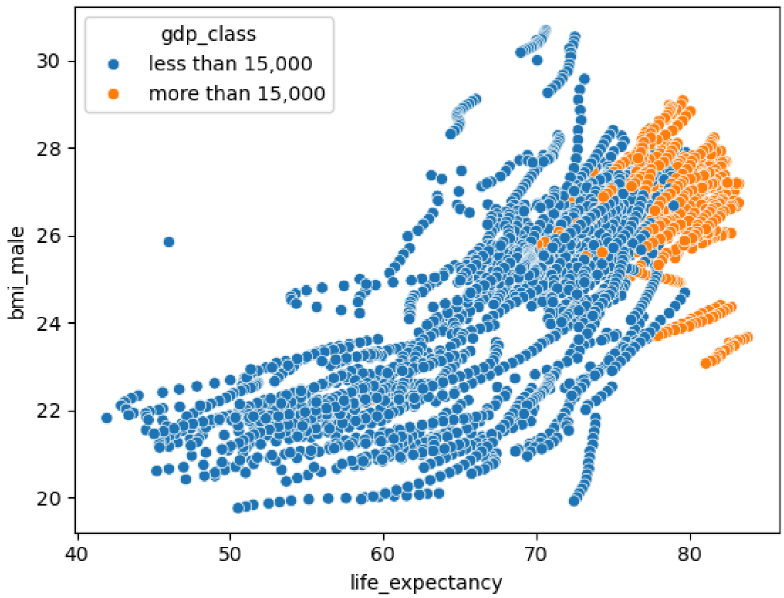


The relationship between life expectancy and GDP per capita reveals a curved, upward-trending pattern, suggesting diminishing returns:

- Countries with lower GDP per capita show a wide spread of life expectancy values.
- Once GDP per capita surpasses approximately $15,000, life expectancy stabilizes above 70–75 years.
- Outliers with extremely high GDP and life expectancy are mostly high-income countries.

4.2.2 Life Expectancy vs. BMI (Male & Female)

Life Expectancy vs BMI Male Scatterplot

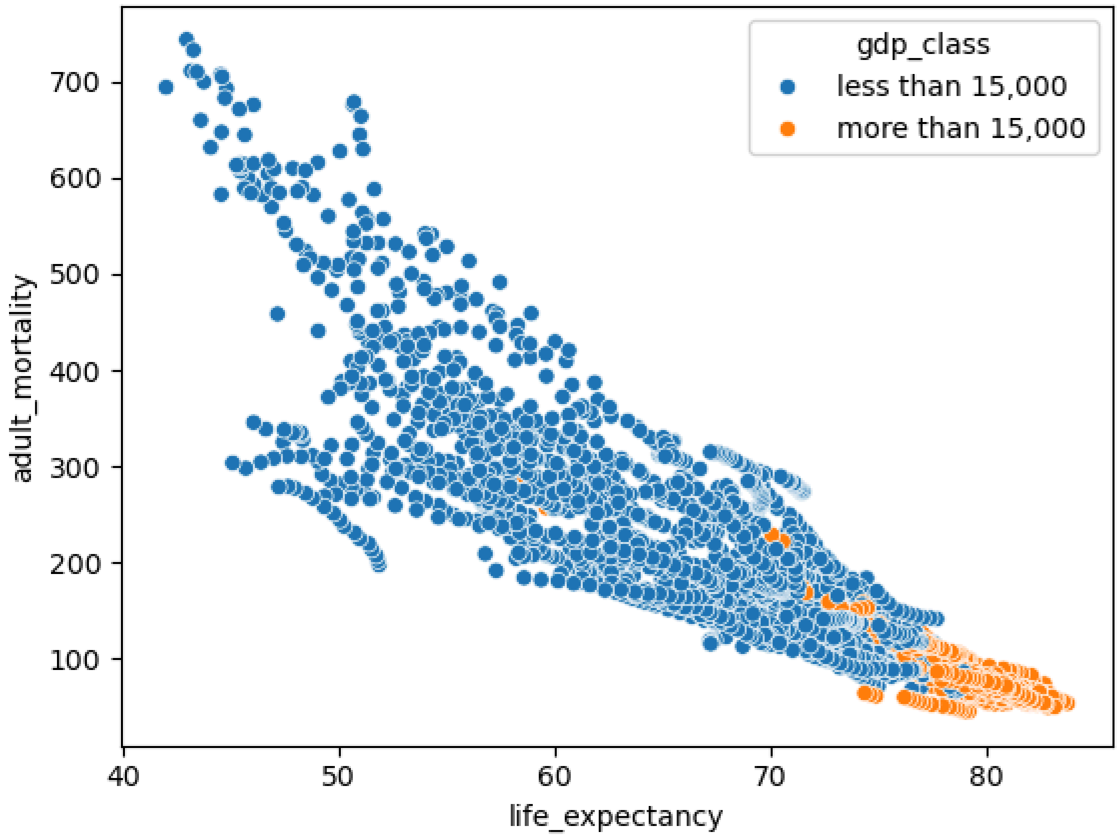


Life Expectancy vs BMI Female Scatterplot

There is a positive linear trend between life expectancy and both male and female BMI:

- Male BMI rises more steeply with increasing life expectancy than female BMI.
- A clear separation appears when stratifying by GDP class — higher GDP nations tend to cluster at the upper end of both BMI and life expectancy.
- Female BMI shows slightly more dispersion but maintains a positive trend overall.

4.2.3 Life Expectancy vs. Adult Mortality

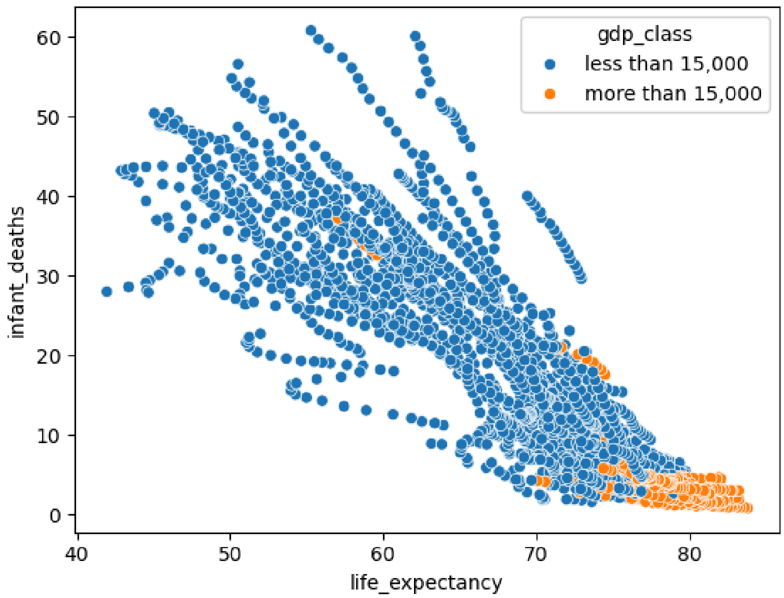


As expected, adult mortality shows a strong inverse relationship with life expectancy:

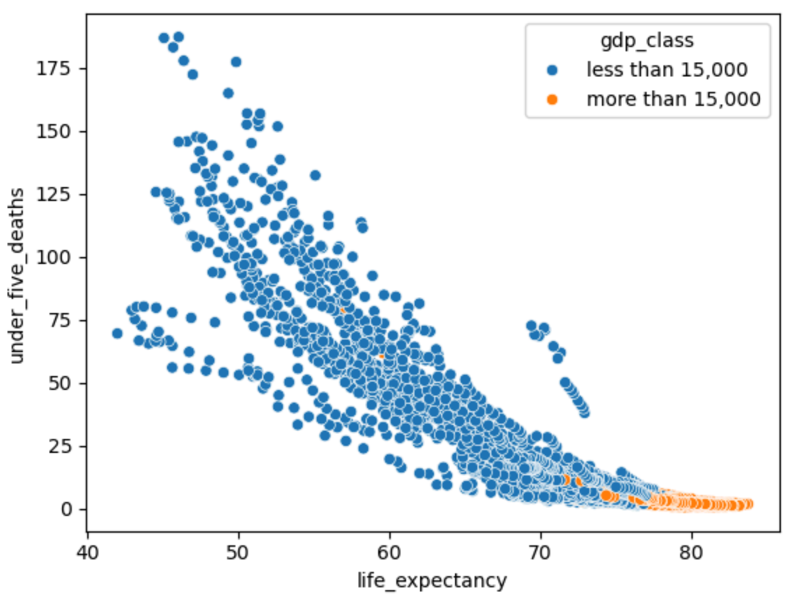
- Most countries with low life expectancy have adult mortality rates above 300 per 1,000.
- Countries with high life expectancy consistently report adult mortality rates below 150.
- The clustering of high-GDP nations in the lower-right corner reinforces this negative association.

4.2.4 Life Expectancy vs. Infant Mortality & Under-5 Mortality

Life Expectancy vs. Infant Mortality Scatterplot



Life Expectancy vs. Under-5 Mortality Scatterplot



Both indicators show a clear negative exponential relationship:

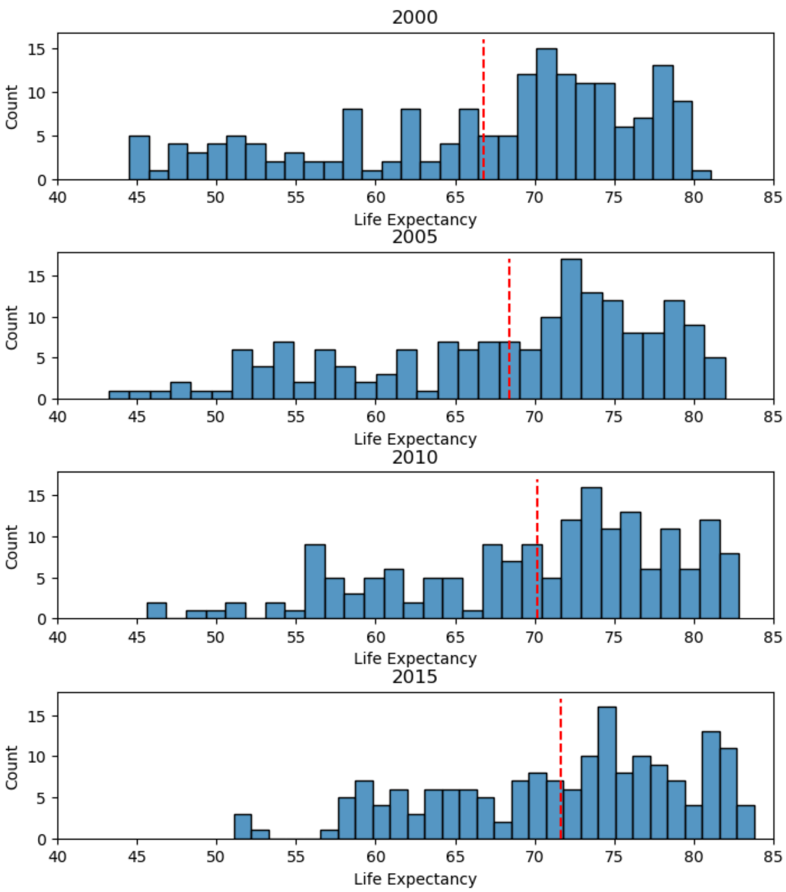
- Countries with higher infant and under-5 death rates exhibit significantly lower life expectancy.
- High-GDP countries form a dense cluster with low child mortality and high life expectancy, whereas lower-GDP countries spread widely, indicating divergent health outcomes.

4.3 Distribution Analysis

Distribution diagrams are powerful tools for understanding how values are spread within a dataset. The following histograms illustrate the evolution of life expectancy distributions across all countries and within each global region over time (2000, 2005, 2010, 2015). Each distribution includes a red dashed line representing the mean life expectancy for that year and region, helping to highlight changes and central tendencies over time.

Life Expectancy

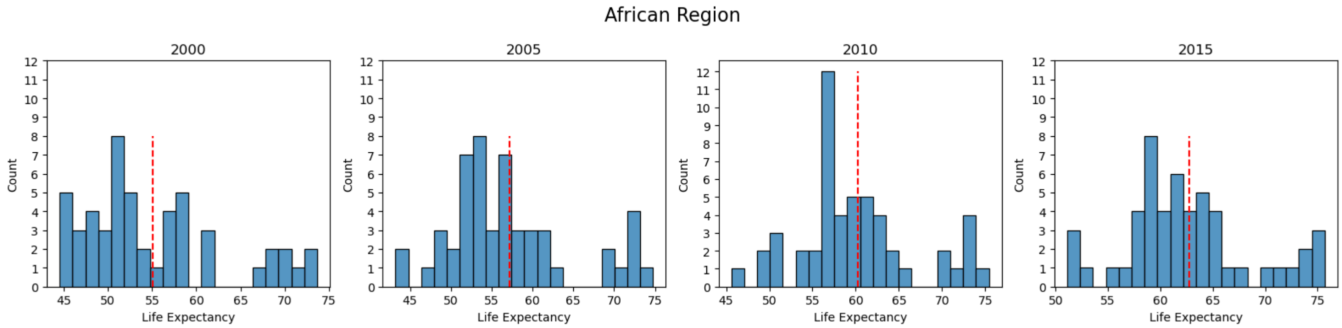
All Countries



The global distribution of life expectancy has steadily shifted to the right from 2000 to 2015, reflecting overall improvement in longevity worldwide:

- In 2000, life expectancy varied widely, with a notable number of countries below 60 years. The mean hovered just above 65 years.
- By 2015, most countries clustered around 70–75 years, with a sharp decline in the number of countries below 60.
- The distribution remains right-skewed, but the gap between low- and high-life expectancy countries has narrowed, suggesting global health convergence.

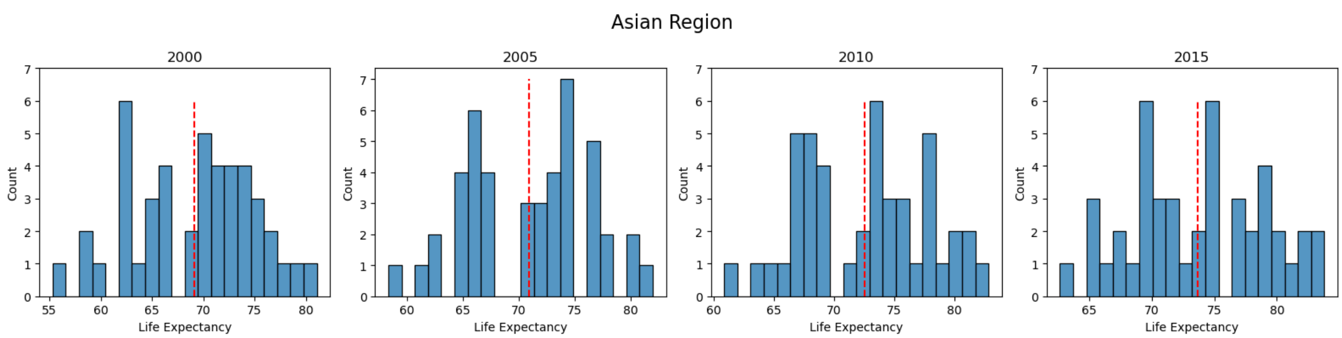
African Region



The African region exhibits consistent improvement, but still lags behind other regions:

- In 2000, the majority of countries had life expectancy between 45 and 60 years, with a mean around 55.
- A gradual rightward shift occurred by 2015, with the mean reaching just above 63 years.
- The shape of the distribution remains wide and slightly bimodal, indicating heterogeneity within the continent, possibly due to differing health infrastructures or conflict-related disparities.

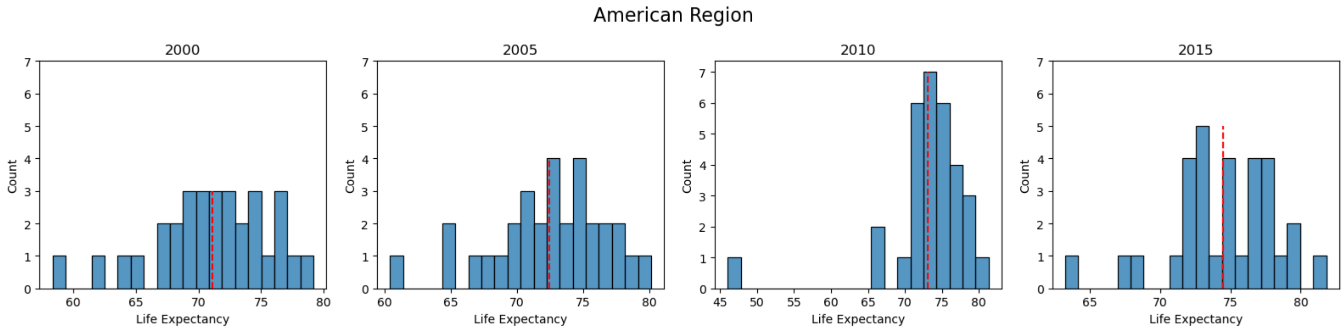
Asian Region



Asia demonstrates steady and relatively high life expectancy:

- The distribution from 2000 to 2015 consistently spans 60–80+ years, with the mean steadily climbing from around 69 to 74 years.
- The shape remains somewhat uniform, reflecting the region's diversity in economic development and healthcare access.
- Improvement is evident across the entire distribution, not just at the top end, indicating broad-based regional health gains.

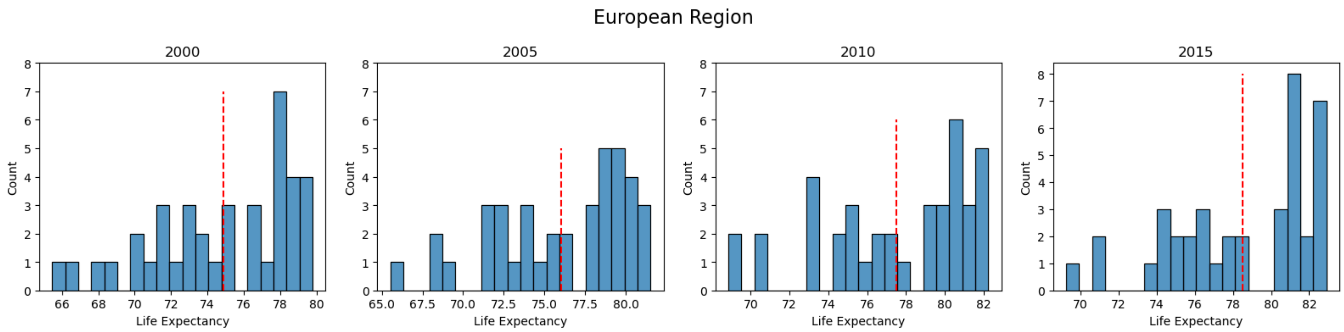
American Region



The Americas display a consistently high and compact distribution:

- From 2000 to 2015, most countries had life expectancy above 65 years, with a strong central tendency between 70 and 75 years.
- The distribution becomes increasingly tight over time, signaling consolidation of health improvements and reduced variance.
- There is less visible skewness, suggesting greater equity in life expectancy across nations in the region.

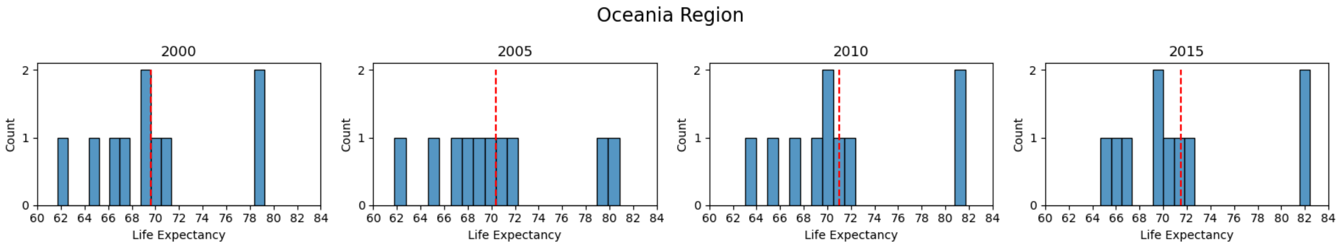
European Region



Europe consistently exhibits the highest life expectancy and least variance:

- The distribution is tightly packed between 70 and 83 years across all four time periods.
- The mean rose steadily from around 75 to nearly 79 years, affirming Europe's position as a global leader in public health and longevity.
- The histogram is close to a normal distribution, reflecting systematic and equitable access to healthcare.

Oceania Region

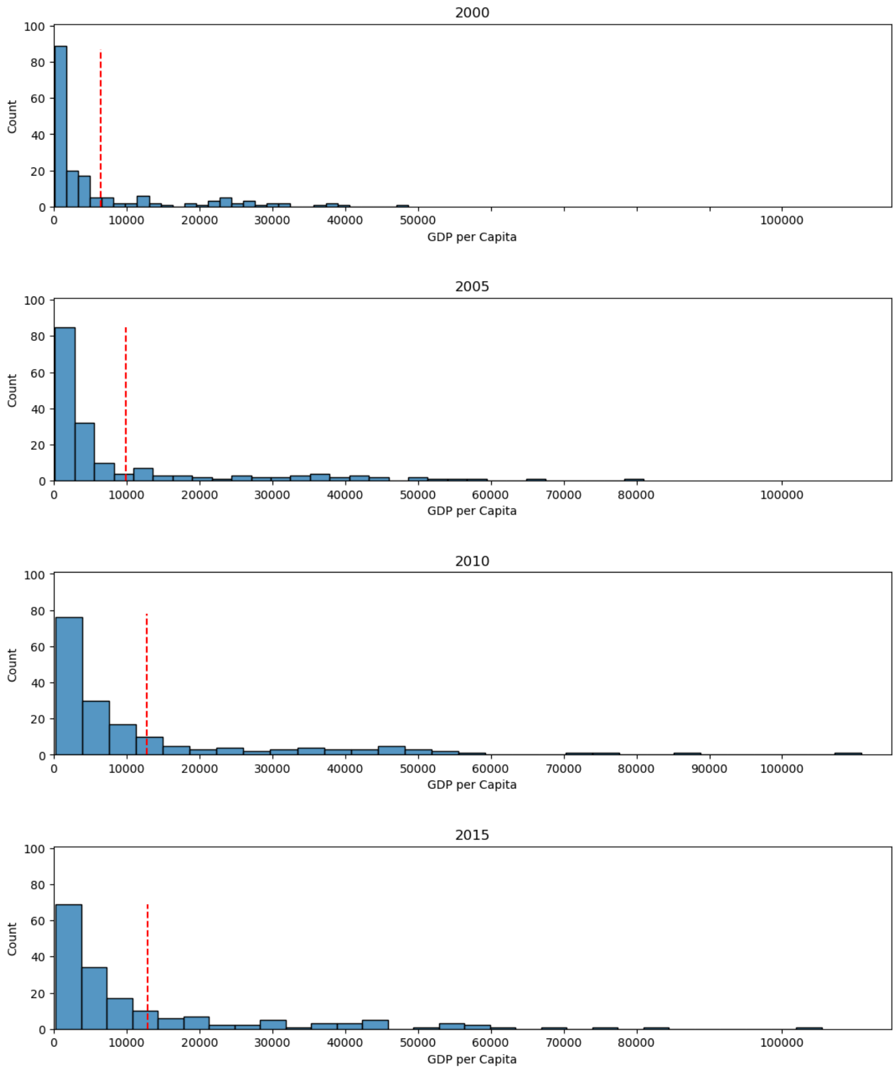


Due to a smaller number of countries, Oceania's distributions appear sparse but stable:

- Life expectancy remains high and consistent across the period, generally ranging between 65 and 82 years.
- The regional mean holds between 70 and 72, with slight increases over time.
- While variance is limited, the small sample size should be considered when interpreting trends.

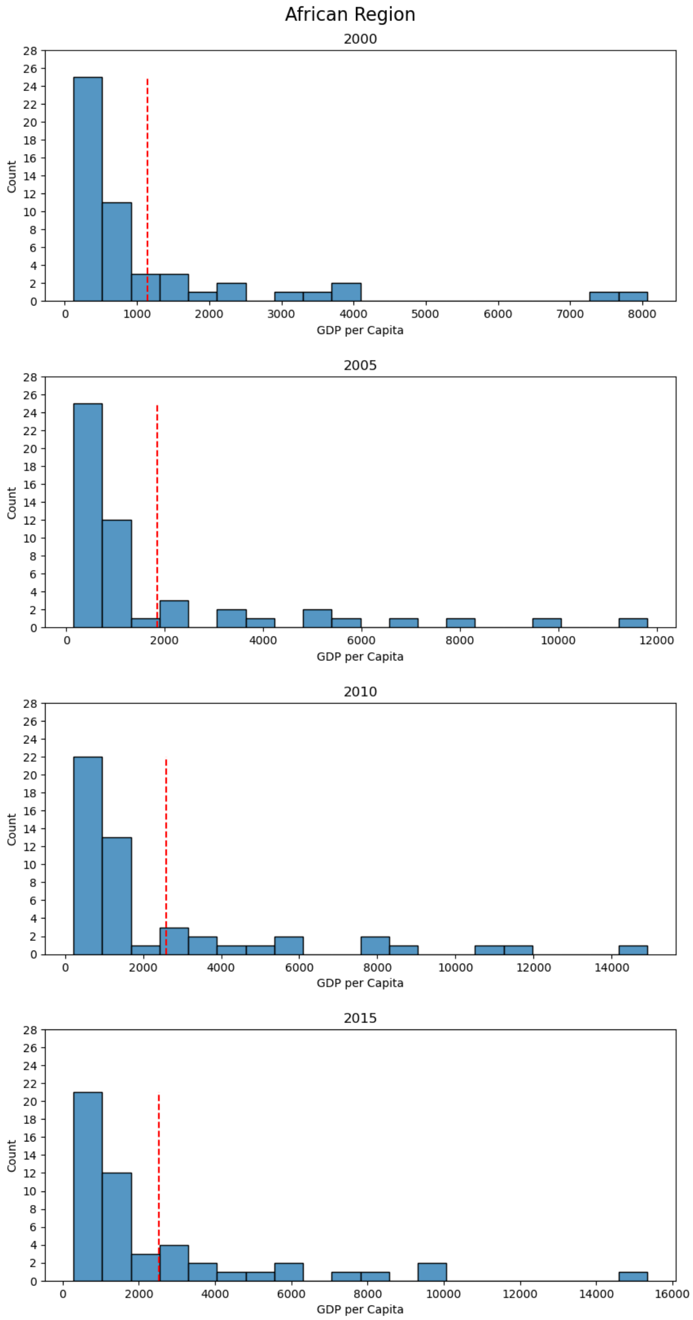
GDP per Capita

All Countries



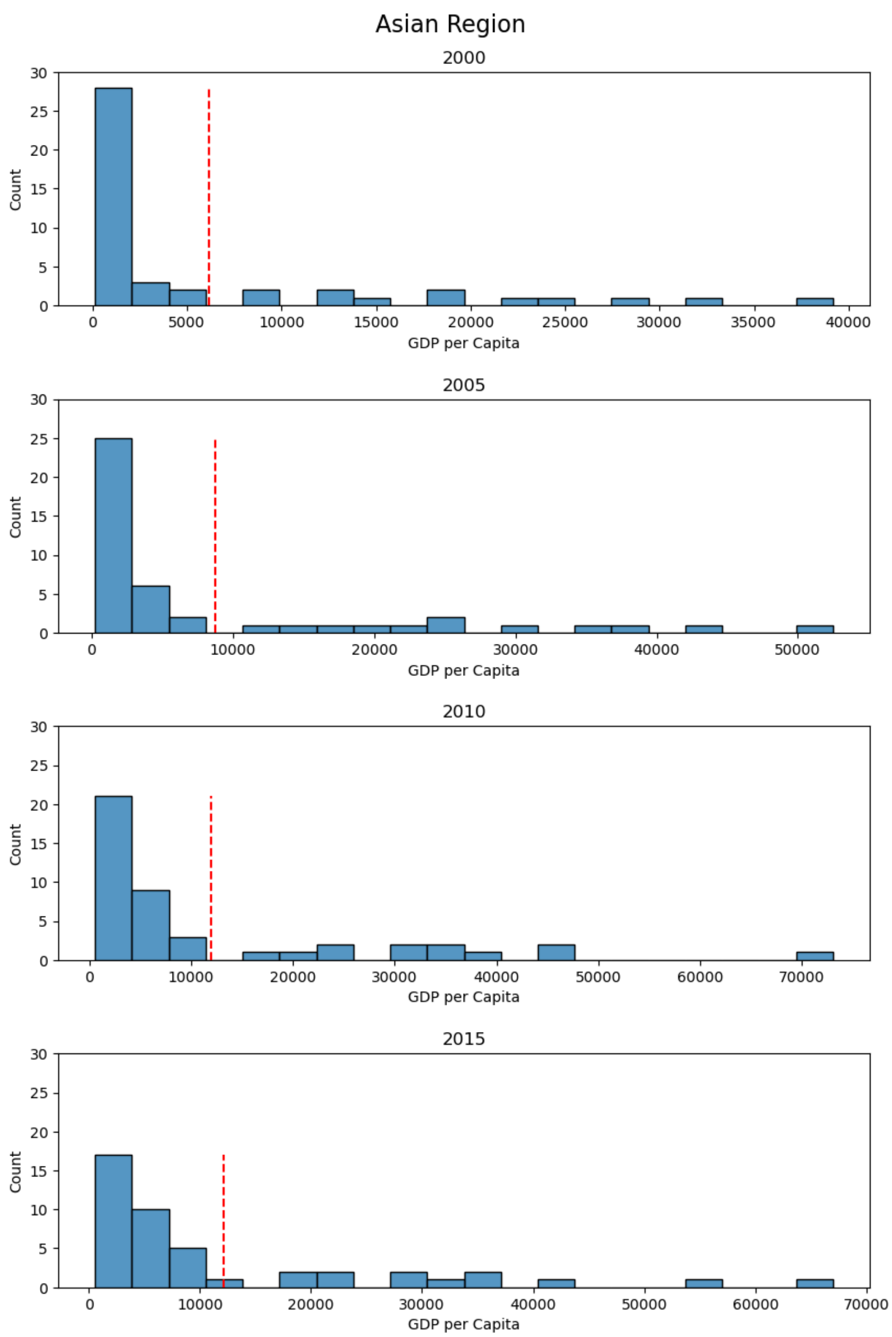
- From 2000 to 2015, the global distribution of GDP per capita remained heavily right-skewed, with most countries concentrated in the lower income range (below $10,000).
- The mean GDP per capita steadily increased over time — however, due to extreme outliers (wealthy countries), it overstates typical country-level income.
- The tail of the distribution grows, indicating that a few nations experienced exceptional economic growth, widening the global wealth gap despite general upward trends.

African Region



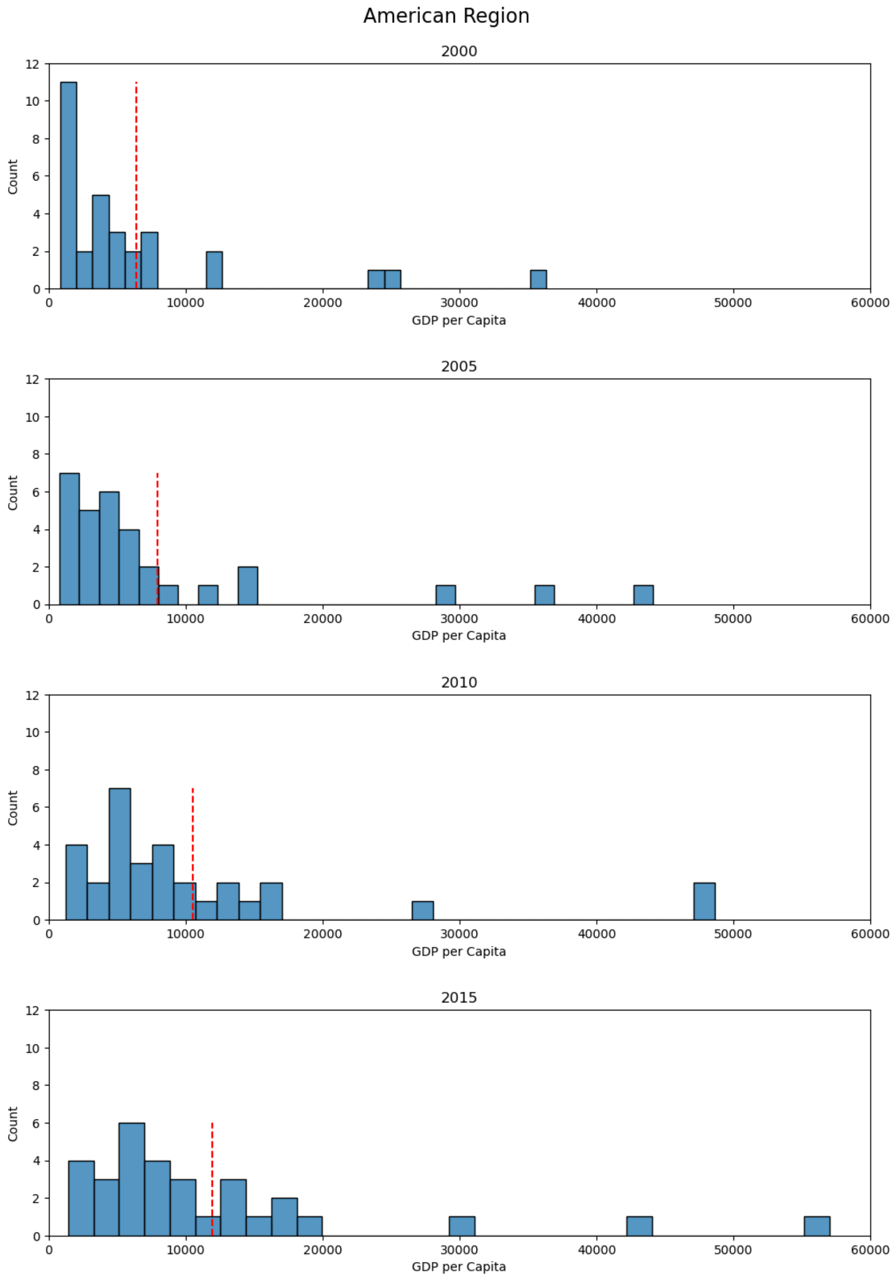
- Africa’s GDP per capita distribution is highly concentrated in the lowest income range, with over half the countries below $2,000 in 2000.
- There is a gradual rightward shift across years, and the mean increases steadily, suggesting slow but positive economic progress.
- Despite improvements, the spread remains narrow, and the skewness persists, highlighting continued economic vulnerability and limited upward mobility across much of the region.

Asian Region



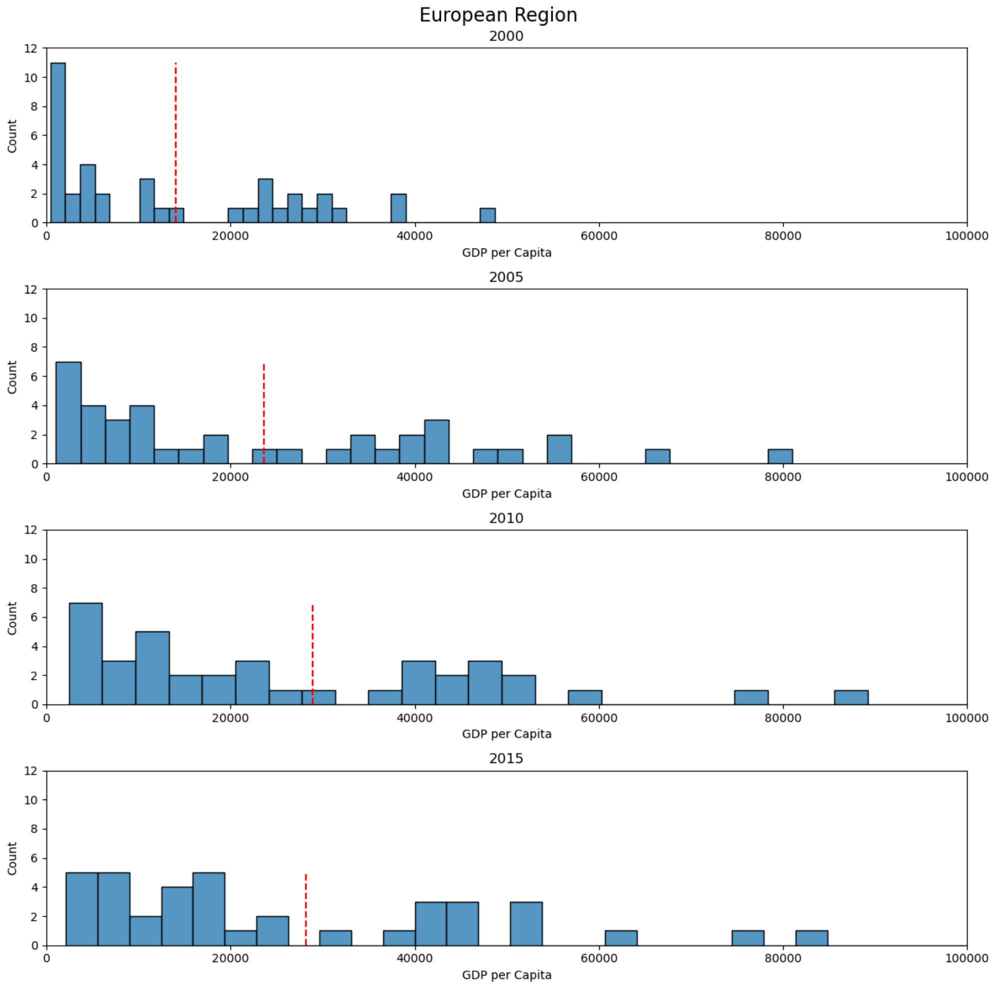
- Asia's distribution begins similarly skewed, but compared to Africa, higher-income outliers are more prevalent.
- The mean rises significantly from 2000 to 2015, mainly due to economic booms in countries like China, South Korea, and Singapore.
- The distribution stretches wider over time, reflecting increased intra-regional inequality, though the overall trend indicates strong upward momentum.

American Region



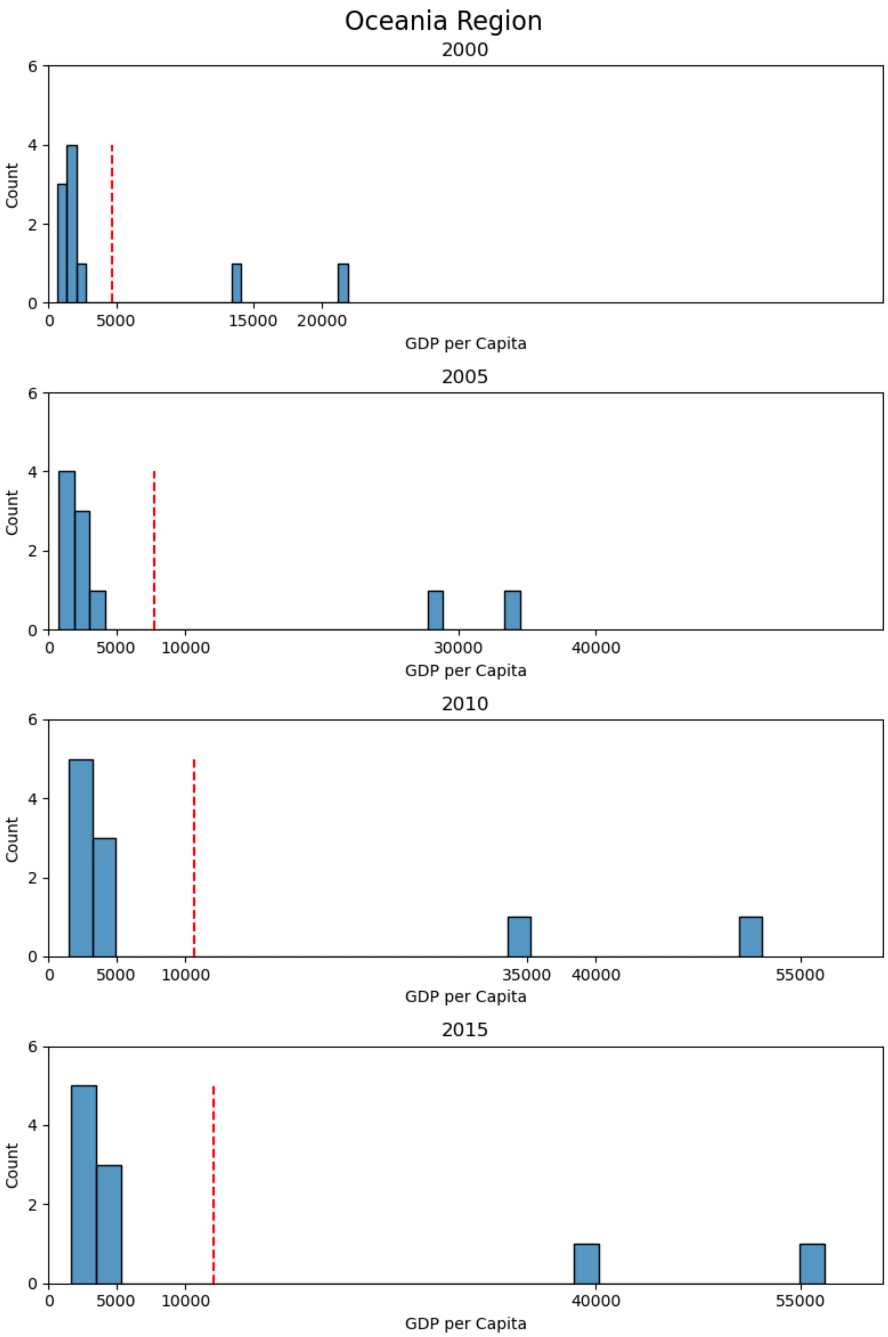
- The Americas show a bimodal-like shape, with clusters of both lower-middle-income countries and wealthier nations.
- Mean GDP per capita gradually increases, though the rise is modest relative to Asia or Europe.
- The spread between the lowest and highest earners remains significant, with a small number of countries (e.g., the U.S. and Canada) driving up the mean, while many countries remain under $10,000.

European Region



- Europe displays the most balanced and broad GDP per capita distribution among all regions.
- While still skewed, the curve stretches more evenly from $10,000 to above$80,000, reflecting high development levels across the continent.
- The average GDP per capita rises steadily over time, and the distribution becomes more uniform, suggesting broad-based prosperity and lower economic disparity within the region.

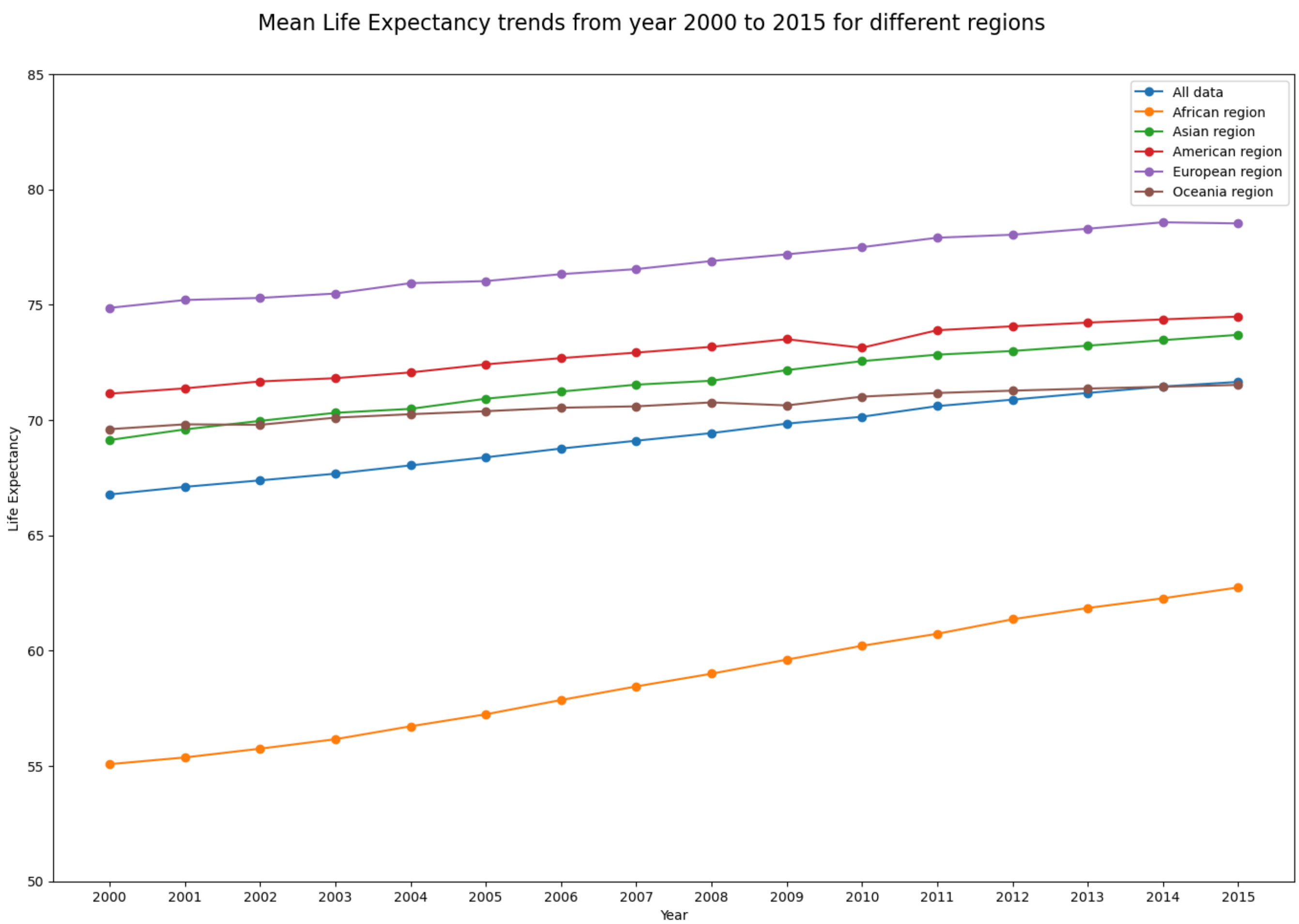
Oceania Region



- With a small number of countries, Oceania's distribution is sparse but consistently high.
- A few countries fall below $10,000, while others like Australia and New Zealand dominate the upper end.
- The mean increases from 2000 to 2015, but the low sample size limits statistical interpretation, and the wide gaps between values remain across the years.

4.4 Line Graph Analysis

Life Expectancy Mean (By Region)

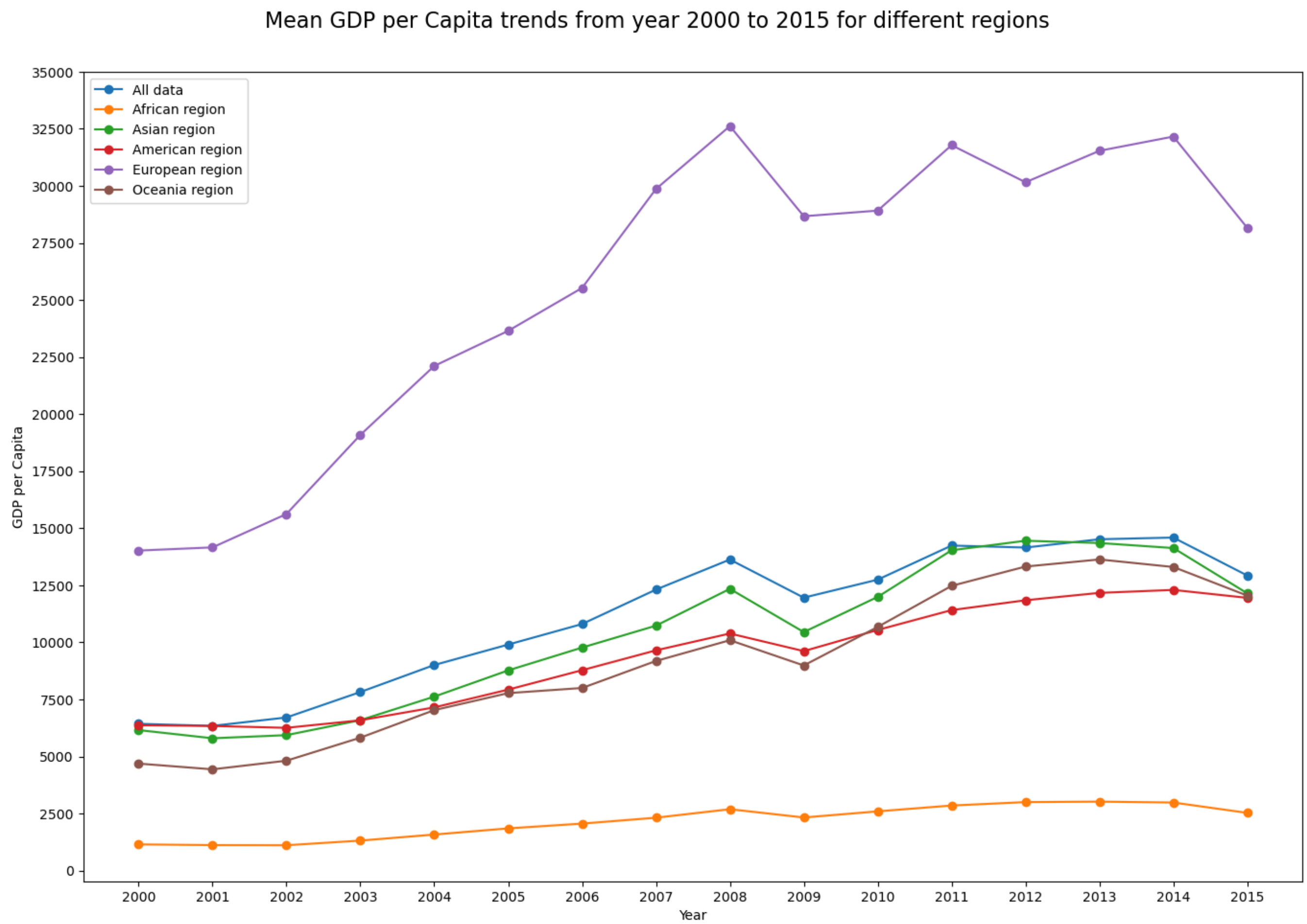


The first line graph illustrates the annual mean life expectancy from 2000 to 2015 for each region, along with the global average:

- General upward trend: All regions show consistent increases in life expectancy over the 15-year period, reflecting overall improvements in healthcare access, nutrition, and public health interventions.
- African Region: This region started with the lowest average life expectancy (around 55 years in 2000) but demonstrated the most notable gains, reaching approximately 63 years by 2015. This improvement suggests impactful health campaigns and reduced mortality rates.
- Asian Region: Life expectancy steadily increased from ~69.5 years in 2000 to over 73 years by 2015. The growth trajectory was relatively stable and consistent across years.
- American Region: Average life expectancy rose from ~71 years in 2000 to nearly 74.5 years in 2015, placing it above the global average throughout the period.

- European Region: Europe consistently had the highest average life expectancy, starting around 75 years in 2000 and climbing steadily to just under 79 years by 2015. This highlights strong healthcare systems and better living standards.
- Oceania: The Oceania region maintained a high average life expectancy, similar to Europe, and saw modest gains over time. The increase from ~69.5 years to 71.5 years is small, partly due to a small and relatively stable population set.
- Global Mean: The world average increased from 67 to over 71 years. The narrowing gap between lower-performing and higher-performing regions over time is indicative of global progress toward health equity.

GDP per Capita Mean (By Region)



The second graph tracks the mean GDP per capita for each region from 2000 to 2015:

- European Region: Europe demonstrated the highest GDP per capita throughout the entire period, with an especially sharp rise between 2003 and 2008 (from ~$15,000to$ 32,500), before stabilizing slightly below $30,000 after 2012. This pattern reflects economic recovery following the global financial crisis.
- African Region: Despite a low starting point (~$1,200), GDP per capita doubled by 2015. The growth was modest but steady, indicative of developmental efforts in infrastructure, trade, and agriculture.

- Asian Region: Asia showed strong economic growth, with GDP per capita increasing from ~$6,000 in 2000 to ~13,000 by 2014. A slight dip is visible in 2015, likely due to regional fluctuations or global economic slowdowns.
- American Region: The Americas experienced a gradual climb from ~$6,500to$ 12,200 by 2015, closely following the Asian region's trajectory with slightly lower growth.
- Oceania: Despite a smaller dataset, Oceania displayed a strong upward trend from ~$4,800to$ 13,000, aligning with the global average by the end of the period.
- Global Mean: GDP per capita for all countries combined shows a generally upward trend from ~ $6,500in2000to$ 13,000 in 2015, reinforcing the pattern of economic growth across most regions.

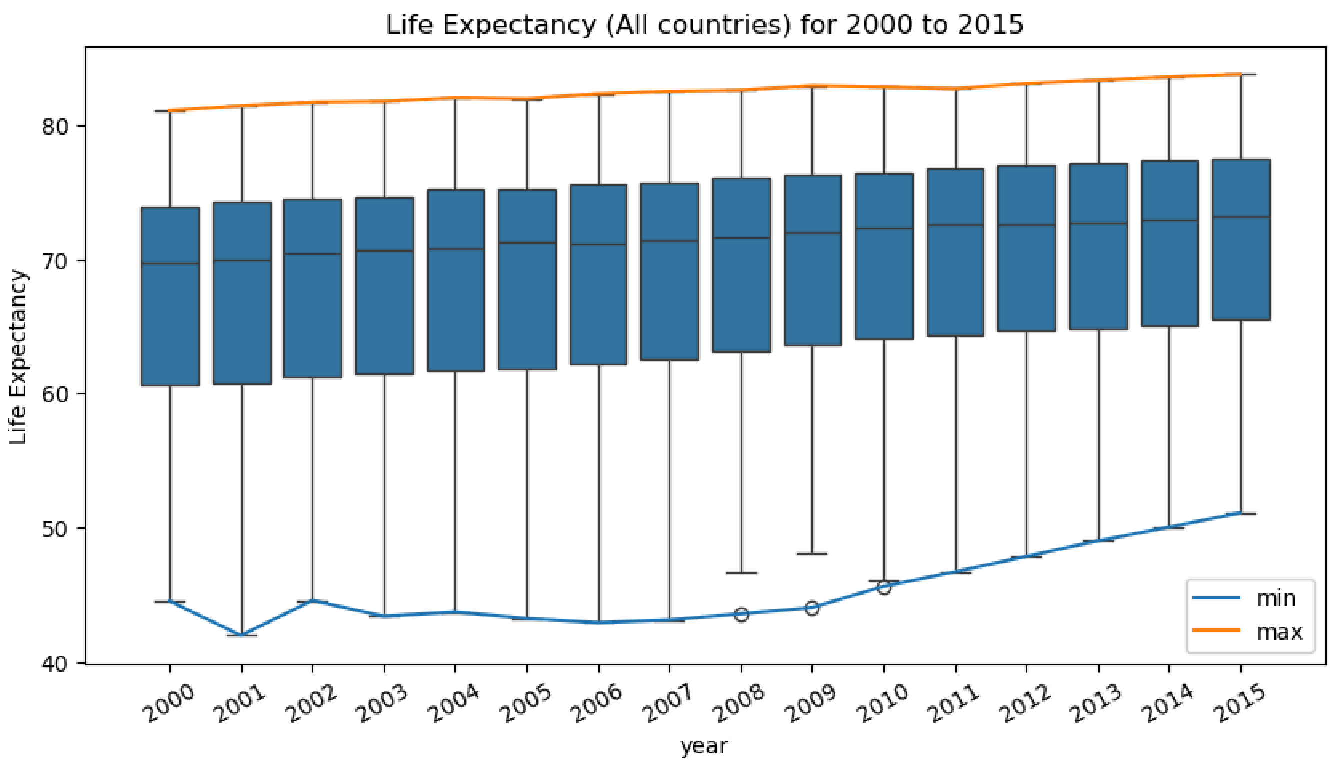
Key Observations and Implications

- Parallel Growth: Most regions showed simultaneous growth in life expectancy and GDP per capita, supporting the study's hypothesis of a positive association between the two indicators.
- Convergence Trends: Lower-income regions like Africa and Asia not only improved in health outcomes but also experienced significant GDP growth, suggesting narrowing development gaps.
- Economic Shocks Reflected: Temporary dips in GDP (particularly around 2009 and again in 2015) are likely associated with global financial and geopolitical events, yet life expectancy trends remained largely unaffected.
- Europe's Dominance: The European region stands out with both high GDP per capita and high life expectancy, reinforcing the benefits of sustained investment in public services, education, and healthcare.

4.5 Boxplots for Comparison

4.5.1 Life Expectancy

All Countries



General Trend:

- Life expectancy showed a steady and consistent upward trend from 2000 (median ≈ 69.5 years) to 2015 (median ≈ 73.5 years).

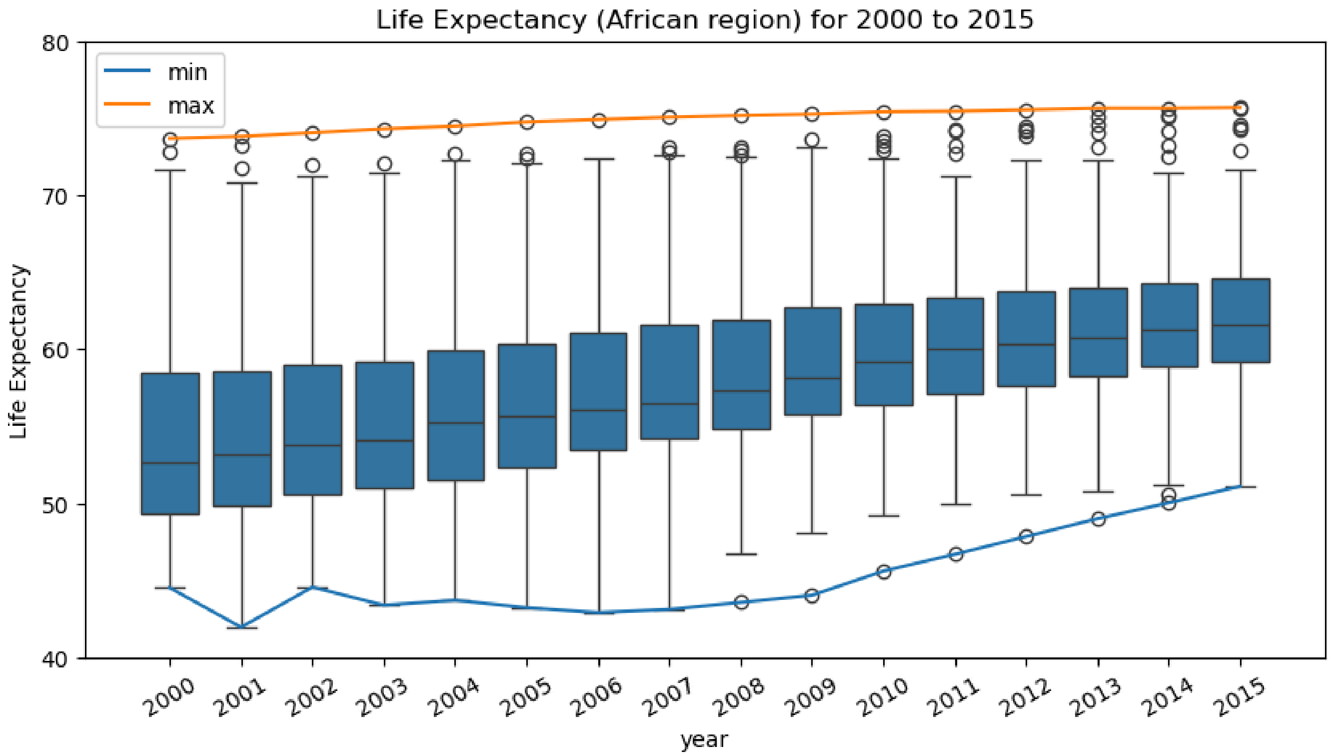
Spread and Outliers:

- The interquartile range (IQR) remained stable over time but narrowed slightly, indicating global convergence.
- Outliers were largely observed on the lower end, representing countries with public health crises or conflict (e.g., Zimbabwe in early 2000s).

Max and Min Trends:

- The max line remained relatively flat, increasing slightly from ≈ 81.0 to 83.5 years, reflecting the biological ceiling in developed nations.
- The min line rose from ≈ 42.0 to over 51.0 years, showing significant health improvements in lower-income countries.

African Region



General Trend:

- Life expectancy improved markedly from ≈ 51.5 years in 2000 to ≈ 63.5 years in 2015 (median).
- This indicates substantial public health progress, particularly post-2005.

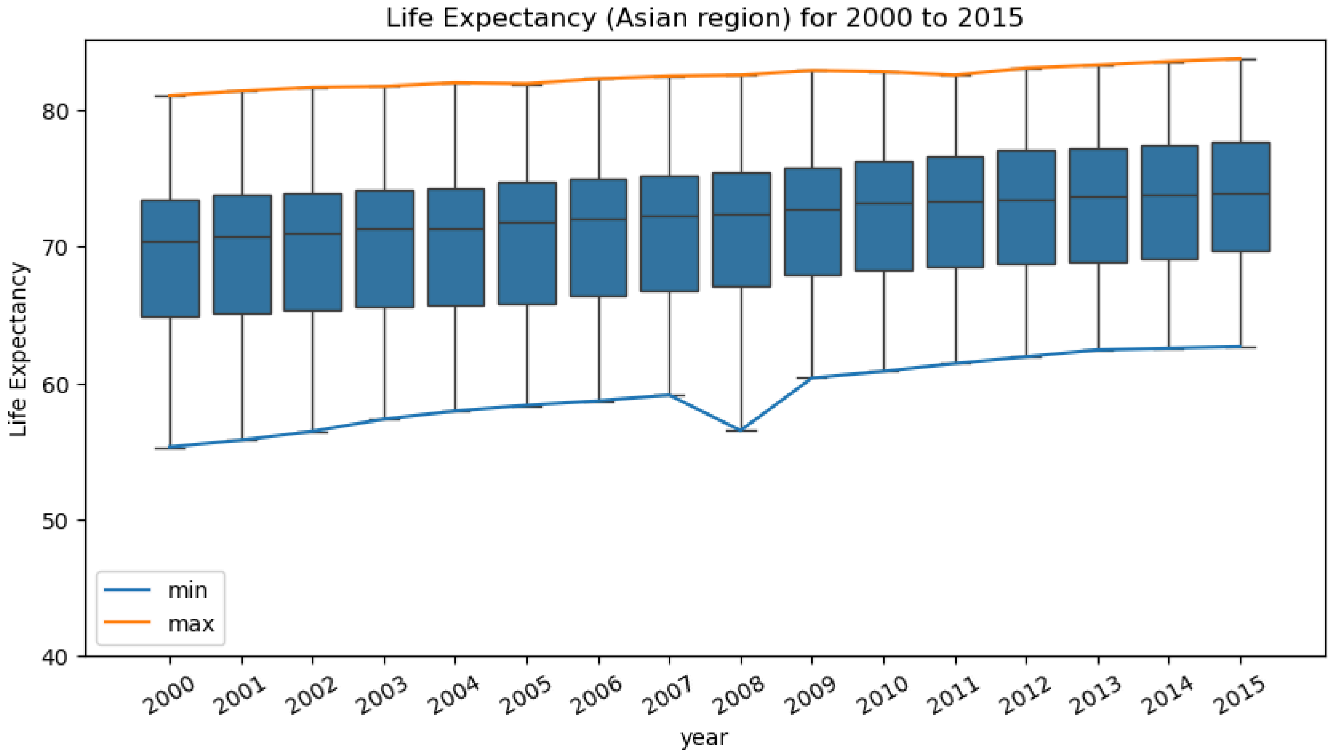
Spread and Outliers:

- The IQR remained wide, reflecting divergent development levels across the continent.
- Numerous outliers persisted throughout the period, suggesting ongoing instability or healthcare disparities in certain nations.

Max and Min Trends:

- Max life expectancy climbed from ≈ 73 to 76 years, while the minimum increased steadily from ≈ 42 to 51 years.
- This narrowing gap reflects slow but notable regional convergence.

Asian Region



General Trend:

- Life expectancy rose gradually from a median of ≈ 70 years in 2000 to ≈ 74.5 years in 2015.

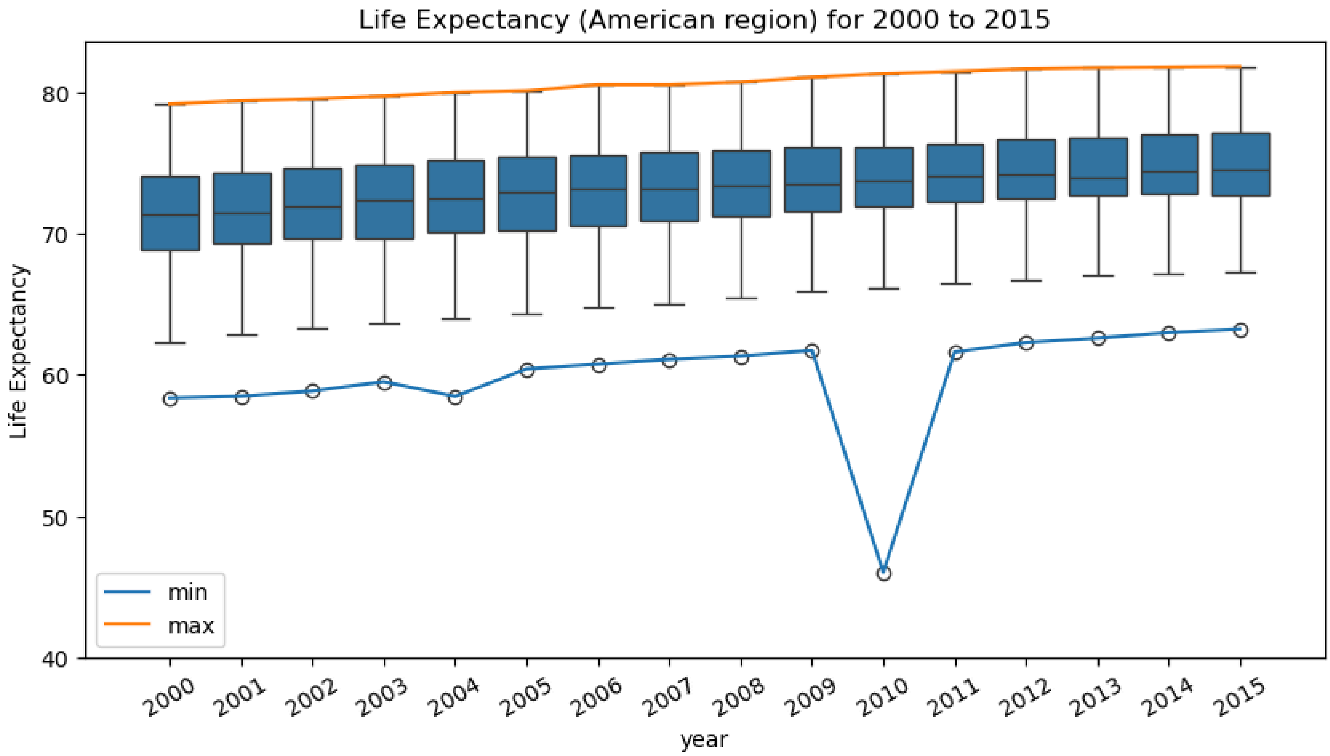
Spread and Outliers:

- IQR stayed moderate throughout the years, showing moderate variation between nations.
- A few persistent low-end outliers (e.g., Afghanistan, Yemen) highlight ongoing regional disparities.

Max and Min Trends:

- Max life expectancy increased from ≈ 81.5 to 83.5 years, while the min line steadily rose from ≈ 55 to 63 years, reflecting broad regional improvement.

American Region



General Trend:

- Median life expectancy remained relatively high and stable, increasing slightly from ≈ 71 to 75 years.

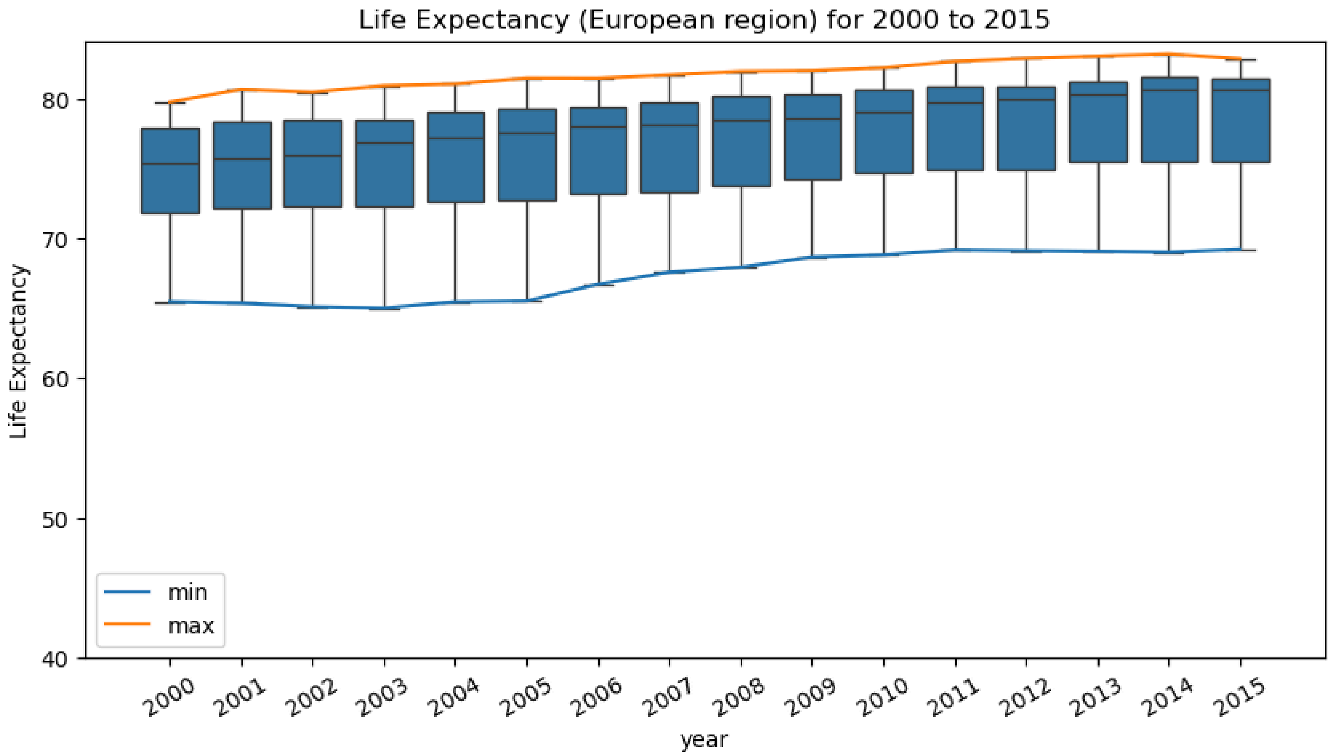
Spread and Outliers:

- The IQR was narrower than most regions, indicating relatively consistent health outcomes across North, Central, and South America.
- A few lower outliers emerged (e.g., Haiti or conflict zones).

Max and Min Trends:

- Max increased slightly to around 82 years.
- The min showed a notable dip in 2010, likely due to a specific country-level health crisis, then returned to an upward trend.

European Region



General Trend:

- Europe had the highest life expectancy median across all regions, increasing from ≈ 75 to 79 years between 2000 and 2015.

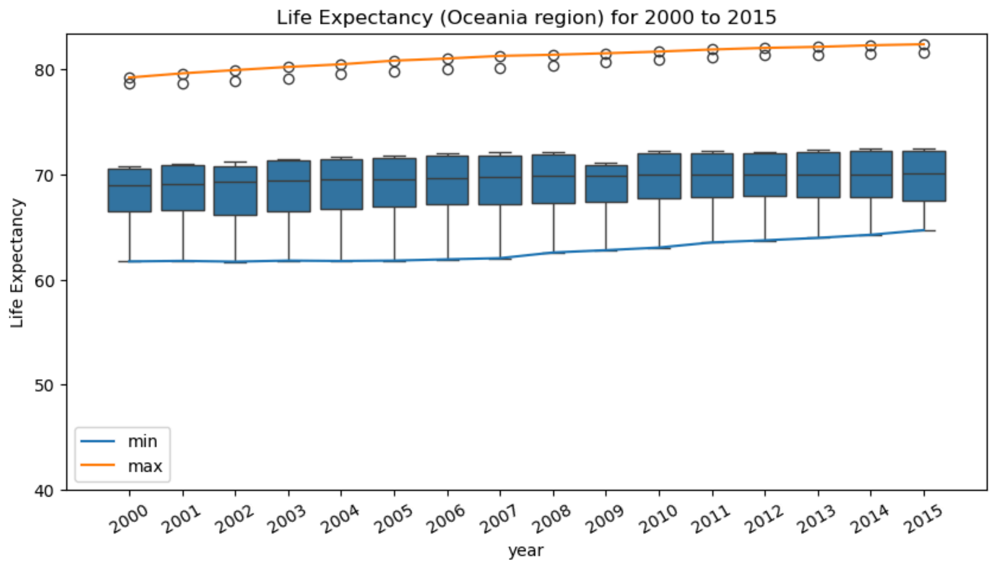
Spread and Outliers:

- The IQR narrowed over time, showing that Eastern and Western European countries are converging in health outcomes.
- Few outliers were present—indicating high regional consistency.

Max and Min Trends:

- Max values remained stable around 83.5 years.
- The minimum rose consistently from ≈ 65 to 69 years.

Oceania Region



General Trend:

- Life expectancy in Oceania increased modestly, with the median hovering between 70.5 and 73 years from 2000 to 2015.

Spread and Outliers:

- A narrow IQR throughout the period suggests strong health parity among the region's nations.
- Outliers were minimal and relatively stable.

Max and Min Trends:

- Max increased slightly from ≈ 78.5 to 82.5 years
- The min rose from ≈ 62 to 65 years, indicating steady improvement in lower-tier countries.

Summary

Overall Growth:

- Life expectancy improved consistently in all regions, with the global median increasing by roughly 4 years from 2000 to 2015.

Regional Comparisons:

- Europe and the Americas maintained the highest life expectancies.
- Africa started lowest but showed the strongest rate of improvement, reducing the global life expectancy gap.

Inequality and Outliers:

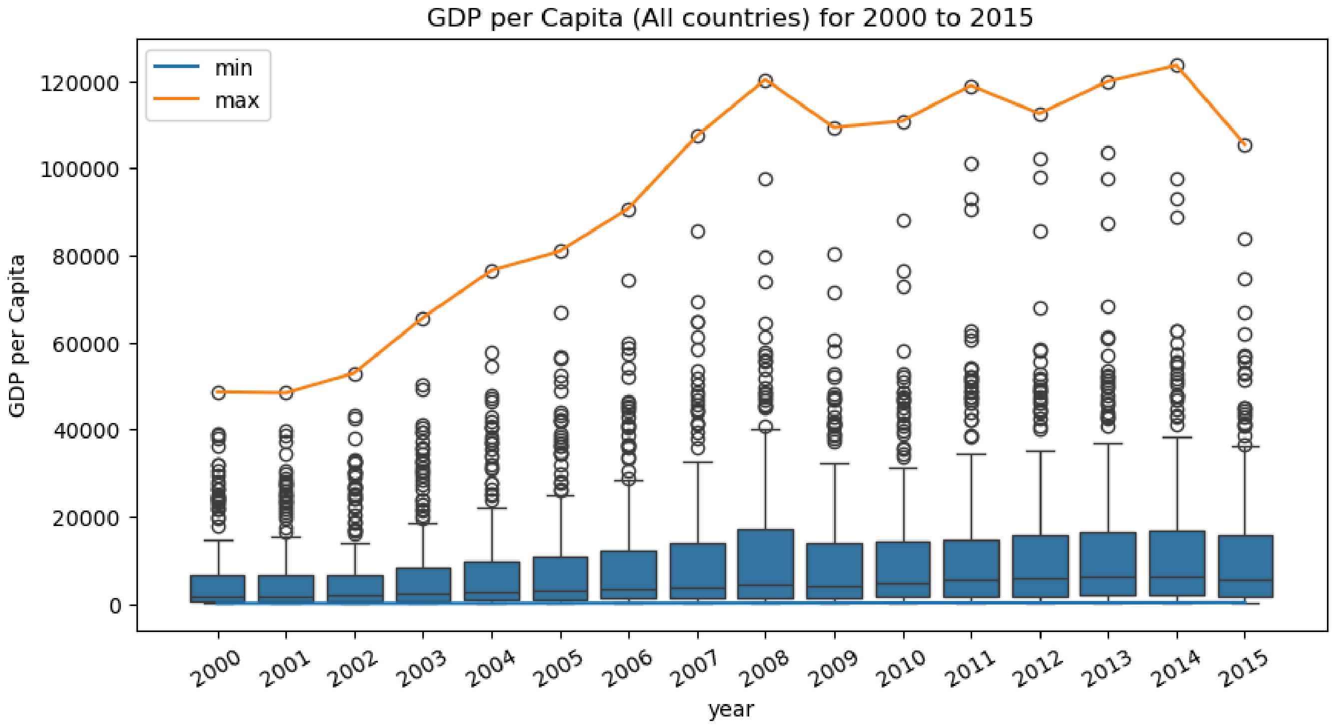
- The number of low-end outliers decreased over time, indicating fewer nations with extremely poor health outcomes.
- Some regions (e.g., Oceania, Europe) displayed tight clustering, while others (Africa, Asia) showed wider spreads.

Conclusion:

The boxplots highlight the effectiveness of global health initiatives over this 15-year period and confirm the positive correlation between income and life expectancy explored further in Section 4.6.

4.5.2 GDP per Capita

All Countries



General Trend:

- GDP per capita shows a clear upward trend from 2000 to 2015.
- The median increased steadily, although fluctuations are visible after 2008.

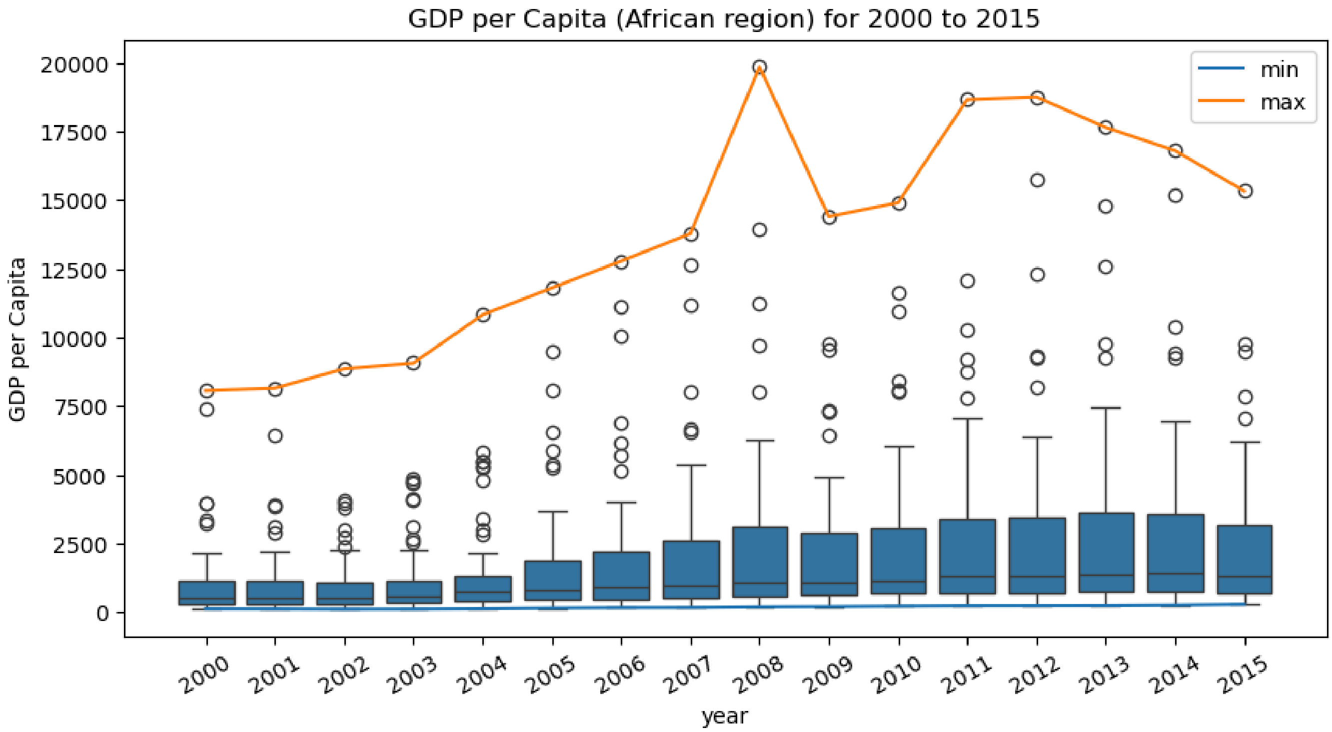
Spread and Outliers:

- A wide interquartile range (IQR) is observed across all years, reflecting global inequality in income.
- Many outliers appear above the upper whisker, corresponding to high-income countries like Luxembourg and Switzerland.

Max and Min Trends:

- Max values peaked around 2008 and again in 2014, exceeding $120,000.
- Min values remained consistently low, indicating persistent poverty in some nations.

African Region



General Trend:

- GDP per capita rose gradually from below $ 1,000 in 2000 to over $ 3,000 in 2015 (median).

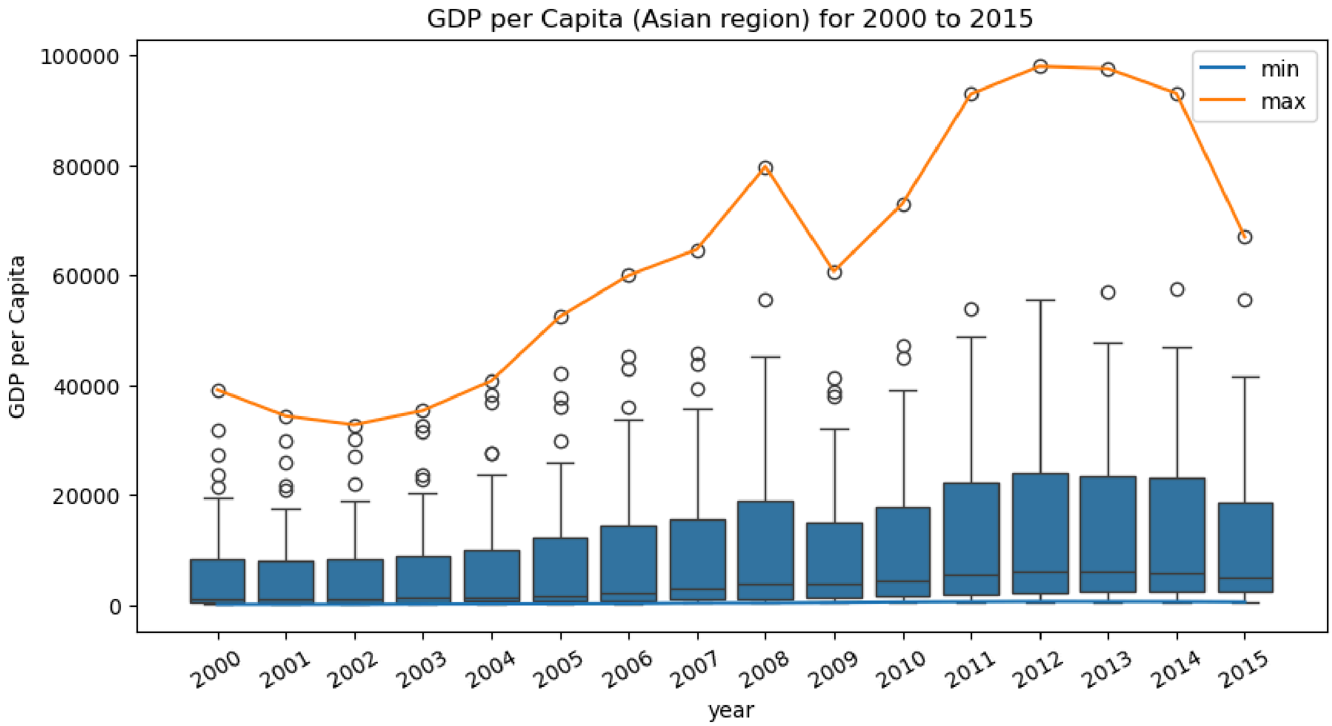
Spread and Outliers:

- The region has a relatively narrow IQR, with many low-end values, reflecting shared economic challenges.
- Outliers above the whiskers represent wealthier African nations like Botswana or Equatorial Guinea.

Max and Min Trends:

- Max values spiked notably in 2008, reaching nearly $ 20,000, likely due to commodity booms.
- The min line remained relatively flat, emphasizing persistent economic hardship for the poorest countries.

Asian Region



General Trend:

- Consistent and strong upward shift in median GDP per capita, especially between 2004 and 2012.

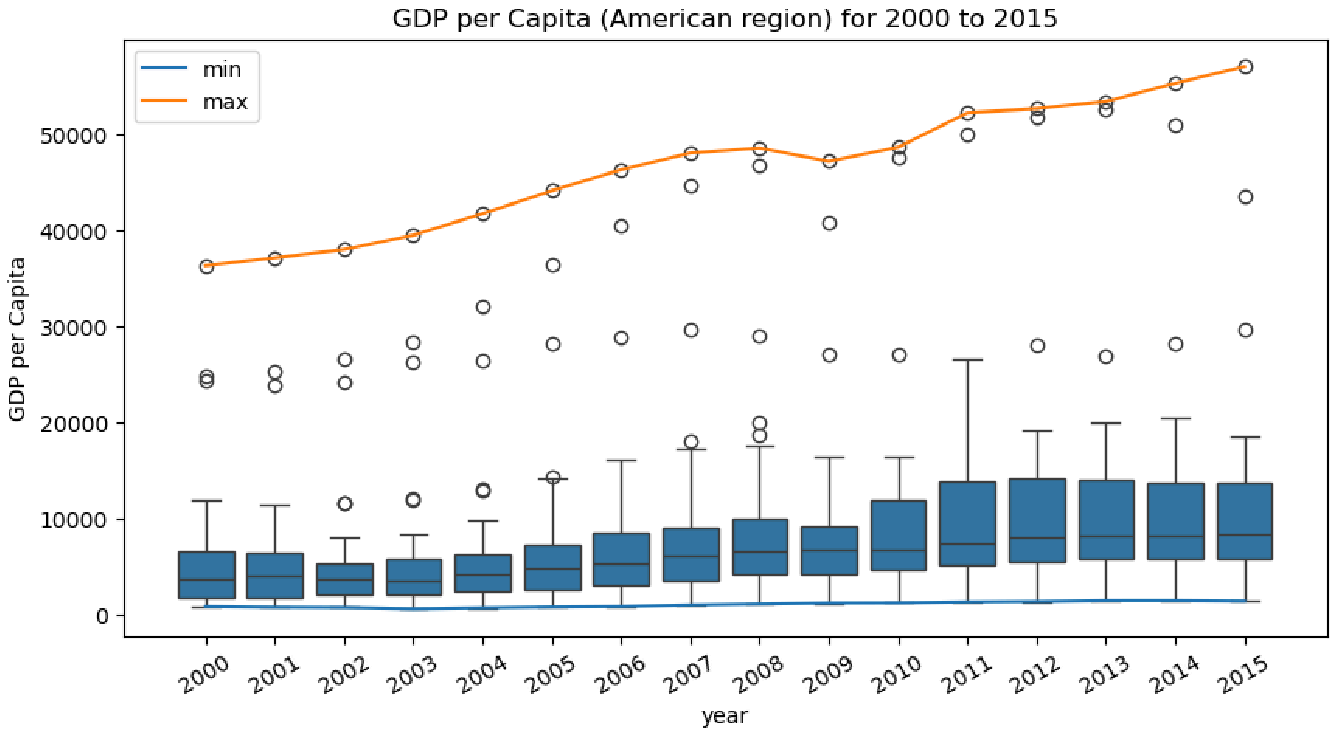
Spread and Outliers:

- A very wide IQR in recent years demonstrates significant economic disparity, with wealthy nations like Singapore and Japan far outpacing others.
- Outliers represent both high-performers and volatile economies.

Max and Min Trends:

- The max line soared past `$` 90,000, driven by rapid growth in East Asia.
- Min values improved marginally, but economic inequality remains substantial.

American Region



General Trend:

- A steady and consistent rise in GDP per capita across the region, with the median improving each year.

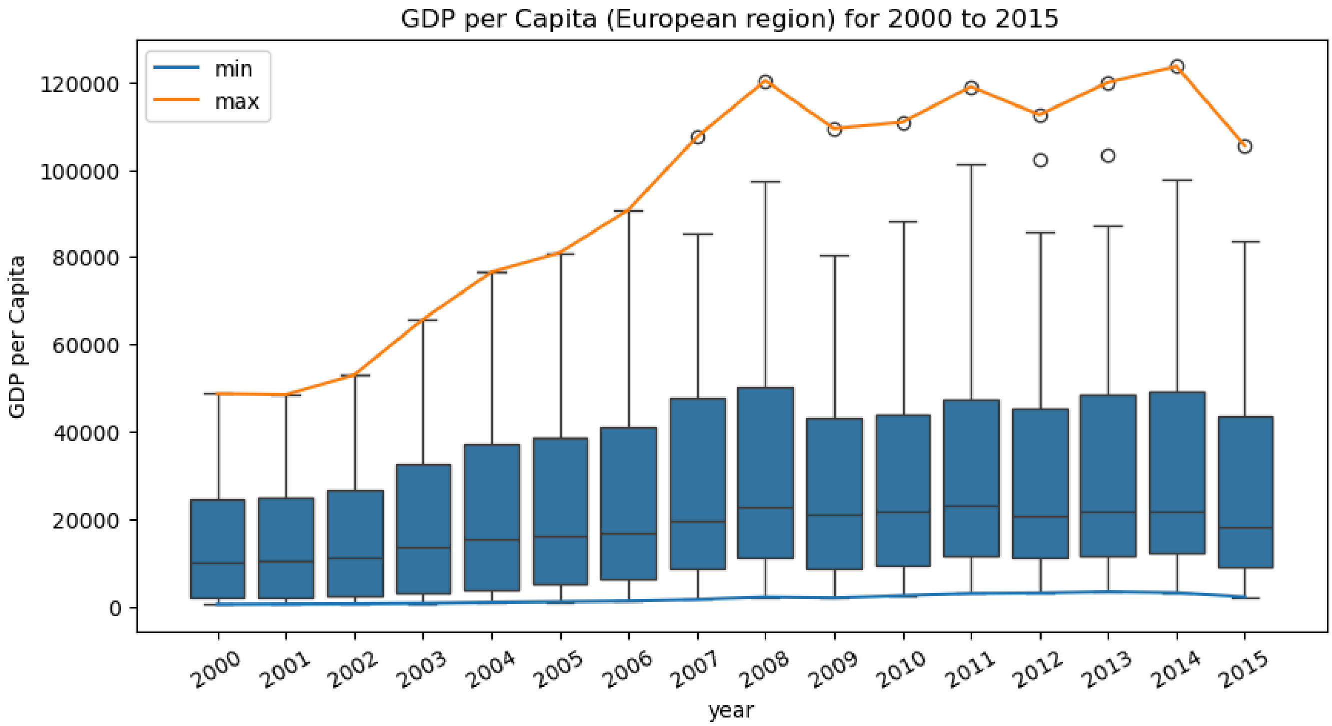
Spread and Outliers:

- A moderate IQR, less extreme than Asia and Europe, suggesting some income cohesion.
- A few outliers at the high end (e.g., U.S., Canada) and low end (e.g., Haiti).

Max and Min Trends:

- Max GDP per capita increased from $36,000 to over $55,000 by 2015.
- Min values gradually rose, indicating broad but unequal regional development.

European Region



General Trend:

- A sharp increase in GDP per capita between 2000 and 2008, followed by fluctuation due to economic crises.
- Median values climbed above $ 25,000, making it the highest among regions.

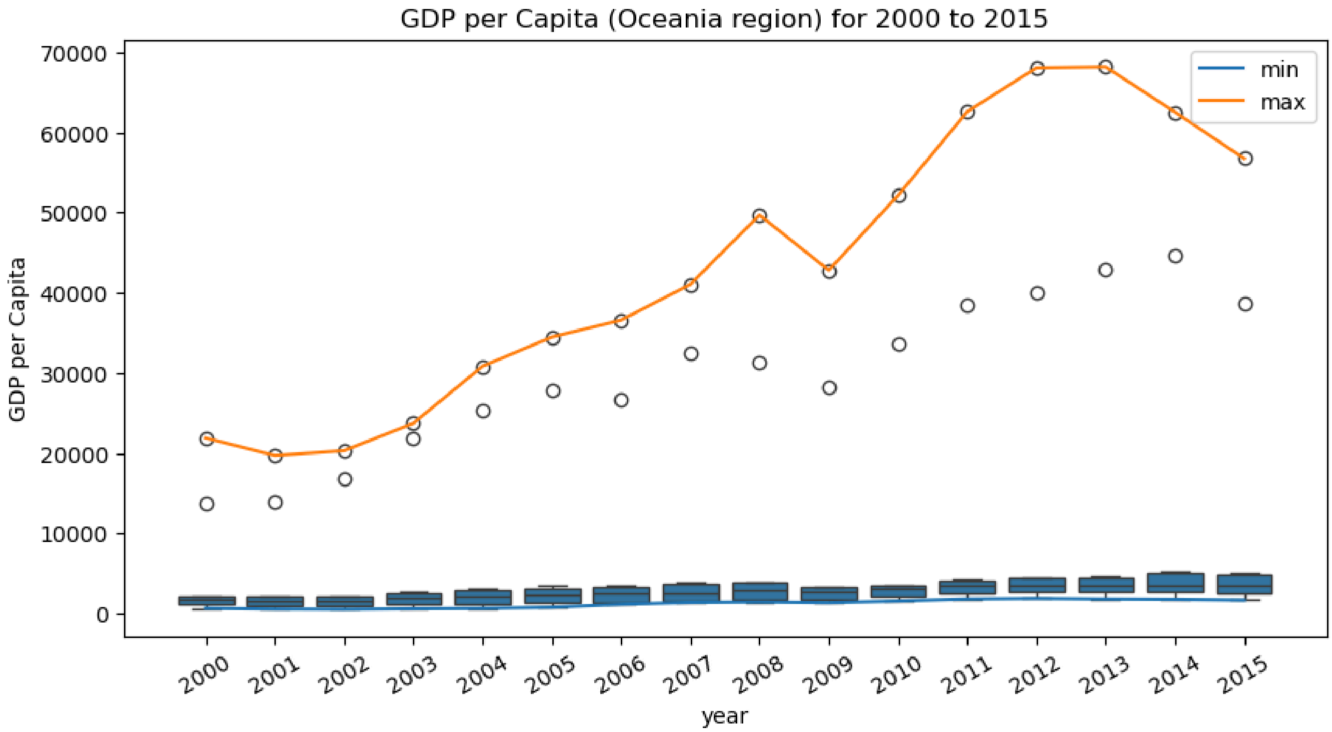
Spread and Outliers:

- A very large IQR with an increasing number of high-end outliers, reflecting Western vs. Eastern Europe divides.

Max and Min Trends:

- Max values peaked at $ 125,000, the highest globally, due to small, high-GDP countries.
- Min values improved only slightly, showing long-term disparity within the region.

Oceania Region



General Trend:

- GDP per capita increased steadily until 2013, then plateaued slightly.
- The median remained consistently above $ 3,000, among the highest.

Spread and Outliers:

- A very narrow IQR, indicating economic uniformity within the region (dominated by Australia and New Zealand).
- Sparse outliers reflect the small number of nations in the dataset.

Max and Min Trends:

- Max values nearly reached $ 70,000, while min values were much higher compared to other regions.

Summary

Overall Growth:

- All regions showed growth in GDP per capita from 2000 to 2015, although at different rates.

Regional Disparities:

- Europe and Oceania had the highest medians and largest max values.
- Africa had the lowest values but showed steady improvement.

Inequality:

- Asia and Europe displayed the widest spreads, indicating sharp contrasts between rich and poor countries within those regions.
- Oceania had the most uniform GDP per capita distribution.

Outliers:

- High-income countries consistently emerged as outliers in global and regional views.

4.6 Deep Dive Into GDP and Life Expectancy

4.6.1 Summary Statistics by Income Class

Low Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	52.3	70.76	Syrian Arab Republic	44.52	Malawi	50.7	6.13	48.13	54.88	6.75
2001	52.92	71.64	Syrian Arab Republic	45.39	Central African Republic	51.18	6.11	49.02	55.1	6.08
2002	53.53	71.94	Syrian Arab Republic	45.41	Central African Republic	51.87	5.96	50.39	55.6	5.2
2003	54.2	72.41	Syrian Arab Republic	45.76	Central African Republic	52.86	5.94	50.87	56.06	5.2
2004	54.99	72.47	Syrian Arab Republic	46.04	Central African Republic	53.48	5.82	51.9	56.49	4.6
2005	55.65	72.77	Syrian Arab Republic	46.43	Central African Republic	54.0	5.78	52.8	57.58	4.78
2006	56.36	73.35	Syrian Arab Republic	46.85	Central African Republic	54.86	5.73	53.65	58.69	5.04
2007	57.0	73.71	Syrian Arab Republic	47.43	Central African Republic	55.55	5.67	54.32	59.18	4.85
2008	57.6	73.55	Syrian Arab Republic	48.02	Central African Republic	56.28	5.52	55.13	59.91	4.78
2009	58.2	73.85	Syrian Arab Republic	48.65	Central African Republic	56.75	5.44	55.76	60.38	4.61
2010	58.72	73.88	Syrian Arab Republic	49.26	Central African Republic	57.19	5.28	56.38	60.75	4.37
2011	59.32	73.31	Syrian Arab Republic	49.95	Central African Republic	57.94	5.11	57.01	61.45	4.44
2012	59.57	67.47	Nepal	50.54	Chad	58.75	4.35	57.37	62.05	4.68
2013	59.94	67.96	Nepal	50.78	Chad	59.74	4.16	57.78	62.56	4.77
2014	60.24	68.08	Nepal	50.57	Central African Republic	60.14	4.15	58.2	62.96	4.76
2015	60.69	67.46	Nepal	51.59	Chad	60.65	3.82	58.46	63.82	5.36

- Steady upward trend: Mean life expectancy increased from 52.3 years in 2000 to 60.69 years in 2015 — a total gain of 8.39 years, which is a 16% improvement over the period.
- Median values closely followed the mean, indicating fairly symmetrical distributions within this group.
- Maximum life expectancy plateaued around 73.88 years in 2010, represented mostly by the Syrian Arab Republic until data gaps occurred from 2012 onwards.
- Minimum life expectancy was lowest in Malawi (2000) and Central African Republic and Chad in later years — values consistently below 50 until after 2008.
- Dispersion (Standard Deviation & IQR):
  - Standard deviation (std) declined from 6.13 in 2000 to 3.82 in 2015, signaling reduced variability in life expectancy across countries in this income group.
  - Interquartile range (IQR) shrank similarly (from 6.75 to 5.36), reinforcing the idea of convergence toward central values.

Insight:

Substantial improvements occurred in life expectancy among low-income countries, but disparities still persist. While the lower bound of life expectancy improved, high variance and some countries lagging far behind indicate ongoing inequality.

Lower-middle Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	61.77	72.46	Viet Nam	44.69	Zimbabwe	62.51	7.39	58.23	67.62	9.39
2001	62.05	72.65	Viet Nam	41.96	Zimbabwe	63.18	7.59	58.2	67.92	9.72
2002	62.42	72.8	Viet Nam	44.56	Zimbabwe	63.57	7.46	58.67	68.36	9.69
2003	62.77	72.98	Viet Nam	43.39	Zimbabwe	64.15	7.46	59.11	68.35	9.24
2004	62.93	73.14	Viet Nam	43.71	Lesotho	64.66	7.31	58.66	68.22	9.56
2005	63.41	73.27	Viet Nam	43.23	Lesotho	64.92	7.3	59.64	68.76	9.12
2006	63.84	73.32	Viet Nam	42.91	Lesotho	65.24	7.16	60.06	69.09	9.03
2007	64.19	73.44	Viet Nam	43.12	Lesotho	65.76	7.07	60.44	69.56	9.12
2008	64.42	73.59	Jordan	43.57	Lesotho	66.26	6.98	60.48	69.94	9.45
2009	64.88	73.8	Jordan	44.03	Lesotho	66.64	6.64	61.25	69.63	8.38
2010	65.11	74.0	Jordan	45.6	Lesotho	67.04	7.01	61.29	70.63	9.34
2011	65.79	74.26	Cabo Verde	46.69	Lesotho	67.4	6.26	61.74	70.81	9.07
2012	66.17	74.48	Cabo Verde	47.84	Lesotho	68.0	6.04	62.3	70.92	8.63
2013	66.52	75.06	Cabo Verde	49.0	Lesotho	68.52	5.86	62.62	71.08	8.45
2014	66.86	75.24	Cabo Verde	50.03	Lesotho	69.04	5.73	63.0	71.18	8.18
2015	67.14	75.01	Jordan	51.1	Lesotho	69.28	5.59	63.23	71.25	8.02

- Mean life expectancy grew from 61.77 in 2000 to 67.14 in 2015 — a 5.37-year increase.
- The median followed suit, with a steady incline each year, reaching 69.28 in 2015.
- Maximum values stayed consistent, around 75 years, with Viet Nam and later Jordan and Cabo Verde leading.
- Minimum values were low in early years due to Zimbabwe and Lesotho, with life expectancies dropping as low as 41.96. By 2015, the minimum had improved to 51.1 years.
- Standard deviation dropped from 7.39 to 5.59, and IQR declined from 9.39 to 8.02, signaling growing cohesion and progress among countries.

Insight:

Life expectancy improvements in this class were moderately paced but steady. Most countries moved forward consistently, though early-decade challenges (HIV/AIDS in southern Africa, for example) caused setbacks in a few.

Upper-middle Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	69.23	77.59	Costa Rica	51.01	Botswana	70.82	5.68	67.26	72.78	5.53
2001	69.44	77.54	Costa Rica	50.68	Botswana	71.09	5.87	67.36	73.21	5.85
2002	69.6	77.98	Costa Rica	50.63	Botswana	71.35	6.0	67.32	73.26	5.94
2003	69.76	78.07	Costa Rica	50.95	Botswana	71.54	6.12	67.21	73.69	6.48
2004	70.03	78.33	Costa Rica	51.25	Namibia	71.78	6.12	67.75	73.71	5.96
2005	70.3	78.5	Costa Rica	51.79	Namibia	72.15	6.09	67.88	74.07	6.19
2006	70.62	78.51	Costa Rica	52.66	Namibia	72.38	5.96	68.48	74.29	5.81
2007	70.96	78.46	Costa Rica	53.68	Namibia	72.6	5.78	68.96	74.61	5.65
2008	71.32	78.44	Costa Rica	54.65	Namibia	72.71	5.59	69.16	74.95	5.8
2009	71.69	78.67	Costa Rica	55.5	Namibia	73.13	5.38	69.63	75.12	5.5
2010	71.98	78.67	Costa Rica	56.02	Namibia	73.32	5.19	70.0	75.33	5.33
2011	72.25	79.35	Costa Rica	56.55	Namibia	73.9	5.08	70.0	75.42	5.42
2012	72.58	79.28	Costa Rica	57.64	Namibia	74.17	4.93	70.38	75.66	5.28
2013	72.85	79.4	Costa Rica	58.69	Namibia	74.24	4.79	70.68	76.18	5.5
2014	73.02	79.42	Maldives	59.6	Equatorial Guinea	74.47	4.64	71.2	75.95	4.75
2015	73.25	79.69	Maldives	60.1	Equatorial Guinea	74.51	4.51	71.51	76.24	4.73

- Mean life expectancy started at 69.23 and rose to 73.25 in 2015, indicating a steady annual growth.
- Costa Rica consistently had the highest life expectancy, surpassing 79 years by the end of the period.
- Minimum values began at 51.01 (Botswana, 2000) and rose steadily, exceeding 60 years by 2015, with Equatorial Guinea at the lower end.

- Standard deviation decreased slightly (from 5.68 to 4.51), and IQR dropped from 5.53 to 4.73, showing modest convergence.
- Distribution remained tighter than in lower income classes, suggesting greater baseline stability and health system reach.

Insight:

Upper-middle income countries demonstrated relatively high and consistent life expectancy values, with narrow variability. Improvements reflect stable economic growth and investments in public health infrastructure.

High Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	76.08	81.08	Japan	69.1	Trinidad and Tobago	76.59	2.85	74.09	78.24	4.15
2001	76.43	81.42	Japan	69.65	Trinidad and Tobago	76.84	2.85	74.4	78.63	4.24
2002	76.59	81.69	Japan	70.49	Trinidad and Tobago	77.07	2.85	74.61	78.71	4.1
2003	76.76	81.76	Japan	69.73	Trinidad and Tobago	77.22	2.88	74.6	78.94	4.34
2004	77.2	82.03	Japan	71.08	Trinidad and Tobago	77.67	2.84	75.18	79.37	4.19
2005	77.36	81.96	Japan	71.25	Lithuania	78.07	2.93	75.12	79.64	4.52
2006	77.6	82.32	Japan	70.87	Latvia	78.37	3.02	75.45	79.96	4.52
2007	77.79	82.51	Japan	70.9	Lithuania	78.56	3.04	75.53	80.17	4.64
2008	78.07	82.59	Japan	71.47	Trinidad and Tobago	78.77	2.96	75.72	80.49	4.76
2009	78.37	82.93	Japan	72.58	Bahamas	78.97	2.88	75.94	80.67	4.73
2010	78.65	82.84	Japan	72.72	Trinidad and Tobago	79.42	2.86	76.38	80.95	4.58
2011	78.99	82.7	Switzerland	72.56	Bahamas	80.0	2.88	76.74	81.18	4.44
2012	79.16	83.1	Japan	72.75	Bahamas	80.12	2.79	76.99	81.4	4.41
2013	79.37	83.33	Japan	73.02	Bahamas	80.4	2.84	77.19	81.74	4.56
2014	79.67	83.59	Japan	73.23	Seychelles	81.08	2.88	77.46	81.91	4.45
2015	79.66	83.79	Japan	73.1	Bahamas	80.78	2.79	77.54	81.96	4.43

- Mean life expectancy rose from 76.08 to 79.66 from 2000 to 2015 — a smaller numerical increase (3.58 years) but from an already high baseline.
- Japan led with the highest values across the entire period (reaching 83.79).
- Minimum life expectancy began at 69.1 and steadily rose to 73.1, with Trinidad and Tobago, Latvia, Seychelles, and Bahamas often at the lower end.
- Standard deviation remained low throughout (hovering around 2.85–3.0), and IQR remained consistently narrow (~4.1–4.7), indicating homogeneity among high-income nations.

Insight:

High-income countries showed limited variance in life expectancy and maintained a consistently high standard across the years. Gains were marginal compared to lower-income classes due to already near-maximal baseline longevity.

Overall Reflection

Magnitude of Progress:

- Greatest improvements in absolute terms occurred in the low income class.
- Upper and high income countries maintained a stable, high life expectancy throughout, with minimal variation.

Reduction in Dispersion:

All income classes showed declining standard deviations and IQRs, indicating convergence in life expectancy and reduction in disparities.

Income Class Gradient:

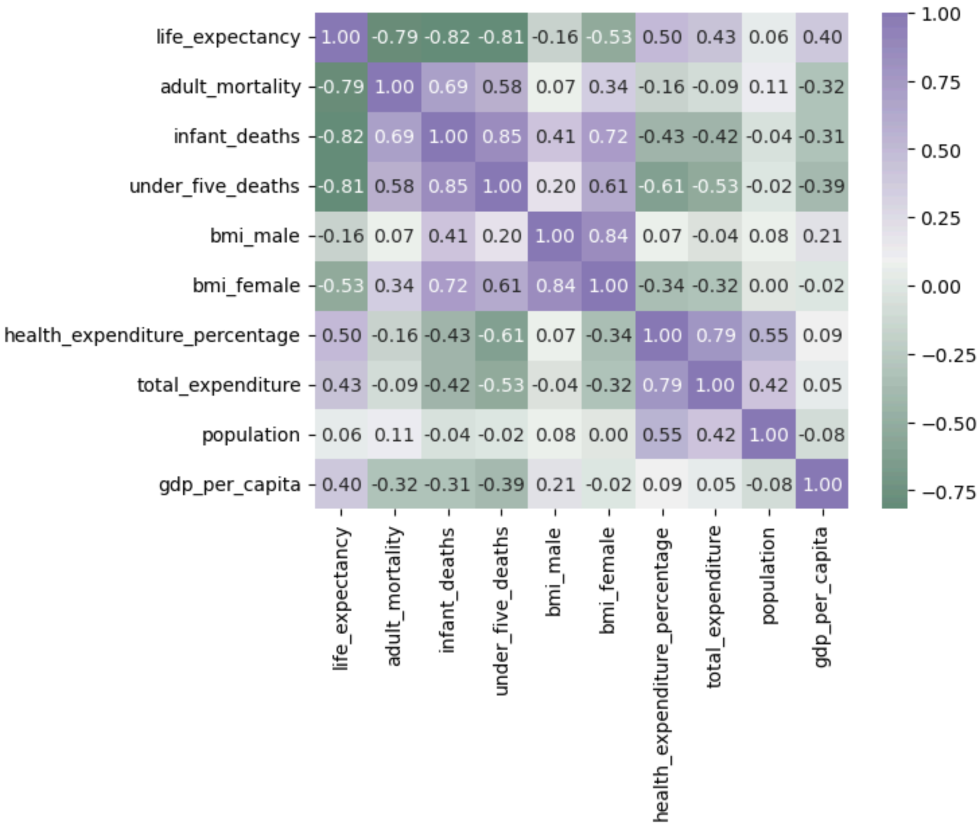
At every point in time, higher income classes had greater median life expectancy values, confirming a positive correlation between income class and longevity.

Equity Gains Over Time:

While disparities persist, lower-income countries have made strides in closing the gap, as evident from steeper gains in mean and median life expectancy.

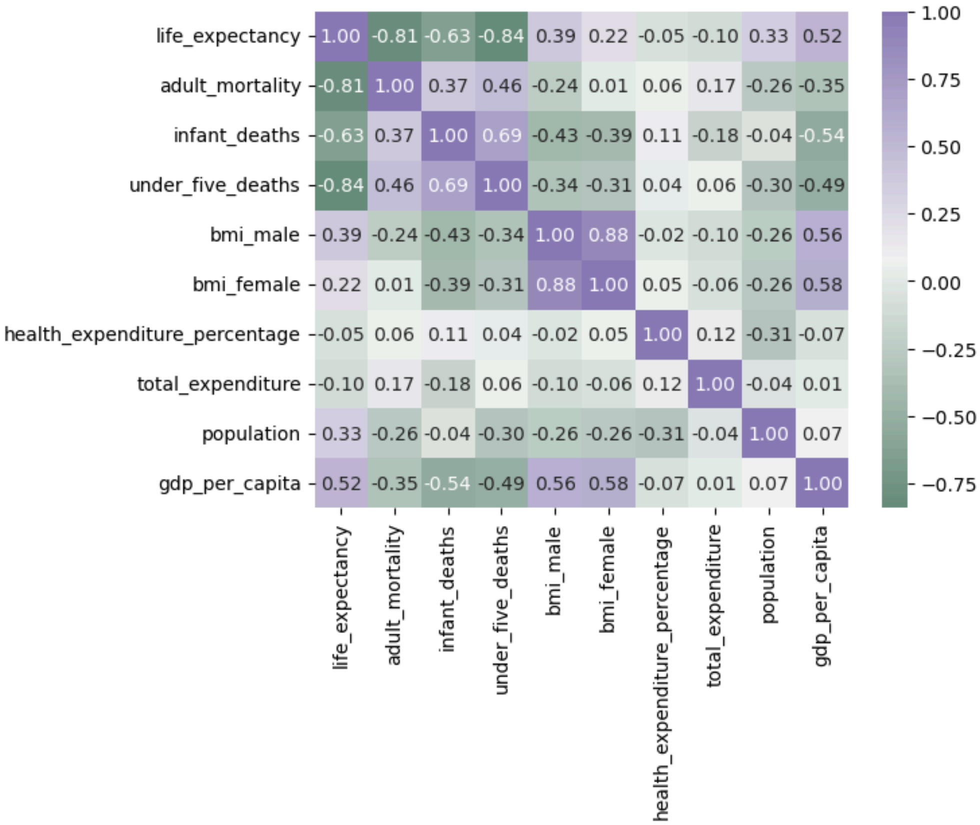
4.6.2 Correlation and Heatmap Analysis

Top 20 GDP Countries



- Life expectancy has a strong positive correlation with GDP per capita (0.84) and population (0.57), suggesting that wealthier, populous nations also achieve higher life spans.
- Mortality indicators such as adult mortality (-0.83) and under-five deaths (-0.81) remain strongly negatively correlated.
- The relationship with health expenditure percentage (0.37) and total expenditure (0.50) shows that investment in health systems contributes positively but not as strongly as economic power.

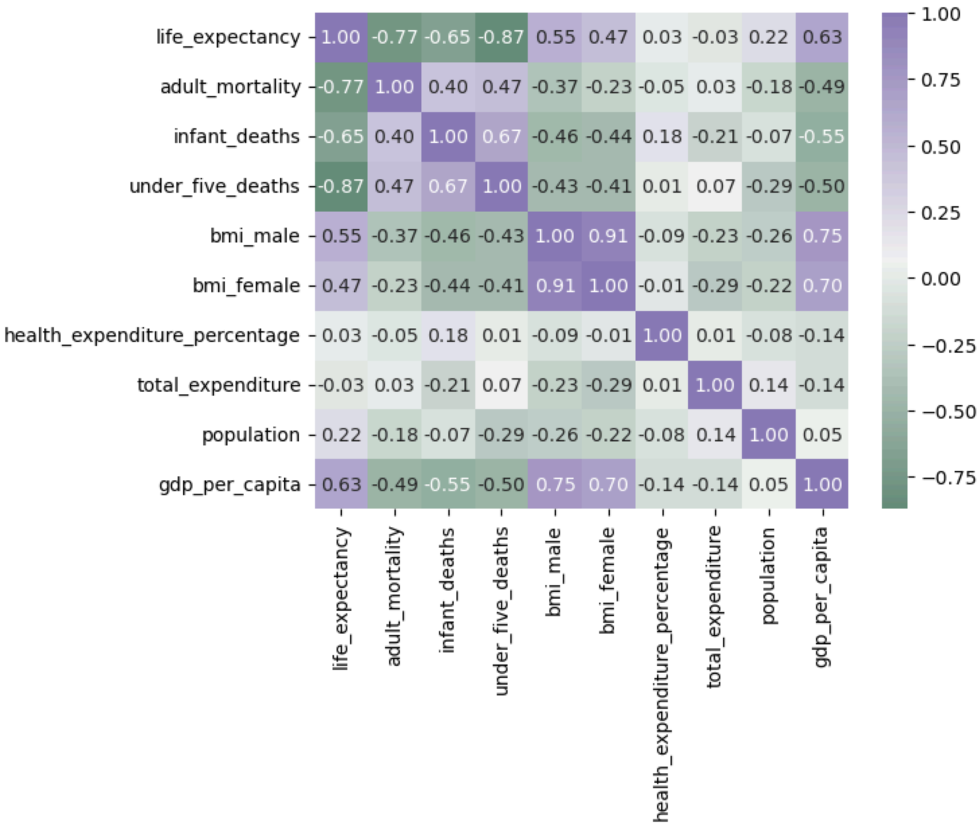
Bottom 20 GDP Countries



- Life expectancy still shows strong negative correlations with under-five deaths (-0.84) and adult mortality (-0.81).
- However, GDP per capita has a weaker correlation (0.52) with life expectancy, indicating that in low-GDP countries, wealth alone is not a decisive driver.
- Health expenditure percentage and total expenditure both show near-zero or negative correlation, implying limited effectiveness of healthcare investments possibly due to poor infrastructure or access issues.

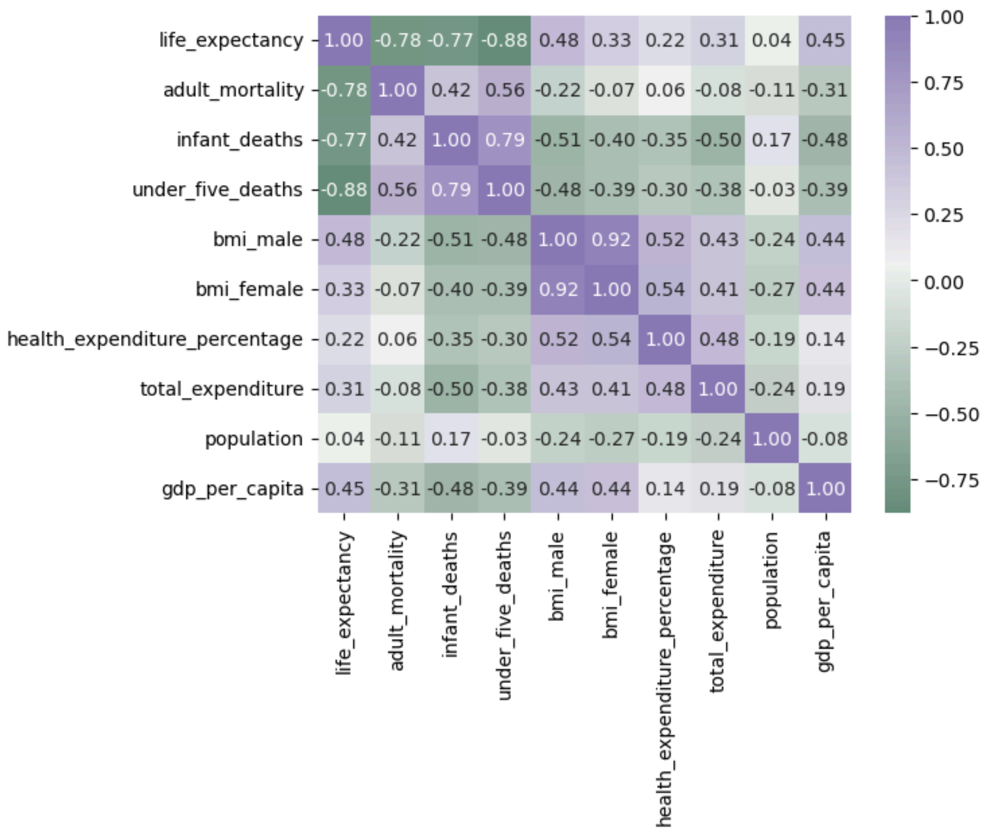
By Income Class

Low-Income Countries



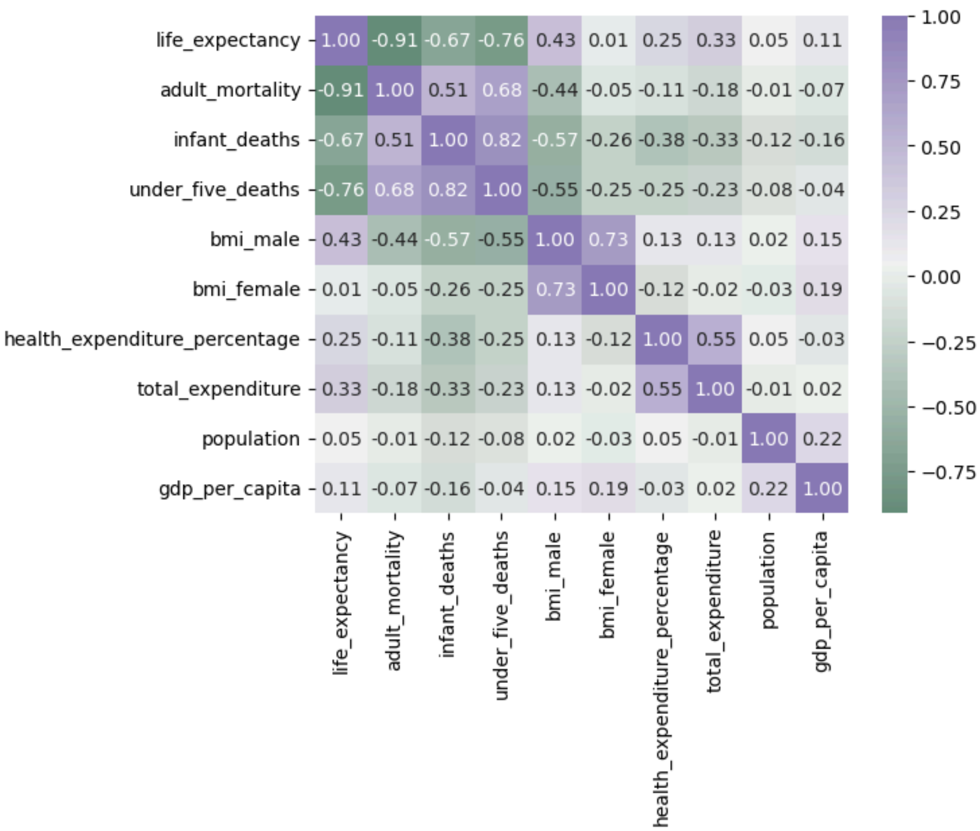
- Life expectancy is most strongly (negatively) correlated with under-five deaths (-0.87) and infant mortality (-0.65).
- Economic indicators such as GDP per capita (0.63) and total expenditure (-0.03) show minimal correlation, implying deep health and economic structural issues.
- Health spending variables offer little to no correlation, suggesting poor return on health investment or misallocation of resources.

Lower-Middle Income Countries



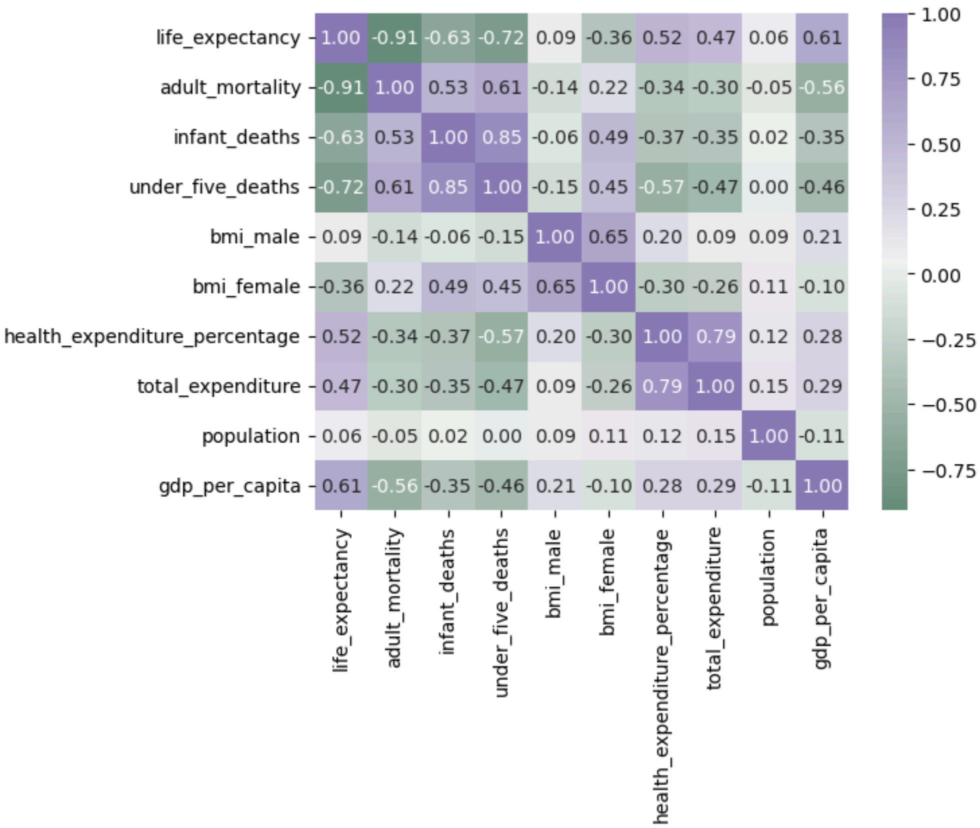
- Mortality indicators continue to dominate, with under-five deaths (-0.88) being the strongest negative correlate.
- GDP per capita (0.45) and total expenditure (0.31) show some moderate positive relationships.
- Health expenditure percentage (0.22) and BMI scores (0.48 for males) also reflect a slight influence.

Upper-Middle Income Countries



- Economic and nutritional factors begin to gain significance:
  - GDP per capita (0.71)
  - Health expenditure percentage (0.52)
  - Total expenditure (0.47)
- Mortality remains influential (adult mortality: -0.91), but the improved correlation with health system investment indicates that upper-middle income countries start to see the payoff of GDP and health spending on life expectancy.

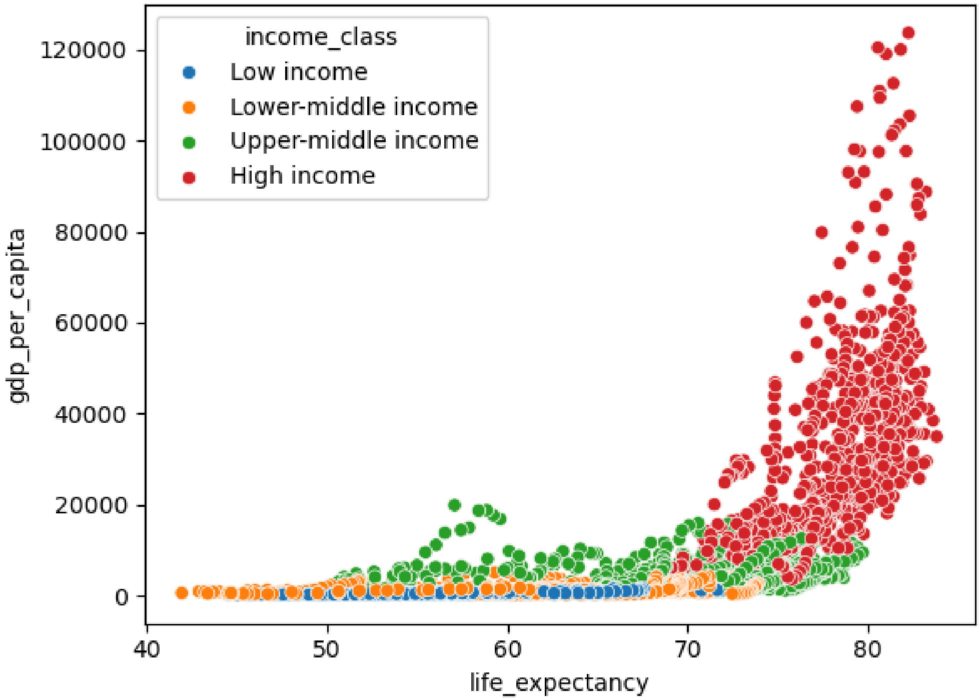
High-Income Countries



- Life expectancy and GDP per capita maintain a strong positive correlation (0.61).
- Health expenditure percentage (0.52) and total expenditure (0.47) also remain significantly linked.
- Mortality indicators like adult mortality (-0.91) and under-five deaths (-0.72) are still relevant, but the system-wide wealth and spending are more influential compared to lower-income counterparts.

4.6.3 Life Expectancy vs GDP per Capita by Income Class

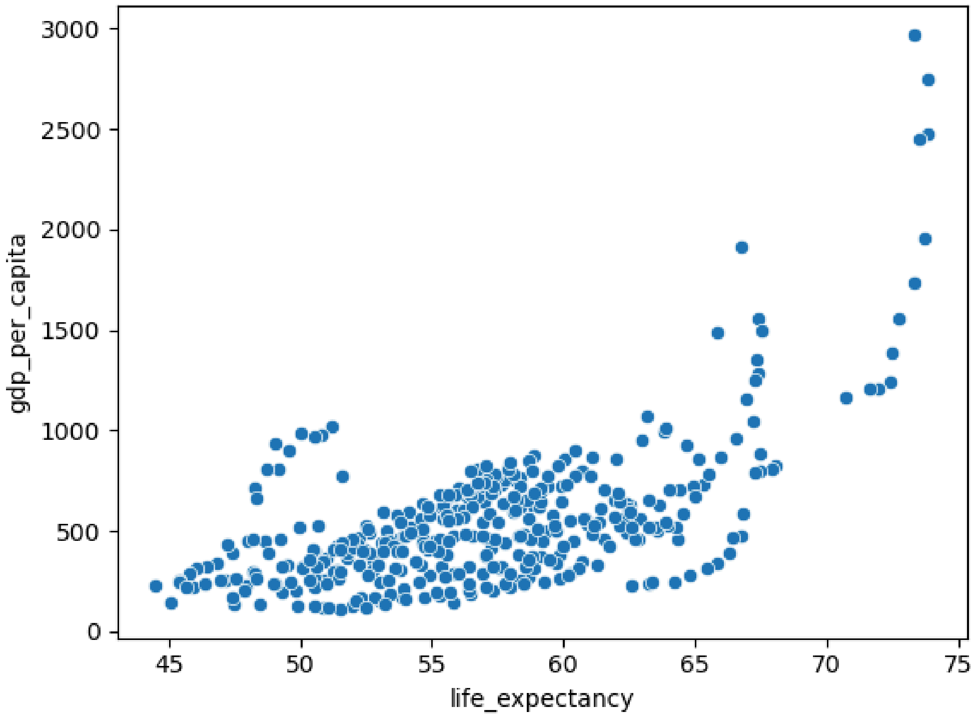
All Countries Grouped by Income Class



This visualization captures the stark stratification between income levels:

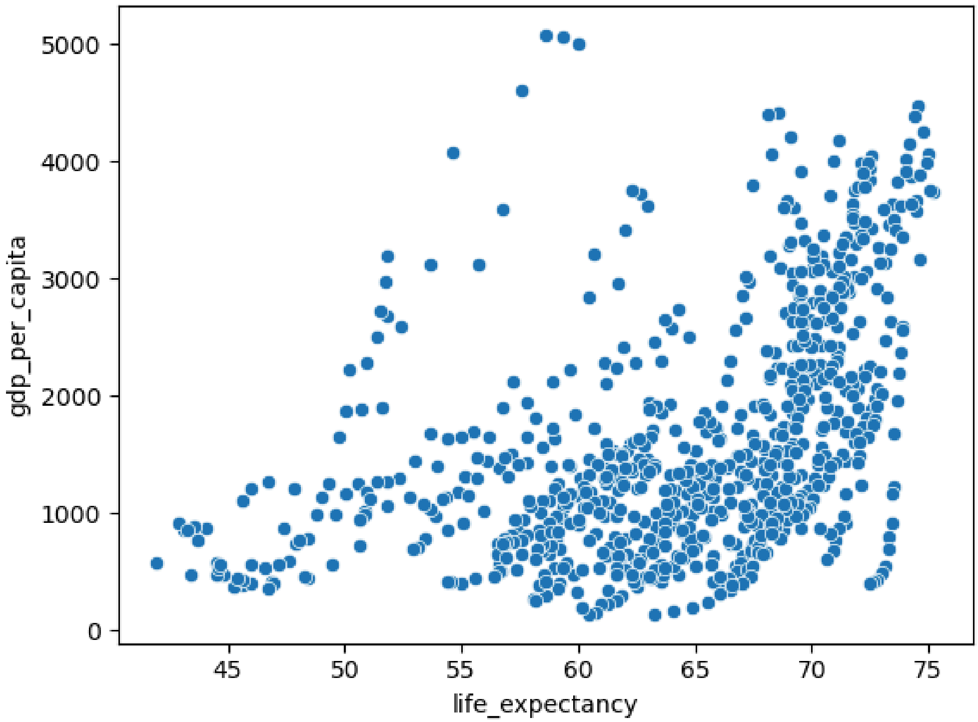
- Low-income and lower-middle-income countries are concentrated in the bottom-left quadrant (low GDP, low life expectancy).
- Upper-middle-income countries exhibit a wider spread.
- High-income countries dominate the top-right quadrant (high GDP, high life expectancy), underscoring the clear income gradient.

Low Income Countries



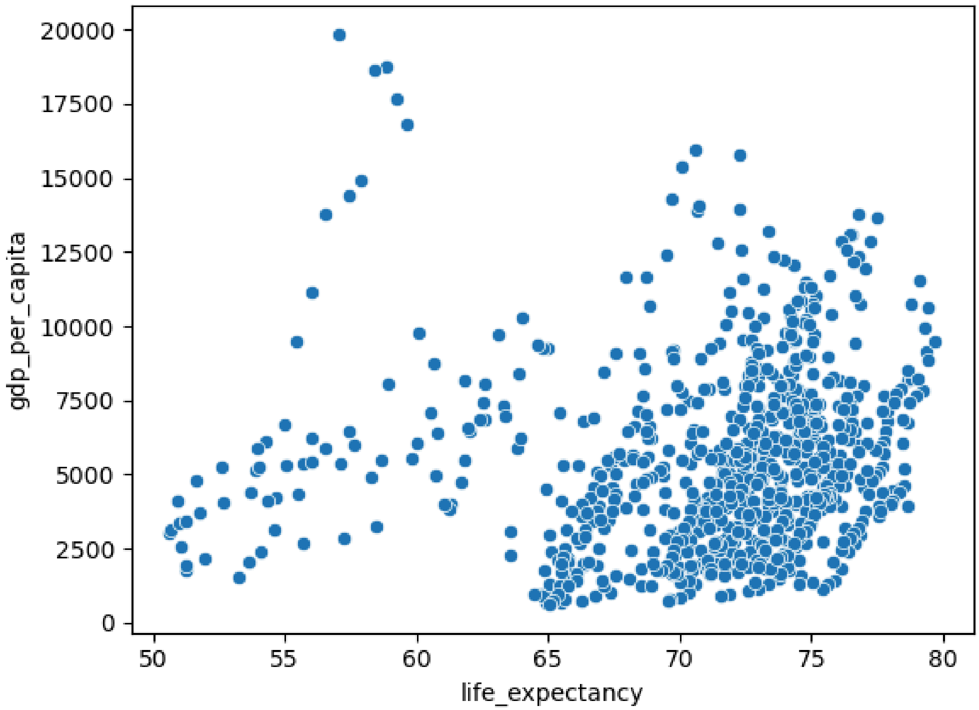
- There is a positive but modest correlation between GDP and life expectancy.
- The spread is tight, with GDP per capita rarely exceeding $3,000 and life expectancy typically between 50 and 65 years.
- This indicates limited variation in outcomes, possibly constrained by structural poverty.

Lower-Middle Income Countries



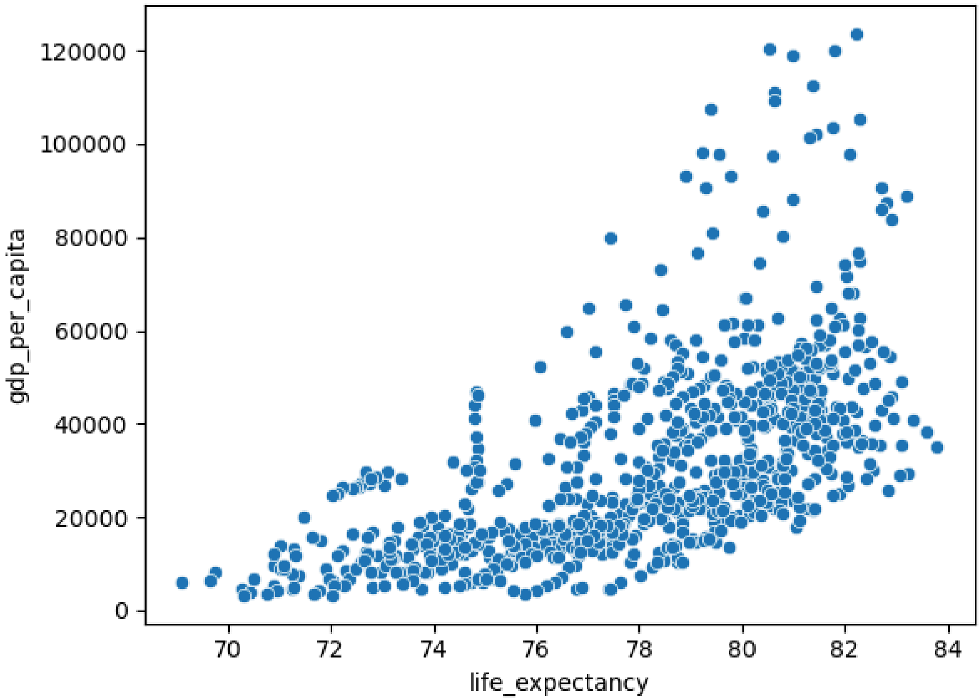
- A more noticeable upward trend is seen.
- Countries with GDP per capita nearing $5,000 tend to have life expectancies above 65 years.
- The relationship remains scattered, suggesting economic growth alone may not be sufficient without targeted health interventions.

Upper-Middle Income Countries



- A stronger upward trend becomes evident.
- Life expectancy in this group often surpasses 70 years, particularly for nations with GDP per capita above $10,000.
- Variability reduces, indicating more consistent health outcomes among economically comparable countries.

High Income Countries



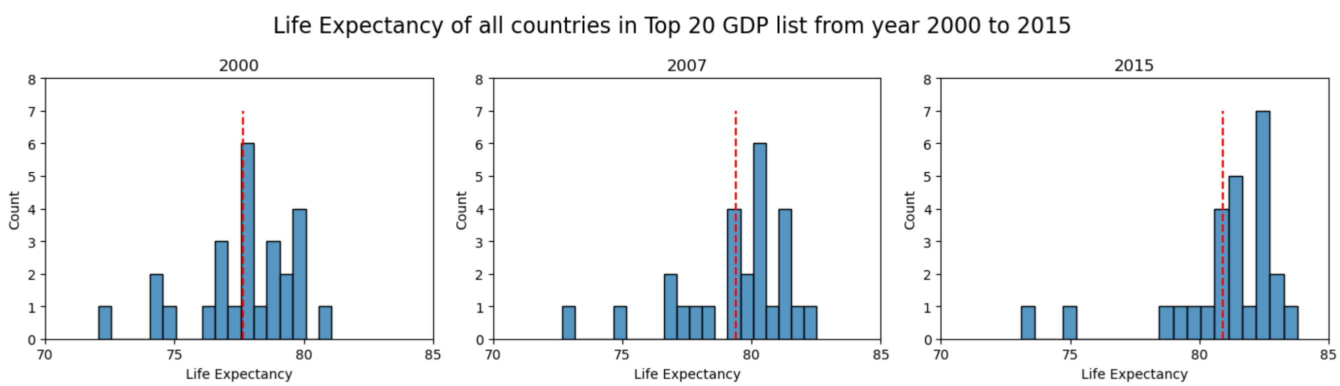
- The plot shows a tightly packed curve, demonstrating that most high-income countries have both high life expectancy (above 75) and GDP per capita well above $20,000.
- While GDP continues to rise, gains in life expectancy appear to plateau — supporting the hypothesis of diminishing returns on economic inputs.

4.6.4 Data Distribution Analysis (Life Expectancy vs GDP)

Life Expectancy vs GDP (Top vs Bottom GDP Countries)

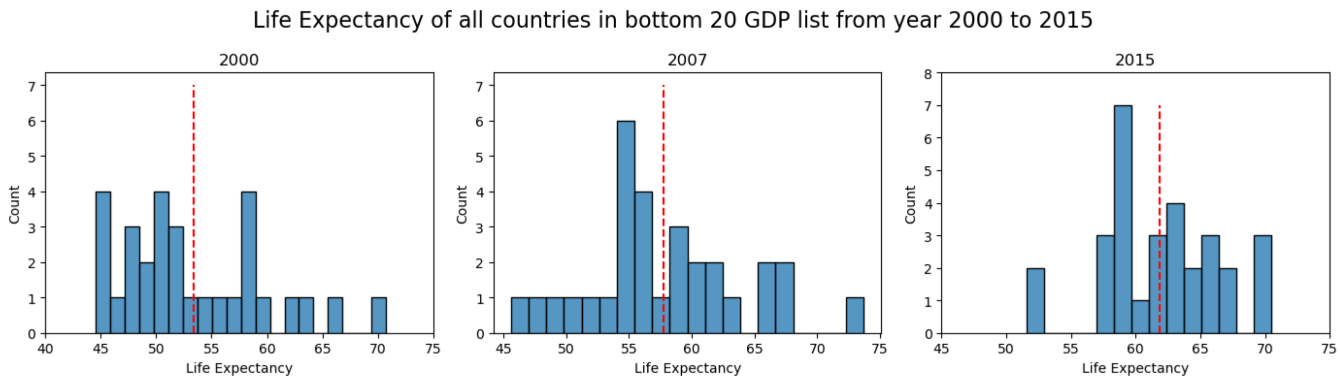
To deepen the understanding of the relationship between GDP and life expectancy, distribution diagrams were created for countries within the top 20 and bottom 20 GDP per capita rankings from 2000 to 2015. The results further underscore the association between economic strength and population health outcomes.

Top 20 GDP Countries



- Across 2000, 2007, and 2015, the top 20 countries maintain consistently high life expectancy, with most nations clustered tightly between 78 and 83 years.
- The mean life expectancy increases slightly over time (from ~78 in 2000 to ~82 in 2015), showing that already developed nations continue to make incremental health gains.
- The shape of the distribution becomes narrower and more concentrated, indicating greater consistency in high life expectancy outcomes among wealthy nations.

Bottom 20 GDP Countries

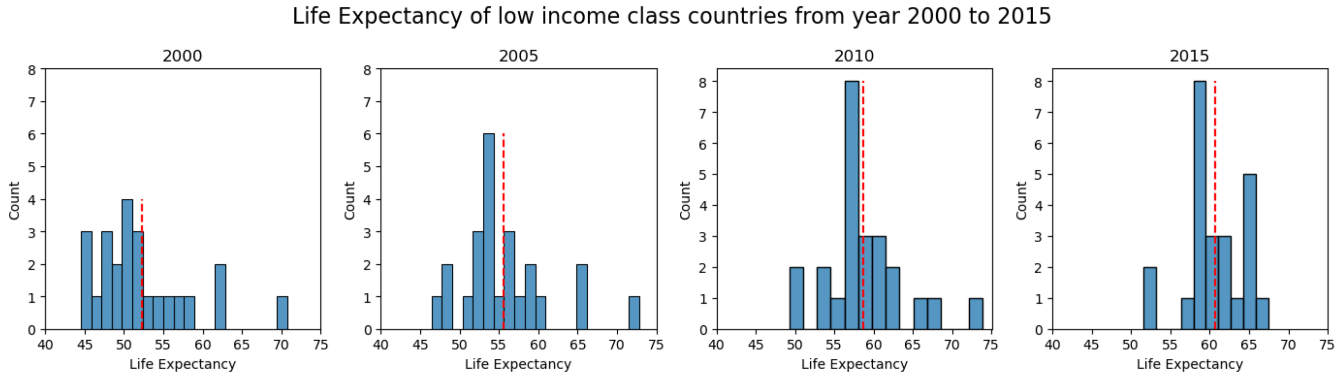


- In contrast, countries in the bottom 20 GDP bracket begin with significantly lower life expectancy in 2000, centered around 50–55 years.
- Over time, there is a clear rightward shift, with the mean increasing from ~53 in 2000 to ~62 in 2015 — a noteworthy improvement.
- However, the distribution remains broad and uneven, reflecting ongoing disparities and slower progress relative to high-income counterparts.
- These findings suggest that while lower-income countries have made substantial health strides, a considerable development gap remains in both economic and health terms.

By Income Class

To further explore the relationship between economic status and health outcomes, we analyze the distribution of life expectancy across four World Bank income classes — Low Income, Lower-middle Income, Upper-middle Income, and High Income — over the period 2000 to 2015. This breakdown provides clearer insights into how life expectancy varies among countries with differing levels of national wealth and development.

Low Income Countries



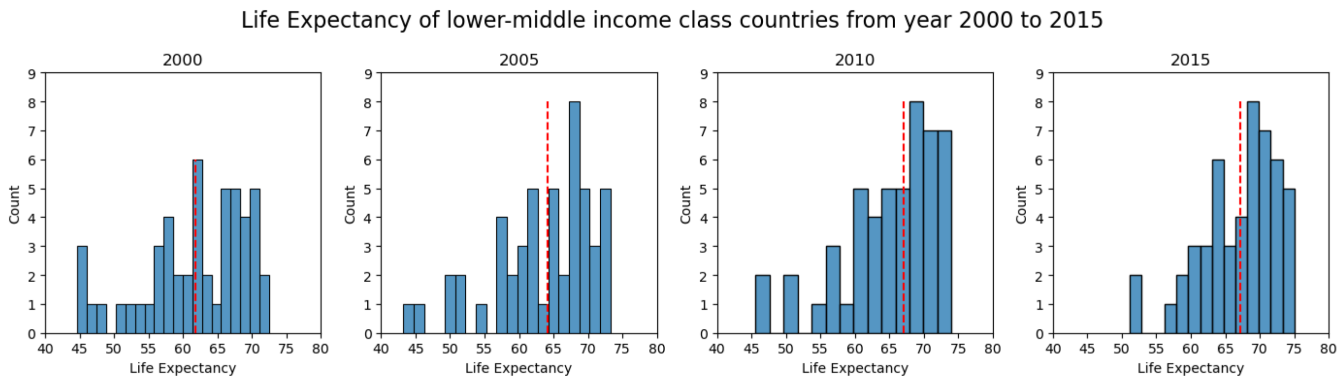
In 2000, life expectancy among low income countries was low and widely dispersed, with many countries falling below 55 years. The distribution showed a significant left skew, indicating a concentration of countries with very low life expectancies.

Over time, the distribution steadily shifts rightward:

- By 2005, the central tendency increases slightly, though still centered around the mid-50s to low 60s.
- In 2010 and 2015, a notable narrowing in distribution is observed, with more countries clustering around 60–65 years, suggesting modest but consistent improvement in life expectancy and reduction in disparity among these countries.

Despite this progress, low income countries remain substantially below global and high-income averages, underscoring continued gaps in healthcare access, infrastructure, and socio-economic development.

Lower-middle Income Countries

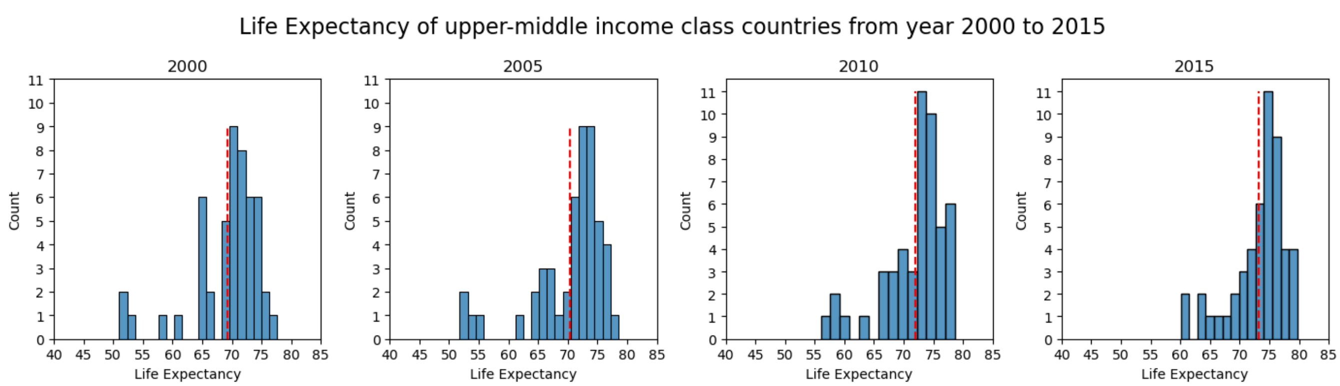


The life expectancy distribution in lower-middle income countries shows clear improvement over time:

- In 2000, the range was wider, with many countries still experiencing life expectancies between 55–65 years, and some falling lower.
- By 2005, a noticeable right shift occurs, and by 2010–2015, the majority of countries cluster tightly around 65–70 years.

This progression reflects the benefits of expanding public health initiatives, education, and infrastructure investment in countries transitioning toward more stable economies. The distribution also becomes more symmetric over time, indicating growing consistency in life expectancy outcomes across countries in this income group.

Upper-middle Income Countries

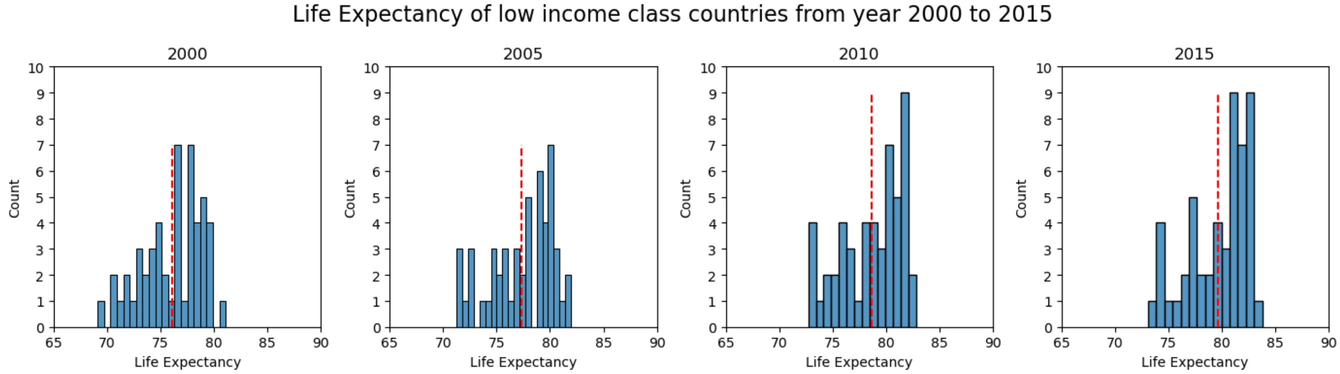


Countries in this category exhibit relatively high life expectancy from the start, with most countries already above 65 years in 2000.

- The 2000 distribution is relatively compact, centered around 70–75 years, and becomes even more concentrated by 2015.
- By 2015, life expectancy in this group converges strongly around 75 years, with most countries achieving sustained improvements in public health, nutrition, and healthcare access.

The narrow spread and high central tendency suggest equitable life expectancy outcomes and lower intra-group variability compared to lower income classes.

High Income Countries



High income countries show consistently strong life expectancy outcomes throughout the analysis period:

- In 2000, most countries in this group already enjoyed life expectancies above 75 years, with distributions showing minimal skew.
- By 2015, life expectancy increases further, with values tightly concentrated between 80–84 years.

This group exhibits the smallest variability and highest average life expectancy of all classes, reflecting the compounded effects of long-term investments in healthcare, education, sanitation, and economic stability. The high degree of clustering also indicates a near-universal achievement of longevity across these countries.

Key Insight: There is a clear, positive relationship between income class and life expectancy. As income class rises:

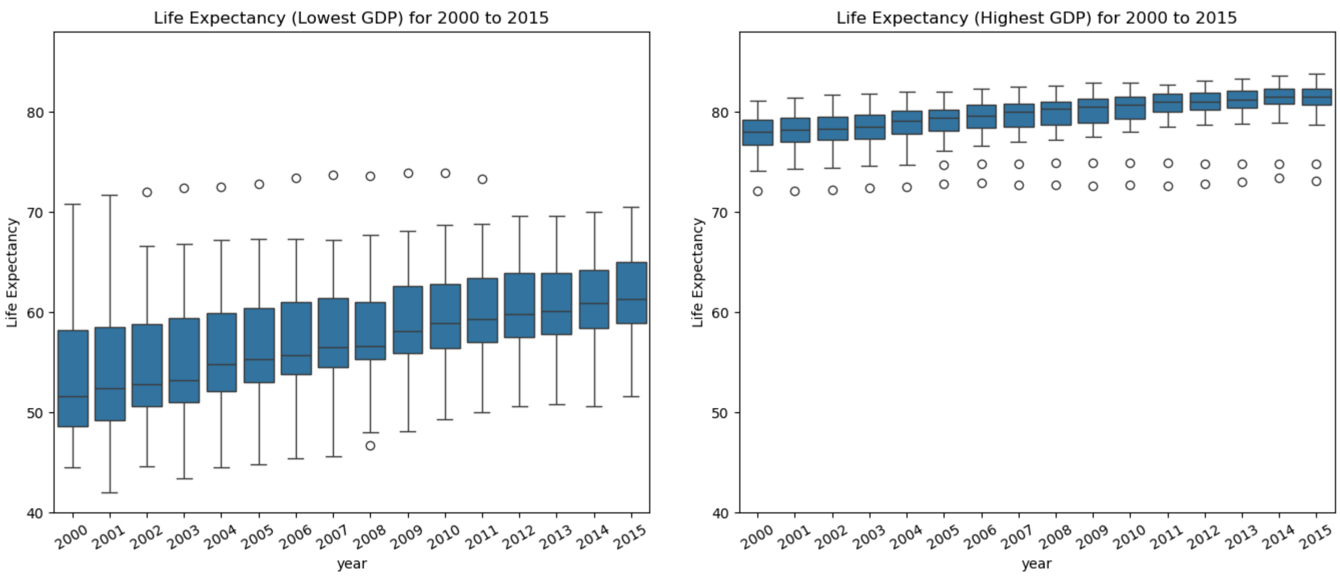
- Life expectancy increases,
- Distributions shift rightward and become more concentrated,
- Within-group variability decreases.

This supports the hypothesis that economic prosperity plays a crucial role in improving and equalizing life expectancy outcomes, making income classification a meaningful lens through which to evaluate global health inequality.

4.6.5 Boxplot Analysis (Life Expectancy vs GDP)

To complement earlier correlation, distribution, and trend analyses, a set of boxplots was produced to further explore the relationship between GDP per capita and life expectancy—focusing on GDP groupings and income classifications over time. These visualizations provide a clear view of central tendencies, dispersion, and variability within GDP categories and across income levels. The following insights were observed:

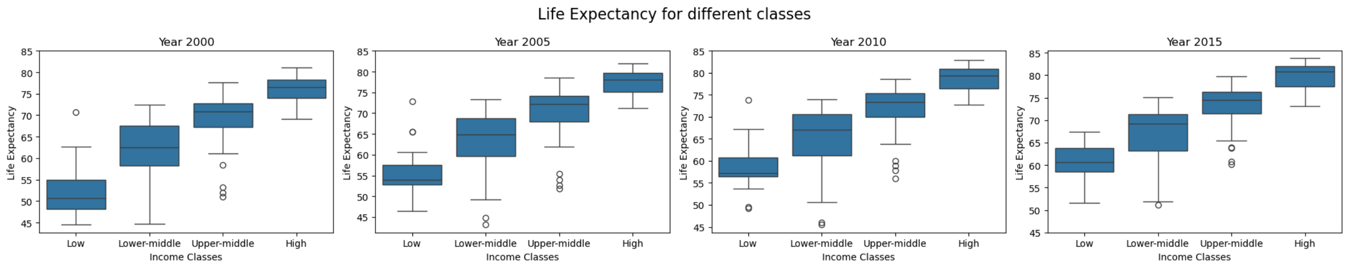
Top and Bottom GDP Countries



- A stark contrast is evident between the two groups across all years from 2000 to 2015:
  - Top 20 GDP countries consistently exhibit higher life expectancy medians, tighter interquartile ranges (IQR), and fewer outliers, indicating stable and equitable health conditions.
  - Bottom 20 GDP countries display wider IQRs and lower medians, often coupled with numerous outliers, signaling more variability and poorer average health outcomes.
- A steady upward shift in medians for the bottom group suggests gradual improvement in life expectancy, yet the gap with the top group remains persistent over time.
- These findings visually affirm the hypothesis: greater economic output is associated with longer and more consistent life expectancy.

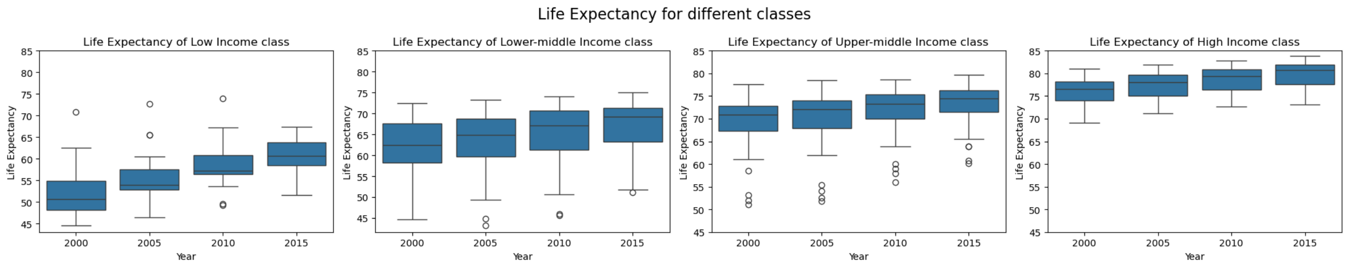
By Income Class

Boxplots by Year – All Income Classes



- This set of visualizations compares life expectancy by income class for four benchmark years (2000, 2005, 2010, 2015).
- Key trends observed:
  - Each higher income class consistently outperforms the lower ones in terms of life expectancy, creating a clear visual gradient.
  - The interquartile range tightens as income increases, reflecting greater healthcare equity and consistency in high-income nations.
  - From 2000 to 2015, the lower classes (low and lower-middle income) show notable upward shifts in their medians, indicating progress despite economic constraints.
  - High-income countries already near the upper bounds of life expectancy show modest but steady gains, confirming plateauing in developed regions.

Boxplots by Income Class – Yearly Trends within Each Class



- These longitudinal boxplots provide a breakdown of how life expectancy evolved within each income class from 2000 to 2015:
  - Low-income class shows the most pronounced improvement over time, with median life expectancy rising and lower whiskers gradually lifting—though dispersion remains wide.
  - Lower-middle income countries display a similar upward trajectory, but with slightly less variability.
  - Upper-middle income nations exhibit relatively strong and steady gains, showing increasing health stability over time.
  - High-income countries show less dramatic changes, affirming that most gains had already been realized by 2000, though continued small improvements are still visible.
- These visualizations confirm that economic stratification correlates closely with population health outcomes, and that life expectancy improves in parallel with income class mobility.

Summary of Boxplot Insights

- The collective boxplot analysis reinforces the foundational assumption of the project: higher GDP and income levels correspond to longer and more equitable life expectancy.
- While high-income and high-GDP countries enjoy more consistent health outcomes, the most compelling visual narrative emerges from the gains made by lower-tier economies, which show gradual but meaningful progress.
- These trends set the stage for deeper synthesis in the final report section, where all analytical threads will converge around the evolving relationship between national wealth and public health.

5. Synthesis of Analytical Insight

This section consolidates the analytical journey undertaken throughout the data wrangling and exploratory data analysis phases, synthesizing key findings that laid the foundation for the final visualization. Drawing on statistical summaries, correlation analysis, trend exploration, and visual comparisons across various groupings, we distill the evidence that supports our overarching hypothesis:

Countries with higher GDP per capita generally exhibit higher life expectancy.

5.1 Recap of Key Trends from EDA

The exploratory analysis revealed a robust and consistent positive association between GDP per capita and life expectancy. This pattern emerged through multiple visual and statistical techniques:

- Scatterplot analyses showed clear upward trends between GDP per capita and life expectancy, with stronger clustering in higher-income regions.
- Heatmaps of correlation matrices, both globally and by income class, consistently highlighted moderate to strong positive correlations (typically ranging from 0.5 to 0.7) between economic indicators (GDP per capita, health expenditure) and life expectancy.
- Line graphs over time demonstrated rising life expectancy trends across all income classes, with the steepest increases among lower-income groups—indicating some degree of global health convergence.
- Distribution analyses of GDP and life expectancy illustrated the stark disparities between income classes, as well as the changing nature of these distributions from 2000 to 2015.
- Boxplots across regions and income classes further highlighted that life expectancy medians, interquartile ranges, and outlier behavior vary significantly depending on economic standing.

These trends, supported by both visual and quantitative evidence, strongly suggested that economic prosperity is a key enabling factor for improved health outcomes across the globe.

5.2 Outlier Identification and Interpretation

While the overall relationship was clear, several countries deviated from the general trend—offering deeper insight into the interplay of additional socio-political and healthcare-related factors.

- Japan consistently appeared as an outlier with very high life expectancy relative to its GDP per capita. This is likely attributable to strong healthcare access, lifestyle factors, and aging population structure.
- United States showed lower life expectancy than expected for a high-GDP country, suggesting that GDP alone is not sufficient to guarantee high health outcomes—factors such as healthcare inequality and chronic disease burdens play significant roles.
- Libya and Syrian Arab Republic demonstrated sharply declining or stagnated life expectancy trends post-2011, likely due to political instability and conflict, reinforcing the importance of contextual interpretation.
- Zimbabwe and Afghanistan appeared as low-income outliers with fluctuating trends, emphasizing challenges in healthcare infrastructure, access, and historical data consistency.

By identifying and acknowledging these outliers, we avoid oversimplifying the GDP-life expectancy relationship and highlight the importance of non-economic determinants such as political stability, healthcare quality, and public health systems.

5.3 Building the Narrative

The process of data wrangling and exploratory analysis gradually shaped a clear narrative: while there are exceptions, the global pattern strongly supports the notion that economic growth facilitates health improvements.

- **Data Cleaning and Transformation (Section 2 & 3):**
  Missing data—particularly for GDP, health expenditure, and life expectancy—were handled using customized imputation strategies, allowing for meaningful comparisons across all 176 countries (175 countries with Somalia excluded). This ensured the statistical and visual outputs in EDA were based on consistent and representative data.

- **Categorical Transformations (Section 2.5):**
  Variables such as income class and region were converted to ordered categorical types, enabling group-wise analysis and comparison, which became crucial for understanding disparities across groups.

- **Descriptive and Group Statistics (Section 3):**
  Statistical aggregates by income class and region highlighted key disparities and served as a precursor to the visual comparisons developed later. In particular, mean and median values showed sharp contrasts between low-income and high-income countries throughout the 2000–2015 period.

- **Exploratory Analysis (Section 4):**
  Each EDA sub-section revealed consistent reinforcement of the hypothesis:
    - Correlation heatmaps confirmed quantitative strength.
    - Scatterplots and line graphs visualized progressive shifts.
    - Boxplots and distribution analyses made disparities visually intuitive.
- **Income Class Focus:**
  During the course of the EDA, it became clear that income class was the most reliable and interpretable grouping for understanding global trends in life expectancy. This insight directly motivated the design of the final boxplot visualization—comparing life expectancy across income classes over time.

Conclusion of Section

Together, the insights from sections 3 and 4 provide strong foundational evidence for the final visualization, which brings these findings together in a clear, comparative format. The relationship between GDP per capita and life expectancy is not only statistically evident but also visually compelling. The next section will present this culminating visualization and analyze how it encapsulates the patterns and implications synthesized here.

6. Final Visualization and Interpretive Analysis

This section presents and interprets the most pivotal visualization in the project: a boxplot comparing life expectancy trends across income classes from 2000 to 2015. This graph was selected as the final visualization because it distills all prior exploration—descriptive statistics, correlations, and group-wise comparisons—into a single view that clearly communicates the primary hypothesis of this study:

Countries with higher GDP per capita generally exhibit higher life expectancies.

6.1 The Final Visualization: Life Expectancy by Income Class (2000–2015)

The boxplot below presents life expectancy distributions grouped by World Bank income classifications: Low Income, Lower-Middle Income, Upper-Middle Income, and High Income, across four selected years (2000, 2005, 2010, and 2015). These intervals were chosen to show progressive change over time while maintaining visual clarity.

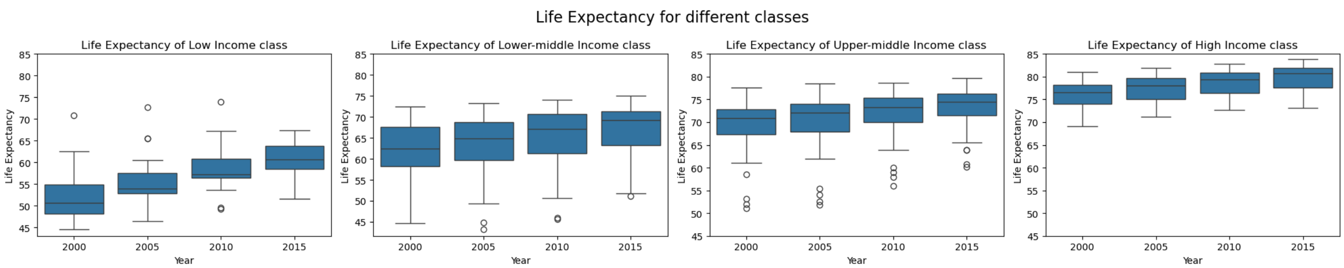


Figure 6.1: Life Expectancy Distribution by Income Class (2000–2015)

Boxplots comparing life expectancy across income classes for the years 2000, 2005, 2010, and 2015. Visualizes disparities and progress over time.

6.2 Insights from the Final Visualization

The boxplot highlights several critical insights:

- Clear and consistent ranking: For each year, the order of life expectancy from lowest to highest aligns exactly with income class ranking. Low income countries consistently show the lowest median life expectancy, while High income countries demonstrate the highest. This visually confirms the hypothesis.
- Upward shift over time: All income classes exhibit a general increase in life expectancy from 2000 to 2015, with noticeable upward shifts in median values and narrowing interquartile ranges (IQRs) in most classes—particularly in lower-middle and upper-middle income groups.

- Narrower distributions among wealthier nations: High income countries have tightly clustered distributions with minimal outliers, indicating relatively consistent health outcomes. In contrast, lower income groups show wider ranges and more variation—reflecting greater inequality in access to healthcare and socio-economic conditions.
- Reduction of extreme outliers: Over time, the prevalence of extremely low life expectancy values—primarily seen in low income countries—diminished slightly, suggesting some convergence toward higher baseline life expectancy globally.
- Subtle convergence patterns: While the absolute gap between high and low income medians remains substantial, the life expectancy gap has slightly narrowed from 2000 to 2015. This supports the notion that global health outcomes are improving, especially in emerging economies.

6.3 Hypothesis Validation

The boxplot provides strong and direct support for the hypothesis:

Countries with higher GDP per capita generally exhibit higher life expectancy.

This relationship is not only statistically validated in earlier correlation and regression analyses but is now also visually compelling in this final comparative visualization. The clear stratification of life expectancy by income class offers a straightforward and interpretable summary of how economic development—represented here by GDP-based classification—correlates with improved health outcomes.

Furthermore, the visualization reinforces the multi-dimensional nature of development, showing that wealthier nations tend to provide more consistent access to healthcare, nutrition, and public health infrastructure, leading to longer and more equitable life spans.

The slight narrowing of gaps over time provides additional insight into the progress of global health equity, even though substantial disparities still persist.

Conclusion of Section

This final visualization ties together the findings from the entire project, transforming abstract correlations into an intuitive and powerful visual narrative. It effectively communicates how economic class influences life expectancy and supports the broader understanding that policy, economic development, and public health are deeply intertwined.

In the next section, we reflect on the broader implications of these findings, along with potential limitations and areas for future research.

7. Conclusion and Discussion

This final section consolidates the findings of the life expectancy and GDP per capita analysis, offering a reflection on the broader implications of the results, the study's methodological limitations, and possible avenues for future research. The project aimed to examine the relationship between a country's economic strength and the health outcome of its population, using life expectancy as the central health metric and GDP per capita as the economic indicator.

7.1 Summary of Findings

Through an extensive process of data preparation, statistical summarization, and exploratory visualizations, the analysis yielded clear evidence supporting the initial hypothesis:

Countries with higher GDP per capita tend to have higher life expectancy.

Key findings that substantiate this conclusion include:

- **Positive Correlation:**
  Correlation matrices showed a moderate to strong positive relationship between GDP per capita and life expectancy across regions and income classes.
- **Visual Validation:**
  Scatterplots and line graphs reinforced the upward trend of life expectancy with rising GDP levels, particularly within middle- and high-income countries.
- **Disparity Across Classes:**
  Boxplots revealed stark differences in life expectancy distributions by income class, with high-income countries consistently outperforming others across all years studied (2000–2015).
- **Improving Global Trends:**
  All income classes experienced growth in life expectancy over time, with low- and lower-middle income countries showing the most pronounced improvements—albeit from lower baselines.
- **Outlier Insight:**
  Noteworthy deviations, such as Japan's longevity despite moderate GDP and the U.S.'s comparatively low life expectancy relative to its wealth, highlighted the complex interplay of healthcare policy, access, and social factors beyond economic strength alone.

Overall, the final visualization (Figure 6.1) encapsulated the evidence gathered, offering a holistic and interpretable illustration of how economic development is strongly associated with improved health outcomes globally.

7.2 Policy Implications

The findings from this study offer several insights that can inform **public policy and international development strategies**:

- **Targeted Health Investments:**
  For low- and lower-middle income countries, even modest gains in GDP per capita are associated with substantial improvements in life expectancy. This suggests that strategic investments in public health infrastructure, nutrition, sanitation, and education can yield high returns in human development.
- **Beyond GDP:**
  The study also reveals that GDP alone is not sufficient to ensure high life expectancy. Countries with high GDP but lower-than-expected life expectancy (e.g., United States) underscore the importance of equitable healthcare systems, preventive care, and social safety nets.
- **Closing the Gap:**
  As lower-income countries improve economically, efforts should focus on ensuring inclusive development that reaches vulnerable populations. This includes addressing rural health access, maternal and child health, and communicable disease burdens.
- **Global Cooperation:**
  International collaboration—through aid, knowledge exchange, and policy support—remains vital for accelerating health progress in developing regions and narrowing global inequality in life expectancy.

7.3 Limitations and Future Research

While the results of this study are compelling, several limitations should be acknowledged to contextualize the findings:

- **Data Gaps and Imputation:**
  Some countries had missing data for GDP and health expenditure indicators. Although imputation techniques were carefully implemented (including trend-based extrapolation and country-specific modeling), the results for those countries may carry a degree of uncertainty.
- **Temporal Limitation:**
  The dataset covers the period from 2000 to 2015. More recent data could reflect changes due to global events such as economic crises or the COVID-19 pandemic, which have had substantial effects on both GDP and life expectancy.
- **Simplification of Economic Indicators:**
  GDP per capita was used as the sole economic metric. Incorporating complementary indicators such as GNI, wealth inequality (Gini Index), or healthcare spending per capita might offer deeper insights.
- **Causality vs. Correlation:**
  The analysis identifies a strong relationship but does not prove causation. Many other variables—education, environment, healthcare policy, cultural norms—also influence life expectancy.

Future Research Recommendations

- **Extending Time Horizons:**
  Incorporate data from 2016 onward to assess whether the observed trends have continued or shifted in recent years.
- **Causal Modeling:**
  Employ regression models or machine learning techniques to isolate the impact of specific factors on life expectancy.
- **Region-Specific Deep Dives:**
  Perform focused analyses on particular regions (e.g., Sub-Saharan Africa or Southeast Asia) to explore regional health dynamics in greater depth.
- **Multivariate Analysis:**
  Include broader socio-demographic factors such as literacy rates, fertility rates, healthcare access indices, and environmental conditions.

Final Reflection

This project demonstrates that data-driven analysis can offer powerful narratives about global development. By carefully preparing, exploring, and synthesizing publicly available data, it becomes possible to uncover meaningful patterns, challenge assumptions, and support evidence-based policy thinking. The results underscore the importance of continuing international efforts to reduce health inequality and promote sustainable development worldwide.